# Simulation of Blood Flow:

## Mathematical Models, Numerical Methods, and an Idealized Stenosis Case Study

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**Abstract.**

This report reviews mathematical models and numerical evidence for blood-flow simulation, with stenotic flow used as the organizing example. The review is structured around the quantities that make a hemodynamic claim interpretable: the retained state, constitutive and wall closures, boundary data, discretization, observation operator, and evidence standard. It compares resolved three-dimensional CFD and FSI descriptions, axisymmetric and two-dimensional reductions, distributed one-dimensional area–flow systems, lumped networks, geometrical multiscale couplings, and learned surrogate or operator maps. The central synthesis is that model dimension is not an accuracy ladder. Each tier changes the state space, closure burden, native observables, and data requirements, so pressure, flow, velocity, WSS, and FFR-style quantities must be interpreted only after their mathematical and observational contracts are declared.

The report then applies this framework to an idealized stenosis case study. The case study specifies a reduced one-dimensional model, closure choices, finite-volume realization, manufactured-solution verification evidence, geometry-rest equilibrium test, and a plane–tetrahedron observation operator for diagnostic comparison with available resolved velocity data. Its role is methodological: it shows how verification, equilibrium preservation, reproducibility evidence, and cross-model observation maps constrain interpretation. It does not establish externally validated clinical prediction evidence, pressure-ratio evidence, or equivalence to a fully matched transient FSI problem. The principal contribution is therefore a review-led framework for reading mathematical hemodynamics results across model tiers while keeping numerical and empirical claim levels separate.

**Keywords:**

*blood-flow simulation, model hierarchy, stenosis, generalized-Newtonian rheology, reduced-order hemodynamics, multiscale modeling, finite-volume methods, verification and validation, observation operators*

# Contents

# 1 Introduction and Report Methodology

Blood-flow simulation sits between continuum mechanics, numerical analysis, and biomedical interpretation. A stenosis is a useful organizing example because a vessel narrowing changes velocity, pressure, wall loading, and downstream flow in ways that depend on the retained state and the observable being compared. A three-dimensional (3D) Navier–Stokes CFD model, an axisymmetric or two-dimensional (2D) reduction of that continuum description, a reduced one-dimensional (1D) area–flow model, and a lumped zero-dimensional (0D) outlet network therefore do not answer the same mathematical question. Anatomical narrowing and functional hemodynamic significance are also different observables: pressure–flow measurements and coronary imaging studies distinguish lesion appearance from functional assessment [1, p. 4401] [2, Abstract, Methods].

The primary purpose of this report is to review mathematical models and numerical methods used in blood-flow simulation. It organizes the literature around the questions that must be answered before a computed hemodynamic quantity can be interpreted: which continuum description is being solved or reduced, which model class is selected, which closures and data enter the equations, which quantity of interest is reported, and what numerical or empirical evidence supports the claim. The idealized stenosis computation developed later in the report serves as a worked numerical example; it illustrates the review framework but does not delimit the scope of the literature.

## 1.1 Literature Review Scope

The review is narrative and scoping in style rather than systematic. Sources were retained for the role they play in interpreting mathematical hemodynamics: derivation, taxonomy, numerical evidence, clinical or experimental observables, software reproducibility, or learned-model scope. Appendix H lists the source-inventory metadata; the main text uses those entries only to support bounded claims.

The inclusion criterion is relevance to the mathematical reading of blood-flow simulation: what equations are solved, which physics are retained or closed, what data are required, what observables are produced, and what evidence can support a comparison. Sources were retained only when they supported at least one of those questions for the model hierarchy, closures, numerical evidence standard, or 1D–3D observation operator. Derivation papers establish equations and assumptions; benchmark papers establish comparative numerical behavior; clinical and experimental studies address external relevance for specified observables; and software or repository materials support reproducibility rather than scientific truth of a model claim. The review does not claim exhaustive coverage of all clinical or computational hemodynamics.

Table 1: Primary source roles used in the narrative review.

| Source role | What it supports | Representative use |
|---|---|---|
| Derivation and theory | Governing equations, reductions, wall laws, coupling assumptions. | Continuum-to-1D reductions and multiscale coupling [3, 4]. |
| Review and taxonomy | Model classes, state variables, and clinical or numerical context. | 0D/1D hierarchy and pressure–flow network interpretation [5]. |
| Numerical benchmark or software | Scheme comparison, reproducibility, workflow traceability. | 1D benchmark cases and open finite-volume workflows [6, 7]. |
| Clinical or experimental observable | External meaning of pressure ratios, anatomy/function distinction, and measured wall or flow quantities. | Angiographic versus functional stenosis assessment [8, 9]. |
| Learned-model study | Performance within a declared training target, geometry family, and output observable. | PINN, neural differential, and operator-learning scope limits [10]. |

## 1.2 Research Problem and Questions

The research problem is to synthesize how blood-flow models are formulated, closed, observed, and supported by numerical evidence. The report asks three review-oriented questions:

1. What mathematical model hierarchies are used to simulate blood flow, and how do retained state, closure burden, coupling structure, and native observables distinguish 3D CFD, FSI, 2D or axisymmetric, 1D, 0D, multiscale, and learned formulations?
2. How do rheology, wall mechanics, geometry, inlet and outlet data, numerical discretization, evidence standards, and observation operators affect the interpretation of reported hemodynamic quantities?
3. What limitations remain across the literature when retained state, resolved scales, computational cost, practical identifiability, validation status, reproducibility, and uncertainty sources are compared?

## 1.3 Literature Review and Report Roadmap

The report's contribution is a structured review framework rather than a new clinical validation study. It first establishes the continuum vocabulary, then compares model tiers by retained state, closure burden, boundary data, and native observables. It next treats numerical methods through evidence standards: conservation, stability, positivity, well-balancing, verification, validation, reproducibility, and cross-dimensional observation operators. The synthesis chapter turns those ingredients into explicit propositions about what can and cannot be inferred from the literature.

Only after that review and synthesis does the report turn to the idealized stenosis case study. The case is a worked example of how a reduced model contract, closure choices, numerical evidence, and a 1D–3D velocity observation operator constrain interpretation. It illustrates the review framework; it does not expand the report into software documentation or clinical validation.

The main continuum symbols are used in their standard units unless a generated numerical table states otherwise: $\rho$ denotes density, $u_z$ axial velocity, $\eta$ dynamic viscosity, $\nu = \eta/\rho$ kinematic viscosity, $R$ radius, $A$ cross-sectional area, and $Q$ volumetric flow. The case study uses the physical variables $A_{\text{phys}}$ and $Q_{\text{phys}}$ when comparing with continuum quantities, but the implemented finite-volume state stores $a = R^2$ and $q = Q_{\text{phys}}/\pi$. Thus $A_{\text{phys}} = \pi a$, $Q_{\text{phys}} = \pi q$, and the section-mean velocity is $\bar{u} = q/a$. Appendix C collects the full symbol and unit table used by the report.

Section 2 supplies the continuum definitions used later. Sections 3–5 compare model tiers, closures, observables, discretizations, and evidence standards. Section 6 states the resulting synthesis propositions before Section 7 applies the framework to one idealized stenosis example. The final discussion answers the research questions directly and closes with the bounded review contribution.

# 2 Continuum Description of Blood Flow

This section fixes the continuum vocabulary used by the later model hierarchy. It begins with material motion and transport identities, then derives the balance laws, constitutive stress closures, and incompressible flow equations that serve as the parent continuum description. Resolved, reduced, lumped, multiscale, and learned models differ in which parts of this continuum description they retain, close, average, or approximate.

## 2.1 Continuum Scale and Field Variables

At the large-vessel scale, blood-flow models replace the cellular and plasma constituents of blood by continuum fields defined on the vascular fluid domain. In many such models, the Cauchy stress is closed by a Newtonian constant-viscosity constitutive law. This report reviews Newtonian and generalized non-Newtonian rheological assumptions and uses them to place three-dimensional CFD, fluid–structure interaction, and one-dimensional area–flow models within a common hierarchy. These rheological closures are modeling choices rather than intrinsic properties of blood. In small-vessel, hematocrit-dependent, or low-shear regimes, constant-viscosity Newtonian closure may be inadequate, and additional constitutive assumptions may be required [11, 12, 13].

**Assumption 2.1** (Large-vessel continuum setting)**.** Let $T_{\text{final}} > 0$, $I_T = (0, T_{\text{final}})$, and $\overline{I}_T = [0, T_{\text{final}}]$. For each $t \in \overline{I}_T$, the blood occupies a spatial lumen $\Omega_B(t) \subset \mathbb{R}^3$, with space-time cylinder

$$\mathcal{Q}_T = \{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}.$$

The continuum unknowns are the Eulerian velocity $\boldsymbol{u}$, pressure $p$, density $\rho$, and Cauchy stress tensor $\boldsymbol{T}$ on $\mathcal{Q}_T$.

The continuum fields in Assumption 2.1 provide the common language of the model hierarchy. A resolved continuum tier evolves $\boldsymbol{u}$ and $p$ on $\Omega_B(t)$. A reduced one-dimensional tier stores cross-sectional area, flow, and mean velocity obtained from these fields through averaging and closure assumptions. Dimensional reduction therefore changes the state space and closure requirements; it is not simply a cheaper discretization of the same unknowns.

## 2.2 Flow Maps and Material Transport

The next definitions supply notation rather than a separate modeling choice. They identify how material statements become the Eulerian equations used later. The Eulerian viewpoint observes fields at spatial locations $\boldsymbol{x}$ in the current lumen. The Lagrangian viewpoint follows a material label $\boldsymbol{\xi}$ from a reference configuration.

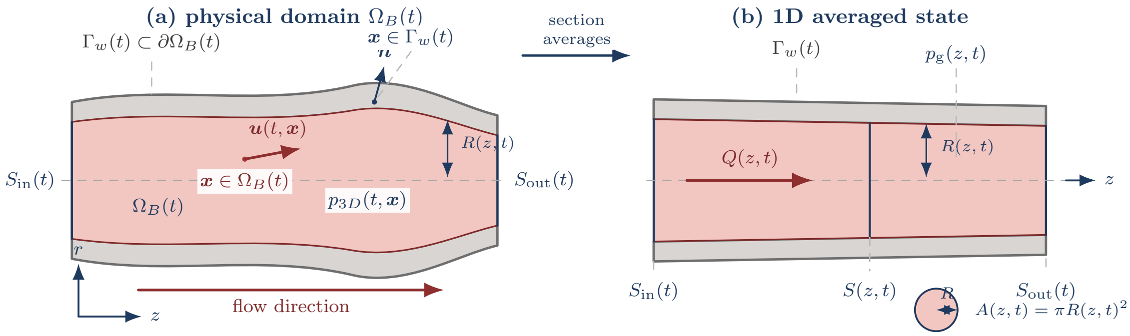

Figure 1: Continuum-to-1D averaging notation for a compliant vessel. The physical lumen carries three-dimensional fields on $\Omega_B(t)$; the reduced description defines section averages such as area, flow, mean velocity, and gauge pressure. The figure is a reference map between model tiers, not a separate wall-traction or secondary-flow model.

**Definition 2.2** (Flow-map vocabulary)**.** A sufficiently regular flow map $\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t)$ sends a material label $\boldsymbol{\xi}$ to its current position $\boldsymbol{x}(t; \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field generates the flow map through

$$\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi})).$$

For an Eulerian scalar field $\phi(t, \boldsymbol{x})$, the Lagrangian pullback is $\widehat{\phi}(t, \boldsymbol{\xi}) = \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$.

The material derivative is the derivative of a field along this trajectory. For a scalar field,

$$D_t \phi(t, \boldsymbol{x}) = \frac{d}{dt}\left[\phi(t, \mathcal{L}_t(\boldsymbol{\xi}))\right] = \partial_t \phi(t, \boldsymbol{x}) + \boldsymbol{u}(t, \boldsymbol{x}) \cdot \nabla \phi(t, \boldsymbol{x}), \qquad \boldsymbol{\xi} = \mathcal{L}_t^{-1}(\boldsymbol{x}).$$

For vector fields it is applied componentwise. This notation is the bridge between material-volume statements and fixed-spatial differential equations.

**Theorem 2.3** (Reynolds transport bridge)**.** *Let $V(t) = \mathcal{L}_t(V(0))$ be a material volume. For a regular scalar field $\phi$,*

$$\frac{d}{dt} \int_{V(t)} \phi \; d\boldsymbol{x} = \int_{V(t)} (D_t \phi + \phi \; div \boldsymbol{u}) \; d\boldsymbol{x} = \int_{V(t)} (\partial_t \phi + div(\phi \boldsymbol{u})) \; d\boldsymbol{x}.$$

This identity explains the transition used below: conservation statements are first written on material volumes and then localized into Eulerian balance laws.

## 2.3 Mass, Momentum, and Stress Closure

The balance laws below are the parent equations for the reduced models reviewed later. Mass conservation states that mass is neither created nor destroyed in material volumes. In Eulerian form it is

$$\partial_t \rho + \mathrm{div}(\rho \boldsymbol{u}) = 0 \quad \Longleftrightarrow \quad D_t \rho + \rho \mathrm{div} \boldsymbol{u} = 0.$$

For constant density $\rho \equiv \rho_0 > 0$, this reduces to the incompressibility constraint

$$\mathrm{div} \boldsymbol{u} = 0.$$

Linear momentum balance compares the rate of change of momentum in a material volume with surface traction and body forces:

$$\frac{d}{dt} \int_{V(t)} \rho \boldsymbol{u} \, dV = \int_{\partial V(t)} \boldsymbol{T} \hat{\boldsymbol{n}} \, dS + \int_{V(t)} \rho \boldsymbol{f_b} \, dV.$$

Using Reynolds transport, mass conservation, and localization gives

$$\rho \left( \partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla) \boldsymbol{u} \right) = \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}.$$

Here $\boldsymbol{f_b}$ is body force per unit mass.

The momentum equation becomes a closed flow model only after a constitutive stress law is selected. With

$$\boldsymbol{D}(\boldsymbol{u}) = \frac{1}{2} \left( \nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top \right),$$

the Newtonian isotropic law is

$$\boldsymbol{T} = -p \, \mathbf{I} + 2\eta \, \boldsymbol{D}(\boldsymbol{u}) + \lambda \, \mathrm{div}(\boldsymbol{u}) \, \mathbf{I},$$

where $\eta$ is dynamic viscosity and $\lambda$ is the volumetric-viscosity coefficient. In the incompressible case this reduces to

$$\boldsymbol{T} = -p \, \mathbf{I} + 2\eta \, \boldsymbol{D}(\boldsymbol{u}).$$

Generalized-Newtonian blood models keep the same stress structure but allow viscosity to depend on a scalar shear-rate measure:

$$\dot{\gamma} = \sqrt{2 \boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}, \qquad \boldsymbol{T} = -p \, \mathbf{I} + 2\eta_{\mathrm{eff}}(\dot{\gamma}) \boldsymbol{D}(\boldsymbol{u}).$$

The Newtonian model is recovered when $\eta_{\text{eff}}$ is constant [12, Part I, Ch. 1, Secs. 2.7–3.1]. This large-vessel Newtonian baseline is a closure assumption for averaged macroscopic behavior, not a microscopic statement about cells suspended in plasma. The scale caveat matters: apparent viscosity can be nearly constant in some large-vessel operating ranges, while diameter-, hematocrit-, and shear-dependent behavior becomes more important in small-vessel or low-shear settings [11, Sec. 1.1.3, Fig. 1.14] [13]. Patient-specific stenosis computations add an observable-specific caveat: Liu and coauthors reported little influence of rheology choice on a translesional pressure ratio in their intracranial stenosis setting, but observable differences in low-WSS regions, especially during diastole [14]. Rheology sensitivity is therefore neither globally negligible nor globally necessary; it must be stated with the vessel class, flow regime, and observable.

The displayed generalized-Newtonian form states the constitutive stress law used in the review, not a theorem-level regularity statement for every scalar viscosity curve. Positivity of $\eta_{\text{eff}}(\dot{\gamma})$ alone is not used here as a standalone parabolicity claim; such a claim would require stated lower-bound, growth, and monotonicity hypotheses for $2\eta_{\text{eff}}(\sqrt{2D : D})D$.

## 2.4 Incompressible Navier–Stokes Forms

Combining constant density, incompressibility, and constant-viscosity Newtonian stress gives the standard incompressible Navier–Stokes tier:

$$\begin{cases} \rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \eta\Delta\boldsymbol{u} + \rho\boldsymbol{f_b}, \\ \operatorname{div}\boldsymbol{u} = 0. \end{cases}$$

Equivalently, with kinematic viscosity $\nu = \eta/\rho$,

$$\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \nu\Delta\boldsymbol{u} + \boldsymbol{f_b}, \qquad \operatorname{div}\boldsymbol{u} = 0.$$

The generalized-Newtonian incompressible form replaces the constant-viscosity Laplacian by the divergence of the effective-viscosity stress:

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \mathbf{div}(2\eta_{\text{eff}}(\dot{\gamma})\boldsymbol{D}(\boldsymbol{u})) + \rho\boldsymbol{f_b}, \qquad \operatorname{div}\boldsymbol{u} = 0.$$

These equations are the full-order continuum reference for cardiovascular models; reduced arterial models inherit their conservation structure after geometric averaging and closure choices [15, Sec. 6].

## 2.5 Integral, Weak, and Moving-Domain Forms

The continuum equations also determine which numerical form is being discretized. A finite-volume method starts from a control-volume balance: for a conserved state $U$, flux $F(U)$, source $S(U)$, and fixed control volume $K$,

$$\frac{d}{dt}\int_K U\,d\boldsymbol{x} + \int_{\partial K} F(U)\cdot\hat{\boldsymbol{n}}\,dS = \int_K S(U)\,d\boldsymbol{x}.$$

A mixed finite-element incompressible-flow method instead starts from a weak velocity–pressure statement. On a fixed fluid domain with Dirichlet boundary part $\Gamma_D$ and traction boundary part $\Gamma_N$, a schematic Newtonian form asks for $(\boldsymbol{u}, p)$ in trial spaces satisfying the velocity boundary data such that, for velocity tests $\boldsymbol{v}$ and pressure tests $r$,

$$\begin{aligned}\int_{\Omega_B}\rho\,(\partial_t\boldsymbol{u}+(\boldsymbol{u}\cdot\nabla)\boldsymbol{u})\cdot\boldsymbol{v}\,d\boldsymbol{x}+\int_{\Omega_B}2\eta\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{v})\,d\boldsymbol{x}-\int_{\Omega_B}p\,\mathrm{div}\boldsymbol{v}\,d\boldsymbol{x}&=\int_{\Omega_B}\rho\boldsymbol{f_b}\cdot\boldsymbol{v}\,d\boldsymbol{x}+\int_{\Gamma_N}\boldsymbol{t}_N\cdot\boldsymbol{v}\,dS,\\ \int_{\Omega_B}r\,\mathrm{div}\boldsymbol{u}\,d\boldsymbol{x}&=0.\end{aligned}$$

The pressure space or boundary data must declare how the pressure gauge is fixed when pressure is determined only up to an additive constant. In a moving domain or ALE statement, the convective velocity is replaced by the relative velocity $\boldsymbol{u}-\boldsymbol{w}$ against the mesh velocity $\boldsymbol{w}$. A fluid–structure interface also imposes kinematic and traction compatibility, schematically

$$\boldsymbol{u}_f=\partial_t\boldsymbol{d}_s,\qquad \boldsymbol{T}_f\hat{\boldsymbol{n}}_f+\boldsymbol{T}_s\hat{\boldsymbol{n}}_s=\boldsymbol{0}.$$

Appendix E records these as bridge forms, not as a completed discretization or solver claim.

## 2.6 Cross-Sectional Bridge to Area–Flow Models

The distributed one-dimensional models reviewed later retain the axial conservation structure of the continuum equations but replace the full three-dimensional field by cross-sectional variables. This subsection names the averages and closures needed for that reduction. Let $S(z,t)$ denote the lumen section at axial coordinate $z$, with wall motion incorporated into the moving section. The physical area, physical flow, and section-mean axial velocity are

$$A(z,t)=\int_{S(z,t)}1\,dA,\qquad Q(z,t)=\int_{S(z,t)}u_z(z,r,\theta,t)\,dA,\qquad \bar{u}(z,t)=\frac{Q(z,t)}{A(z,t)}\quad(A>0).$$

When a section pressure is used later, it denotes a declared representative such as

$$\bar{p}(z,t) = \frac{1}{A(z,t)} \int_{S(z,t)} p(z,r,\theta,t)\, dA,$$

or a pressure reconstructed from a wall law. It is not a pointwise pressure unless the observation map explicitly says so. Applying mass conservation to a short vessel slice, with impermeable wall motion represented by the moving section, gives the reduced continuity law

$$\partial_t A + \partial_z Q = 0.$$

The axial momentum equation is not closed by $A$ and $Q$ alone. Its convective term contains the second axial-velocity moment, commonly written through the momentum-flux correction factor

$$\alpha(z,t) = \frac{A(z,t)}{Q(z,t)^2} \int_{S(z,t)} u_z(z,r,\theta,t)^2\, dA,$$

where the expression is interpreted through the corresponding velocity-profile closure when $Q = 0$. Suppressing model-specific closure details, the section-averaged axial balance has the structure

$$\partial_t Q + \partial_z \left( \alpha \frac{Q^2}{A} \right) + \frac{A}{\rho} \partial_z \bar{p} = \mathcal{S}_{\text{wall}} + \mathcal{S}_{\text{fric}} + \mathcal{S}_{\text{geom}} + \mathcal{S}_{\text{body}},$$

where $\bar{p}$ is a section pressure representative. The reduced model is therefore defined as much by its closures as by its conservation form: $\alpha$ and the friction term encode the assumed velocity profile and rheology, the pressure–area relation encodes wall mechanics, geometry terms encode axial variation of the reference radius or coordinate map, and inlet/outlet conditions close the finite vessel problem [3, 5, 16]. Formaggia, Lamponi, and Quarteroni also show why there is no single canonical area–flow system: different structural simplifications for the vessel wall lead to different one-dimensional differential operators [3]. Averaging loses transverse velocity, secondary flow, wall traction, and local stress information; those quantities can only re-enter through reconstruction, profile closure, or an observation operator. A velocity-profile closure supplies missing sub-section structure inside the reduced equations, whereas an observation operator maps a computed state to a reported pressure, flow, velocity, or wall quantity. Appendix E gives the supporting transport identities. The model hierarchy in Section 3 uses this cross-sectional reduction to explain why distributed 1D models inherit both conservation structure and closure obligations from the continuum tier.

## 2.7 Fixed and Moving Domains, Boundary Data, and Observables

A fixed-domain blood-flow model treats the lumen geometry as prescribed. In the simplest rigid-wall setting, $\Omega_B(t) = \Omega_B$ and the wall boundary carries a no-slip or no-penetration condition while inlet and outlet portions carry velocity, flow, pressure, traction, or reduced-network data. A moving-domain or FSI setting instead evolves $\Omega_B(t)$ through the wall motion and couples the fluid velocity to the wall kinematics and stresses. Dimension-reduced models replace parts of this boundary motion and stress information by section area, wall-law, profile, friction, and rheology closures [11, 15]. Three maps are being distinguished: the fluid material map follows blood particles, the structural map follows the wall or membrane reference state, and an ALE or mesh map moves computational coordinates. Pressure conventions must be declared with the same care. A pressure–area law may use gauge pressure, transmural pressure relative to an external pressure, or an absolute pressure offset; it may refer to reference area/radius or current area/radius. Wall shear stress is likewise a boundary traction projection with a wall-normal and projection convention, not a generic near-wall scalar [17].

The principal quantities of interest depend on the selected model tier and on the observation operator used to extract a scalar, profile, section average, boundary quantity, or time history from the computed field. Table 2 identifies which continuum outputs survive reduction as native states and which become closures, reconstructions, or observation maps when a continuum field is compared with a reduced stenosis model.

Table 2: Principal continuum observables and the corresponding reduced quantities used in stenosis modeling.

| **Continuum quantity** | **Typical observation** | **Reduced counterpart** |
|---|---|---|
| Velocity $\boldsymbol{u}$ | Point, profile, section mean, or resolved field. | Mean axial velocity $\bar{u} = Q_{\mathrm{phys}}/A_{\mathrm{phys}}$ and reconstructed profile summaries. |
| Pressure $p$ | Local pressure, pressure drop, or pressure history. | Section-averaged pressure or gauge pressure supplied by a pressure–area law. |
| Flow through a section | $Q_{\mathrm{phys}} = \int_{S(z,t)} u_z \, dA$. | Stored flow state and physical-flow reconstruction. |
| Wall traction and shear | Boundary stress or wall-shear measure. | Friction, profile, and wall closures rather than a resolved traction field. |
| Current geometry | Lumen area, radius, and wall displacement. | Section area, reference geometry, and wall-law relation. |

**Illustration from the idealized stenosis study.**

The idealized stenosis example later in the report uses this section only as a foundation. Its 1D state is meaningful because area, flow, mean velocity, and pressure are reductions of continuum fields. Its closure choices are necessary because transverse velocity, wall traction, wall motion, and rheological detail are not all solved directly. Its comparison quantities are continuum-derived observables: physical flow, section-mean axial velocity, pressure when supplied by the selected closure, and velocity-profile summaries.

The continuum definitions therefore constrain every lower model tier. A lower tier may be appropriate for a pressure–flow, pulse-wave, or screening question, but any non-native quantity has to be named as an average, constitutive state, reconstruction, or postprocessed observation before it can support a comparison.

# 3 Mathematical Model Hierarchy

The central organizing device of the review is the model hierarchy. The hierarchy is not a ladder of accuracy. It is a way to ask what a formulation retains, what it closes, what it can observe natively, and what it must map before a comparison is meaningful. The major review sources contribute different parts of that frame. Quarteroni and Formaggia place cardiovascular simulation inside a continuum and multiscale modeling program; Formaggia, Lamponi, and Quarteroni identify the variables and closures retained by distributed 1D artery models; Shi, Lawford, and Hose organize the 0D/1D network literature around pressure–flow states and terminal coupling; and Koppl–Helmig recast the same landscape as dimension reduction from detailed flow fields to lower-dimensional states [3, 5, 11, 15]. Read together, these sources motivate a hierarchy organized by retained state and observation, not by a simple claim that higher dimension is always the appropriate model. Here *retained state* means the variables advanced or constrained by the mathematical formulation itself. A *native observable* is computed directly from that state before any reconstruction, diagnostic closure, or cross-tier output map is applied.

The dimensional labels below therefore describe retained state and observation contracts, not merely mesh geometry. A 0D model stores lumped pressure, volume, or flow variables and is often used as a terminal or network closure for a distributed or resolved domain. A 1D vessel model advances axial balance laws for section area, flow, mean velocity, and wall-law pressure along a centerline [3, 5, 11]. An axisymmetric or 2D model still solves spatial PDEs, but it retains variation only in selected coordinates or imposes symmetry, suppressing the remaining direction. Resolved 3D CFD advances velocity and pressure fields on the lumen, while resolved FSI adds structural wall degrees of freedom and interface traction or kinematic coupling [15, 18]. Thus 0D, 1D, and 2D tiers are not simply alternative "3D CFD methods"; they are reduced or coupled formulations with different native states and observation maps.

## 3.1 Resolved 3D CFD and FSI Models

The resolved continuum tier starts from conservation of mass and momentum on a three-dimensional lumen. In a fixed-wall CFD model the wall is a prescribed geometric boundary, while in resolved FSI the wall has structural degrees of freedom and transfers kinematic and traction information through a moving interface [15, 18]. These models can expose local velocity gradients, wall traction, pressure fields, recirculation, separation, and geometry-specific flow structures. A WSS quantity, for example, is not just a scalar label: it requires a stress tensor, wall normal, tangent projection convention, and surface output map [17].

The resolved tier also contains distinct uses of "three-dimensional" data. Peskin's numerical analysis of blood flow in the heart is an early example in which geometry, moving boundaries, and numerical representation are already part of the mathematical problem, not merely input data [19]. Quarteroni–Formaggia-style cardiovascular modeling then broadens that perspective across continuum flow and system coupling, while

3D–1D FSI coupling work makes the interface and wall state explicit [15, 18]. By contrast, a fixed-wall resolved stenosis computation used to estimate a pressure or velocity diagnostic may retain the same local velocity and pressure fields while closing wall motion and downstream circulation differently. The source-to-source contrast is therefore not simply 3D versus reduced. It is which local fields are retained, which wall and boundary variables are closed, and which observable is extracted from the resulting field.

### 3.2 Axisymmetric and Two-Dimensional Reductions

Axisymmetric and two-dimensional models occupy an intermediate role. They retain more geometry and local velocity information than a one-dimensional area–flow model, but they remove at least one spatial direction or impose symmetry. Classical idealized stenosis studies use this controlled setting to separate obstruction, pressure loss, and flow-structure effects from patient-specific geometry and boundary uncertainty [20, 21]. Their contribution is therefore not only that they occupy a lower-dimensional category; it is that they show how idealization can isolate a stenosis mechanism.

The same reduction also imposes a limit. Axisymmetric and two-dimensional models remain useful for sectional diagnostics and controlled comparisons only when their boundary conditions, rheology, wall treatment, and observable definitions are declared. Suppressed secondary flow, asymmetric geometry, and wall mechanics cannot later be recovered as native outputs unless an additional closure or observation map supplies them.

### 3.3 Distributed One-Dimensional Area–Flow Models

Distributed 1D models reduce each vessel segment to axial variables, most often cross-sectional area, flow, mean velocity, and pressure. Formaggia, Lamponi, and Quarteroni make this state reduction concrete through area–flow equations, wall laws, and profile assumptions, while Koppl–Helmig emphasize how such models should be read as dimension-reduced representations with explicit closure and coupling choices [3, 11]. This tier preserves pulse propagation and vessel-to-vessel transport at far lower cost than resolved CFD/FSI, but profile, wall, friction, rheology, stenosis, and boundary effects enter as closures rather than resolved fields. It is therefore appropriate for single-vessel stenosis calculations when the target observables are area, flow, mean velocity, and reduced pressure, and when unavailable transverse flow, wall traction, and local stress fields are not required as native outputs.

Stenosis-specific reductions also divide by observable. Pressure-loss formulas and pressure-drop reduced-order models, such as the Lyras–Lee pressure-drop ROM, target obstruction through a flow–loss relation [22]. Extended 1D stenosis formulations instead place narrowed reference geometry and wall response inside an area–flow system [16]. The Canic et al. arXiv preprint reconstructs radial velocity only in post-processing, so that radial field is not a native 1D state. Network-scale coronary models add tree structure and terminal assumptions, so the Duanmu coronary-tree contribution is not just another single-vessel stenosis model; it shows how reduced vessel states become part of a larger pressure–flow network [23]. These sources should therefore be compared by target observable and closure contract, not collected under a single "1D" label.

### 3.4 Zero-Dimensional Networks and Multiscale Coupling

Zero-dimensional and network models replace distributed fields with lumped states and algebraic or ordinary-differential pressure–flow relations. The Shi–Lawford–Hose review is useful here because it treats 0D and 1D models as circulation models with explicit resistance, compliance, junction, and terminal assumptions, rather than as merely cheaper versions of resolved CFD [5]. Their state variables are not local wall or velocity fields. In a single-vessel calculation, these models most often enter through inlet or outlet closure rather than as the main spatial solver.

Geometrical multiscale models connect tiers, for example 3D–1D, 3D–0D, or 1D–0D, through interface conditions that transfer pressure, flow, traction, impedance, or characteristic information [4, 18]. Quarteroni and Veneziani foreground the ODE–PDE coupling problem, while Formaggia, Moura, and Nobile expose the additional stability and interface issues that arise when the coupled tiers include FSI. The mathematical issue is therefore not only that two solvers are coupled; it is that the interface variables must have compatible meanings across dimensions. Coronary FFR comparisons between 1D and 3D models show the same issue on the output side: the reduced and resolved states must be mapped to a common pressure-ratio observable before agreement or disagreement has a mathematical meaning [9].

### 3.5 Learned Surrogates, PINNs, and Operator Maps

Surrogate and scientific-ML models add a second axis to the hierarchy. The general PINN framework constrains an approximate solution or inverse map with residual, boundary, data, and regularization terms, while hemodynamics-specific reviews ask how that framework translates to vascular flow settings [10, 24]. DeepONet and Fourier neural operators shift the claim from one computed solution to a learned map between functions or parameterized fields [25, 26]. Hemodynamics-specific neural differential and deep-operator examples then target reduced blood-flow or stenosed-vessel maps directly. Csala and coauthors report conditional error reductions in an idealized stenosis/waveform setting, while Velikorodny and coauthors derive FFR after learning pressure and flow maps in a simplified stenosis setting [27, 28]. These learned categories do not remove the need to state the underlying model tier: every learned-model claim should identify its training target, input functions or parameters, output observable, geometry family, and evaluation domain.

### 3.6 Native States and Cross-Tier Observation Maps

Tables 3 and 4 summarize the hierarchy in two steps. The first table summarizes retained state, reduction, closure, and data requirements. The second summarizes observables, evidence role, and interpretation limits. The split keeps the reading frame organized around state and observation rather than around individual papers. Read each row as a claim contract: what the tier solves natively, what it closes, what it can report directly, and which evidence role its outputs can support.

Table 3: State, reduction, closure, and data contract for cardiovascular model tiers.

| Tier | Native state and form | Reduction | Closure burden | Data or coupling requirement |
|---|---|---|---|---|
| Fixed-wall 3D CFD | $\boldsymbol{u}, p$ on a prescribed lumen; Navier–Stokes or Stokes PDE. | None in space. | Rheology, wall boundary, numerics. | Geometry and inlet/outlet data. |
| Resolved FSI | Fluid fields plus wall state; coupled fluid/structure PDEs. | Structural idealization only. | Wall material, interface, mesh/ALE. | Wall data and coupling histories. |
| Axisymmetric / 2D | Symmetry-constrained velocity and pressure; reduced spatial PDE. | Suppressed azimuthal/asymmetric modes. | Symmetry, wall, rheology, boundaries. | Radius/profile data. |
| Distributed 1D | $A, Q, \bar{u}$, pressure/wall state; axial balance law. | Section averaging. | Profile, wall, friction, geometry, rheology. | Inlet/outlet, junctions, terminal beds. |
| Lumped 0D | Pressures, volumes, flows; ODE/algebraic network. | Spatial distribution removed. | Resistance, compliance, inertance, valves. | Network topology and terminal data. |
| Geometrical multiscale | Tier-specific states plus interfaces; coupled PDE/ODE systems. | Local/system split. | Interface compatibility and stability. | Pressure, flow, traction, impedance. |
| Learned surrogate | Learned solution, parameter, or operator map; trained objective. | Inherits source model/data tier. | Architecture, loss, training distribution. | Training inputs, geometry family, labels. |

Table 4: Observable, evidence, and interpretation limits for cardiovascular model tiers.

| Tier | Native observables | Reconstructed or diagnostic outputs | Evidence role and limit |
|---|---|---|---|
| Fixed-wall 3D CFD | Velocity and pressure fields. | WSS indices and section means. | Supports local-field verification; wall motion and circulation are closed. |
| Resolved FSI | Fluid fields, wall displacement, and interface traction. | Scalar WSS or pressure diagnostics. | Supports coupled-interface analysis; many data and material inputs must be identified. |
| Axisymmetric / 2D | Symmetry-constrained sectional fields. | 3D asymmetry proxies. | Supports mechanism isolation; suppressed modes are not native. |
| Distributed 1D | Area, flow, mean velocity, and wall-law pressure. | WSS proxies and radial-profile reconstructions. | Supports pulse or stenosis comparison; local traction and transverse flow are closed. |
| Lumped 0D | Global pressure–flow states. | Local fields only by added model. | Supports system interaction; stenosis geometry and WSS are not native. |
| Geometrical multiscale | Tier-native and interface values. | Cross-tier diagnostics. | Supports coupled-domain consistency; interface maps can dominate interpretation. |
| Learned surrogate | Declared learned target. | Derived diagnostics such as FFR. | Supports in-distribution generalization tests; credibility is bounded by the data-generating contract. |

Cross-tier comparisons are therefore observed-output comparisons:

$$Y_{1D} = \mathcal{O}_{1D}(U_{1D}), \qquad Y_{3D} = \mathcal{O}_{3D}(\boldsymbol{u}, p, \Omega_B), \qquad \delta Y = Y_{1D} - Y_{3D}.$$

Unless the comparison design separates them, $\delta Y$ combines model-form, closure, parameter, numerical, data-matching, and observation-operator effects. Blanco's 1D–3D FFR study is read here as an FFR-specific matched-contract comparison with pressure boundary matching and dedicated stenosis/bifurcation losses, not as evidence of general model-tier equivalence [9].

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**Illustration from the idealized stenosis study.**

The case study later in the report sits in the distributed 1D branch. Its mathematical state is a single-vessel area–flow system with an analytic stenosis reference geometry and reduced wall/friction/profile closures. The case therefore belongs beside other 1D stenosis models, including extended variable-radius formulations [16], rather than beside resolved CFD or FSI as a native wall-traction or wall-displacement solver. When it is compared with resolved data, the comparison passes through an observation map from 1D area–flow variables to the chosen velocity, flow, or pressure quantity.

---

# 4 Closure Choices, Data, and Hemodynamic Observables

Choosing a model tier does not close a hemodynamic problem. The mathematical state must be paired with constitutive, wall, geometric, boundary, data, and observable conventions. These choices determine which terms appear in the equations and which reported quantities are native states, derived fields, diagnostics, or cross-tier observation maps. They are therefore part of the mathematical model, not secondary implementation detail.

## 4.1 Rheology and Effective-Viscosity Modeling

Blood is often modeled as an incompressible Newtonian fluid in large-vessel simulations, especially when the target quantities are dominated by moderate to high shear rates and the uncertainty in geometry or boundary data is already large. Non-Newtonian categories remain important in stenotic, small-vessel, or low-shear regimes because viscosity may depend on shear rate, hematocrit, and diameter. Pries, Neuhaus, and Gaehtgens make the point from tube-flow evidence: apparent blood viscosity is not a single universal constant once vessel scale and hematocrit enter the measurement [13]. Carreau and Carreau–Yasuda laws therefore do more than fill a taxonomy slot; they encode shear-thinning with limiting viscosities. Casson laws introduce yield-stress-like behavior, and power-law models express scale-free shear dependence. Computational studies of Carreau–Yasuda and intracranial stenosis rheology then show the consequence for reporting: changing the viscosity law can alter the hemodynamic quantity being interpreted, so the selected rheology must be carried with the observable [14, 29].

**Definition 4.1** (Effective-viscosity closure catalog)**.** A reduced realization that selects a rheology family $\mathcal{R}$ must state the scalar effective dynamic-viscosity law $\eta_{\text{eff}}^{\mathcal{R}}(\dot{\gamma})$ before its friction, pressure, or shear diagnostics can be interpreted. At the review level, the closure contract consists of a dynamic-viscosity law, the shear-rate proxy on which it is evaluated, the conversion $\nu_{\text{eff}}^{\mathcal{R}} = \eta_{\text{eff}}^{\mathcal{R}}/\rho$, and any regularization or bounds used near zero shear. The case-study implementation names these rheology families as

`newtonian`, `carreau`, `carreau-yasuda`, `casson`, `power-law`.

Appendix G.1 gives the case-study parameterization and regularization formulas. The listed families are closure options used to make reduced rheology choices explicit, not universal blood constants.

## 4.2 Wall Laws and Structural Assumptions

Wall modeling is a separate closure axis. A fixed-wall model prescribes the lumen boundary and solves only the fluid problem. A reduced compliant-wall model uses an algebraic or differential pressure–area relation to close a 1D or network state. A resolved FSI model solves structural wall motion and couples it to the fluid at the interface [3, 18]. The source contrast matters. Formaggia, Lamponi, and Quarteroni use a wall law to close reduced pressure–area dynamics, while Formaggia, Moura, and Nobile treat wall motion as an interface-coupled FSI state. Clinical imaging work on coronary distensibility reinforces the same distinction from the observable side: wall compliance can be a measured or estimated quantity rather than a hidden numerical constant [30]. The same rheology can be paired with any of these wall choices, so rheology and wall mechanics should not be merged into a single modeling label.

Membrane terminology belongs to the structural side of this split. A Newtonian or generalized-Newtonian law describes the fluid stress, whereas an isotropic elastic wall law describes how the vessel wall resists deformation. In this report, a membrane model is a thin-wall structural closure: through-thickness stress and bending are collapsed into a surface or centerline relation in which fluid traction or transmural pressure drives normal or radial displacement and elastic stiffness resists it. The quasi-static Canic-style relation used later, $C_0\eta_r = f_r$, is this reduced radial balance in wall-law variables. It can provide a wall-motion scale or structural closure, but it is not by itself a full transient moving-boundary FSI model.

A wall-law statement must also say which configuration and pressure convention it uses. Reference area or radius belongs to the unloaded, imaged, or otherwise declared baseline geometry; current area or radius belongs to the deformed state being solved or reconstructed. Pressure may be absolute, gauge, or transmural, and a transmural law must state the external pressure convention. Thickness, elastic modulus, prestress or residual stress, and viscoelastic memory change the meaning of the same pressure–area curve. Reduced wall laws therefore differ from resolved structural states: the former close pressure, area, and wave speed inside a 1D model, while the latter solve wall motion and stress. Wall parameters may also compensate for geometry or boundary uncertainty, so they are not automatically identifiable from pressure and flow alone.

## 4.3 Stenosis Geometry and Reference-State Choices

A stenosis can enter the mathematics as a resolved lumen narrowing, an axisymmetric radius profile, a 1D reference-area variation, an empirical pressure-loss relation, or a network resistance. Distributed 1D formulations make the reference area and its derivatives part of the balance-law closure, while resolved models represent the same narrowing through the computational domain. The extended 1D stenosis formulation is important here because it argues that variable-radius effects can alter the reduced balance law itself; they are not captured by simply inserting a narrowed healthy-radius profile into an otherwise standard area–flow model [16]. Geometry is therefore not only a mesh or image artifact: it determines whether throat effects appear through domain shape, source terms, pressure loss, or lumped parameters.

Classical idealized stenosis studies occupy the corresponding evidence role. Young and Tsai's steady and unsteady model-stenosis experiments connect controlled narrowing to pressure-loss and flow-structure effects under declared geometry and flow conditions [20, 21]. They are therefore useful evidence for obstruction mechanisms and pressure-loss framing, not patient-specific stenosis validation or clinical severity inference.

Severity must also be defined by radius, diameter, or area rather than treated as interchangeable. A 23% radius reduction, diameter reduction, and area reduction imply different lumen changes. Patient-specific geometry adds another layer: Gambaruto and coauthors found, in one cerebral aneurysm case, that reconstructions differing within imaging resolution produced sensitivity greater than but comparable to rheology choice and WSS differences of 3–18% [31]. That result is case-specific, but it supports the broader reporting rule: geometry reconstruction is part of the model uncertainty. Idealized geometry is useful for mechanism isolation; patient-specific geometry is required for patient-specific inference.

### 4.4 Inlet, Outlet, and Initial Data

Boundary data are model dependent. Inlets may be prescribed as velocity profiles, mean-flow waveforms, pressure histories, or characteristic incoming data. Outlets may be prescribed as pressure, traction, resistance, Windkessel, impedance, reflection, or characteristic conditions. In multiscale models, the same words also name interface contracts between tiers, not only external boundary conditions [4, 5]. Quarteroni and Veneziani treat such data as coupling variables in a geometrical multiscale system, while Shi, Lawford, and Hose emphasize how network and terminal choices close lumped or distributed pressure–flow states. For a single-vessel problem, the boundary is therefore not just a geometric edge. A resolved 3D problem may prescribe velocity, flow, traction, pressure, or wall-coupling information on inlet, outlet, and wall portions, while a 1D model applies incoming information to area–flow or characteristic variables. The case study later states its boundary rule as part of its model contract, because boundary data close the mathematical problem. For subcritical 1D flow, the conceptual characteristic count is one incoming piece of information at each end of a single vessel. Prescribed pressure, prescribed flow, traction, impedance/Windkessel, characteristic, and multiscale coupling data therefore impose different mathematical information. Boundary conditions can encode omitted vascular dynamics, and initial transients must be distinguished from a periodic or statistically steady operating assumption.

### 4.5 Pressure, Flow, Velocity, WSS, and FFR Observables

Observables should be named at the level where they are actually defined. Pressure may be an absolute pressure, gauge pressure, cross-sectional mean pressure, diagnostic pressure reconstructed from a wall law, or terminal compartment pressure. Flow may be a surface integral, a 1D state variable, or a network edge variable. Velocity may be a resolved vector field, a section average, an assumed radial profile, or a sampled centerline quantity. WSS is a traction-derived wall quantity requiring stress and wall geometry, not a generic name for any reduced shear proxy [17]. Displacement belongs to a wall or structural state, so it is native to resolved FSI or a structural model and only indirectly represented by a reduced pressure–area wall law.

Clinical pressure-flow quantities add another observable layer. FFR literature separates anatomical stenosis severity from functional severity. The FAME trial source supplies the clinical contrast between angiographic appearance and functional assessment, while FFR overview literature defines the diagnostic as a pressure-ratio quantity rather than a generic flow-model output [8, 32]. CT-derived FFR sources then shift the same diagnostic into an image-derived computational pipeline, so the observable depends on segmentation, boundary assumptions, solver model, and pressure-ratio extraction [33, 34]. The 1D–3D FFR comparison literature makes the cross-tier version of the point: a pressure-ratio scalar can be compared across model classes only after both states have been observed through compatible pressure and flow conventions [9].

The operator forms used in the report are intentionally explicit:

$$Q = \int_S \boldsymbol{u} \cdot \hat{\boldsymbol{n}}_S \, dA, \qquad \bar{u} = Q/A, \qquad \bar{p} = A^{-1} \int_S p \, dA,$$

where $S$ is the declared section. A pressure drop is $\Delta p = p_{\mathrm{up}} - p_{\mathrm{down}}$, and a pressure ratio is $p_{\mathrm{down}}/p_{\mathrm{up}}$ only after the pressure convention and locations are declared. A WSS vector is the tangential projection of $\boldsymbol{T}\hat{\boldsymbol{n}}$ on the wall, with scalar indices such as AWSS or OSI derived only after the vector convention is fixed. An FFR-style pressure ratio should not be called FFR unless the physiological, boundary, and diagnostic assumptions of the FFR setting are part of the model contract.

Table 5: Closure-contract axes that should remain explicit before comparing stenotic-hemodynamics models.

| Closure | Mathematical location | Required declaration | What this does not establish |
|---|---|---|---|
| Rheology | Stress law or 1D friction source. | $\eta$ or $\eta_{\mathrm{eff}}(\dot{\gamma})$ and proxy. | A viscosity law alone does not determine pressure-ratio or WSS sensitivity; the observable and regime must be stated [14]. |
| Velocity profile | Momentum flux and wall friction. | Profile shape, $\alpha$, shear factor. | A profile closure does not produce a native resolved radial field or wall traction. |
| Wall mechanics | Pressure–area law or structural PDE. | Reference/current geometry, stiffness, external pressure. | A wall law does not identify stiffness, prestress, or material memory unless those data are independently constrained. |
| Geometry/reference state | Domain, reference radius, source terms, loss model. | Segmentation or analytic radius and severity definition. | A shared nominal severity does not remove reconstruction or reference-state uncertainty [31]. |
| Inlet data | Boundary state or interface. | Flow, pressure, velocity profile, characteristic data. | A prescribed inlet does not model omitted upstream dynamics unless it is supplied by a coupling model. |
| Outlet data | Boundary state, terminal ODE, or impedance. | Pressure, traction, resistance, Windkessel, characteristic data. | A terminal label does not determine wave reflection or distal circulation without its parameters and assumptions. |
| Junction law | Network interface. | Conservation and pressure/energy condition. | A junction condition does not make variables compatible across tiers unless pressure, flow, and energy conventions match. |

Table 6: Reporting conventions for reduced and cross-tier observables.

| Observable convention | Required declaration | Affected output | What this does not establish |
| --- | --- | --- | --- |
| Pressure convention | Gauge/absolute/transmural reference. | Pressure drop and ratio. | A pressure convention does not define external pressure or upstream/downstream sites by itself. |
| Flow or velocity map | Section, coordinate mode, quadrature, and averaging rule. | Flow, mean velocity, and discrepancy summaries. | A section mean does not establish resolved secondary-flow or WSS agreement. |
| WSS convention | Stress, wall normal, projection. | WSS, AWSS, OSI. | A scalar WSS index does not identify the vector convention unless the traction projection is declared [17]. |
| FFR-style observation | Upstream/downstream pressures and physiological contract. | FFR or FFR-style result. | A pressure ratio should not be called FFR without the physiological and boundary contract [9]. |

**Illustration from the idealized stenosis study.**

The case study later in the report uses this closure framework in a bounded way: Newtonian viscosity for the principal comparison; a parabolic velocity profile for the main area–flow closure; an elastic pressure–area wall law; a smooth analytic stenosis reference radius; a prescribed reference-area inlet flow; and a fixed-area characteristic outlet approximation. The native states are reduced area and flow, with mean velocity and pressure obtained through declared coordinate and wall-law maps. The resolved-data comparison therefore reads as a velocity-observation comparison, not as a direct WSS or wall-displacement comparison.

# 5 Numerical Methods and Evidence Standards

Numerical evidence for stenotic-hemodynamics models is inseparable from the method that produces the state, the reference data used to check it, and the operator used to observe it. A resolved incompressible-flow or FSI computation, a distributed 1D area–flow network, and a diagnostic postprocessing map can all report velocity or flow, but they do so through different discrete spaces, stability constraints, conservation properties, and measurement conventions. This chapter therefore reviews numerical methods through the evidence standards needed to interpret blood-flow simulations: verification, validation, stability, positivity, conservation, well-balancing, reproducibility, and observation operators.

The notation below separates the mathematical model from the discrete realization:

$$
\begin{aligned}
&\mathcal{M}(U) = 0, && \text{continuous model,}\\
&M_h \dot{U}_h = R_h(U_h), && \text{semidiscrete equations,}\\
&U_h^{n+1} = \Phi_{\Delta t,h}(U_h^n), && \text{fully discrete scheme.}
\end{aligned}
$$

A reported numerical quantity is then

$$Y_{h,\Delta t} = \mathcal{O}_h(U_h^n),$$

with an observation map $\mathcal{O}_h$ that carries its own quadrature, coordinate, and output conventions. The same semidiscrete notation also names the local coupling pattern. For a residual component $R_{h,i}$, define its spatial stencil as

$$\mathcal{S}_i = \{j : \ R_{h,i}(U_h) \text{ depends on } U_{h,j}\}.$$

Thus a stencil is the set of discrete degrees of freedom that enter one residual equation. Finite-difference methods approximate differential equations at grid point values by replacing derivatives with difference quotients, so their stencils are the point values used in those quotients [35]. Finite-volume methods approximate integral conservation laws over cells or control volumes; their stencils are the cell averages and reconstructed interface states used to balance numerical fluxes across cell faces [36]. Finite-element methods approximate weak forms after restricting trial and test functions to finite-dimensional spaces; their stencils are induced by basis functions whose supports overlap in the variational forms [12]. Time integration changes how these residuals are evaluated or solved; it does not by itself change the spatial dependencies encoded in $R_h$, although an implicit step solves the coupled algebraic equations together.

## 5.1 Finite-Element Incompressible Flow and FSI

With the continuum fields and balances fixed in Section 2, a resolved finite-element calculation discretizes the corresponding weak incompressible-flow or FSI problem. The numerical claim is therefore a mixed weak-form and interface-coupling claim, not simply a mesh claim. A finite-element realization chooses discrete velocity and pressure spaces and solves the resulting saddle-point problem. Several numerical contracts have to be kept separate. First, when the continuous problem determines pressure only up to an additive constant, the discrete problem needs a pressure gauge, quotient space, or equivalent null-mode constraint; that is a pressure-uniqueness convention. Second, the velocity–pressure pair must satisfy an appropriate discrete inf-sup condition, or it must include pressure stabilization such as pressure projection, PSPG, or local-projection stabilization. Stable Taylor–Hood-type pairs and suitable divergence-conforming constructions address this mixed-space requirement directly, while equal-order pairs generally need an added pressure-space stabilization contract. Third, advective stabilization such as SUPG addresses convection-dominated transport behavior. Fourth, grad-div terms improve discrete divergence control. SUPG and grad-div terms do not by themselves repair an inf-sup-unstable pressure space. Galdi and coauthors are useful here not as a generic FEM citation, but because their hemodynamical-flow treatment ties continuum equations, mixed discretization, stabilization, and simulation evidence together [12]. Package-facing Gridap wording in this report is therefore bounded to focused contract and smoke-test evidence for continuous P2-velocity/P1-pressure routes. That evidence checks the local mathematical contract; it is not a manuscript-grade reproduction or production-execution claim. At the algebraic level, this local dependency appears as sparsity. For element-local Galerkin terms, a stiffness or coupling block associated with basis functions $\phi_i$ and $\phi_j$ is nonzero only when their supports meet on an element; local stabilization terms may enlarge the patch, but the row pattern is still induced by mesh connectivity rather than by all degrees of freedom at once.

When the wall moves, the numerical contract also includes the fluid–structure interface. FSI formulations must transfer traction and velocity or displacement data across the interface, update the fluid domain or mesh, and control the added-mass and time-lag effects that appear in partitioned coupling. Monolithic and partitioned schemes can both be useful, but the FSI and multiscale coupling sources show why they expose different solver errors, stability limits, and coupling residuals [4, 18]. For a reduced 1D comparison, a resolved FSI or CFD field is therefore not an unqualified comparison target; it is a numerically produced state that still needs its own mesh, timestep, boundary, material, and output-operator specification.

## 5.2 Finite Differences, Finite Volumes, DG, and High-Resolution Schemes

One-dimensional blood-flow solvers are usually written as balance-law solvers, so their numerical evidence must address conservation, sources, and wave propagation together. Finite-difference schemes approximate derivatives at grid points and can be compact and high-order, but conservation and geometric-source balance must be checked at the discrete level. Finite-volume schemes instead advance cell averages using numerical fluxes through cell faces. That structure makes local conservation visible and is well suited to wave propagation, boundary fluxes, and discontinuity-capturing limiters [36]. In arterial networks, Wang's 1D network treatment adds the next layer: each vessel solve must be compatible with node and terminal rules that determine incoming and outgoing characteristic information [37]. The method comparison is therefore not finite volume versus network modeling; it is how a conservative vessel update is embedded in a coupled vascular system.

Stencil language keeps the distinction mathematical. A centered finite difference for a second derivative on a one-dimensional grid uses the familiar three-point stencil $\{i-1, i, i+1\}$. A first-order finite-volume update for cell $I_i = [z_{i-1/2}, z_{i+1/2}]$ uses the interface fluxes at $z_{i-1/2}$ and $z_{i+1/2}$, so the residual for cell $i$ depends on the neighboring cell states used to evaluate those two fluxes and on local source terms. Higher-order finite-volume or finite-difference reconstructions enlarge this set by drawing on additional neighboring cells while retaining a local residual structure.

Discontinuous Galerkin methods combine finite-volume fluxes with local polynomial approximation. They can raise spatial order, expose element-local residuals, and support $p$-refinement studies, but they still require numerical fluxes, limiters or troubled-cell controls, and timestep choices. WENO and other high-resolution reconstructions serve a related purpose for finite-volume and finite-difference methods: they recover high-order accuracy on smooth regions while suppressing nonphysical oscillations near steep gradients. In stenotic area–flow problems, these design choices interact with wall and geometry sources, so smooth-solution accuracy alone is not enough. A scheme can show good manufactured-solution behavior and still fail to preserve a physiologically meaningful rest state if flux and source terms are not balanced.

### 5.3 Stability, Positivity, Conservation, and Well-Balancing

Stability and well-balancing are evidence claims about the discrete operator, not only runtime settings. Explicit wave-propagation schemes carry CFL restrictions: the timestep must stay compatible with the grid scale and characteristic speeds. Semi-implicit and implicit network methods can relax the fastest explicit restriction, especially when wave speed or network coupling is stiff, but they replace it with nonlinear or linear solver tolerances, splitting errors, and coupling-residual criteria. The benchmark and open-solver literature makes this evidentiary rather than only algorithmic: Boileau and coauthors compare numerical schemes on shared 1D arterial cases, OpenBF exposes a finite-volume solver description, and recent well-balanced or Lax–Friedrichs studies show how boundary and coupling policies affect the accepted run. Such workflows specify not only the equation family, but also the timestep policy, boundary assumptions, network coupling, and diagnostic metadata [6, 7, 38, 39].

For compliant-vessel balance laws, three checks are especially direct. First, positivity guards ensure that area or radius states remain physical; any projection or lower-bound correction must be counted because it changes the discrete solution. Second, conservation diagnostics test whether the finite-volume balance is consistent with boundary fluxes and source terms. Third, well-balanced tests ask whether a target equilibrium is preserved by the discrete operator itself. High-order well-balanced schemes make this requirement explicit by reconstructing or integrating flux and source contributions so that selected rest states cancel after discretization, not only in the continuous equations [40]. The 2023 JCP article targets all stationary solutions of a model with discontinuous geometrical and mechanical properties, so the evidence role is the preserved stationary set, not a generic high-order accuracy statement. For the present report, geometry-rest preservation is therefore a core numerical property rather than a postprocessing preference or a cosmetic refinement of an otherwise verified scheme.

## 5.4 Verification, Validation, and Reference Data

ASME-style VVUQ terminology separates credibility claims [41]. Code verification concerns whether the implementation conforms to the mathematical description. Solution verification concerns numerical error in a computed quantity. Validation concerns whether a model represents a specified real application for a specified observable and context of use. Uncertainty quantification concerns the effect of uncertain numerical and physical inputs. These categories are related, but none is a substitute for another.

Verification asks whether the implementation solves the equations it claims to solve. Manufactured solutions are useful because the analyst chooses a smooth state, derives the forcing needed to make that state exact, and then checks the discrete error under grid and timestep refinement. Exact or manufactured states can probe fluxes, sources, time integration, boundary traces, and coordinate maps without requiring physiological data. They also expose whether a nominally high-order method is limited by boundaries, source quadrature, limiters, or operator splitting.

Grid and timestep convergence studies then ask whether changes in resolution change the reported quantities in the expected direction and magnitude. These studies are strongest when spatial and temporal effects are separated; otherwise they are better reported as sensitivity or time-step-insensitivity checks. Benchmark problems provide a broader reproducibility context by comparing multiple solvers on shared model cases. The Boileau benchmark is strongest in that role: it helps interpret scheme behavior under common assumptions, but a benchmark row does not by itself establish accuracy for a different stenosis geometry, wall law, or boundary rule [6].

Open modeling platforms and model repositories occupy a related but distinct evidence role. A platform such as SimVascular supports inspectable model construction, meshing, simulation, and postprocessing workflows, while a vascular model repository provides reusable geometry and model descriptions [42, 43]. Those resources strengthen reproducibility and make reference cases easier to inspect. Their contribution is traceability of workflow and model descriptions, not automatic validation evidence; validation still requires the geometry, boundary data, material assumptions, numerical settings, and observable to match the claim being tested.

External validation is therefore a different evidence category. It requires comparison with an accepted physical or application reference for the target question, such as controlled experimental measurements or clinical measurements with known acquisition conventions. A resolved computational comparator can support a cross-model comparison when geometry, material parameters, boundary history, timestep, and observable are matched closely enough for the intended claim, but it is not automatically validation data merely because it is higher-dimensional. In every category, the observation operator determines what quantity is actually being compared.

## 5.5 Cross-Dimensional Observation Operators

Cross-dimensional comparison is only meaningful after both numerical states are observed in a common data space. A 1D area–flow state supplies section area, flow, and mean axial velocity after a coordinate convention is fixed. A 3D velocity field supplies nodal, elemental, or quadrature-point velocities on a mesh. Comparing them therefore requires an observation operator: a declared map from each numerical state to a common observable.

For section comparisons, the operator must specify the target planes, mesh intersection rule, interpolation of velocity onto the cut geometry, quadrature weights, area normalization, and treatment of empty or invalid cuts. For profile comparisons, it must also specify the radial coordinate, binning rule, profile closure used by the 1D reconstruction, and any azimuthal averaging. Multiscale and FSI literature treats such maps as part of the model interface rather than as after-the-fact plotting choices [4, 18]. Clinical FFR and CT-FFR sources show the same requirement in diagnostic form: overview and CT-derived FFR work make the reported pressure-ratio observable interpretable only after the imaging, model, boundary, and pressure-extraction conventions have been fixed [32, 33, 34]. Comparisons of 1D and 3D FFR estimates make the cross-dimensional version of that requirement explicit [9]. In this report, the plane–tetrahedron quadrature operator plays that role for the available resolved 3D velocity data, while the 1D state is observed through physical flow and area-mean velocity. A reported discrepancy can then include model-form error, closure or parameter differences, numerical error, data-matching error, and observation-operator error. The section geometry used by the observation map must also be declared: reference-coordinate cuts and deformed-coordinate cuts are different output operators even when they are applied to the same velocity file.

## 5.6 Evidence Standard for the Present Report

The review above gives a compact standard for reading the numerical evidence. Each reported result should identify its equation tier, discrete operator, boundary rule, timestep policy, stability and positivity diagnostics, reference data, and observation operator. Table 7 summarizes the evidence roles used by the present manuscript and later applied in the case study.

Table 7: Claim-to-evidence map for numerical-method statements in the report.

| **Claim** | **Minimum evidence** | **Does not establish** |
|---|---|---|
| Implementation reproduces a manufactured smooth solution | Code verification and observed order for the declared forced operator, including flux, source, boundary, and time-integration wiring. | Physical validity or grid/time resolution for the target case. |
| Computed quantity is grid/time resolved | Solution verification with grid and timestep sensitivity or an uncertainty estimate for the reported observable. | Agreement with external data or with another model tier. |
| Rest or steady state is preserved | Named equilibrium family plus a drift, residual, or exact-discrete preservation test. | Preservation of unrelated transients or patient-specific behavior. |
| Two schemes agree | Shared benchmark, matched model contracts, and comparable output quantities. | Correctness if the shared model contract is incomplete. |
| Two model tiers agree | Matched inputs and closures plus a common observation operator. | Native-field equivalence or clinical validation. |
| Model represents physical or clinical reality | External data for the specified observable and context of use. | Transfer to a different observable, population, or decision setting. |
| Result is reproducible | Archived or declared inputs, versions, commands, generated outputs, and source/provenance declarations. | Accuracy or validity without the relevant evidence category above. |

**Illustration from the idealized stenosis study.**

The later case study applies three of these evidence categories. The MMS run is the manufactured-state verification problem: it checks whether the declared finite-volume operator, forcing residuals, boundary traces, and timestepper reproduce a known smooth area–flow solution. The rest-equilibrium run is the well-balanced test: it asks whether the discrete stenosis geometry, wall source, and numerical flux preserve the quiescent state that the continuous balance law admits. The plane–tetrahedron quadrature rule is the observation-operator test: it defines how a resolved 3D velocity field becomes section area, physical flow, and area-mean velocity comparable to 1D outputs. These checks answer different numerical questions, so agreement in one cannot substitute for a limitation or ambiguity in another.

# 6 Literature Synthesis and Open Challenges

The preceding chapters introduced the review axes one at a time: continuum fields, model hierarchy, closure choices, observables, numerical methods, and evidence standards. This chapter compares across those axes. Its purpose is not to add another catalog of sources, but to state what the landscape collectively shows. The central finding is that blood-flow models are interpretable only when retained state, resolved scale, data, closure choices, observables, numerical evidence, and validation status are aligned with the question being asked.

The synthesis can be organized into four proposition families used throughout the remainder of the report:

1. **Model state.** Model hierarchy is a hierarchy of retained state and observation maps, not a monotone hierarchy of accuracy; dimension reduction changes closure requirements and native observables, not merely computational cost.

2. **Closures and observables.** Local stenosis hemodynamics depends on omitted circulation through boundary or multiscale coupling models, and rheology, wall, and geometry sensitivity must be reported relative to a specified observable and regime.

3. **Evidence and validation.** Agreement between 1D and 3D models is interpretable only under matched model contracts and a common observation operator; manufactured solutions, grid studies, equilibrium tests, benchmarks, cross-model comparisons, external validation, UQ, and reproducibility support different claims.

4. **Learned and identifiable maps.** Practical identifiability depends on measured outputs, boundary model, noise assumptions, and experimental design; learned models inherit the geometry family, boundary inputs, source-model assumptions, training distribution, and target observable of their data.

## 6.1 Resolved Scale, Cost, and Retained State

Increasing resolved scale can expose local velocity gradients, wall interaction, and geometry-specific flow features, but it also increases the number of boundary, material, geometric, and numerical assumptions that must be identified. The continuum and FSI sources show what is gained by retaining local flow fields, wall kinematics, and interface work; the multiscale sources show that those fields still have to be connected to lower-dimensional circulation states; Formaggia, Lamponi, and Quarteroni show that structural simplifications change the one-dimensional differential operator itself; and the 0D/1D review literature shows what is deliberately compressed into axial or lumped pressure–flow variables [3, 4, 5, 18]. The reduced tiers are not weaker by definition. They are appropriate when the question concerns pulse propagation, network pressure–flow behavior, rapid screening, or parameter studies for which fully resolved data and boundary conditions are unavailable or unnecessary.

The cost of that compression is interpretive. A 1D model may have fewer state variables than a 3D model, but it can still be underdetermined if wall, profile, friction, rheology, and boundary closures are treated as adjustable without independent evidence. Conversely, a highly resolved forward model can depend on material properties, prestress, inlet histories, outflow impedances, geometry state, and wall motion that are rarely all observed with matching certainty. Parameter-identifiability work for hemodynamics PDE models makes this tension explicit: changing measured outputs and boundary-condition class changed which parameters could be inferred in a 1D pulmonary model [44]. Adding modeling detail is therefore valuable only when the data and outputs can distinguish the resulting parameters. The synthesis point is therefore comparative rather than taxonomic: the right tier is the one whose retained state is identifiable for the claim being made.

## 6.2 Observables and Cross-Model Comparability

Observation operators are part of the model, not post-processing decoration. A 1D area–flow solution does not natively contain the same observable as a resolved 3D velocity field, and a pressure-drop, flow-rate, section-mean velocity, wall-shear, or FFR-style quantity each imposes a different comparison contract. Multiscale coupling work already treats interface quantities and compatibility conditions as mathematical structure [4, 18]; diagnostic 1D–3D comparison needs the same discipline at the output side. A flow-rate comparison, for example, tests a different map than a pressure-ratio or wall-shear comparison even when the underlying geometry is shared.

The clinical FFR literature sharpens the point because it separates anatomical stenosis appearance from pressure-flow function: Tonino and coauthors supply the angiographic-versus-functional contrast, while FFR overview work defines the scalar as a pressure-ratio diagnostic [8, 32]. CT-derived FFR and 1D–3D FFR comparison sources then move that diagnostic into computational chains. Blanco and coauthors compared 20 patients and 29 geometries under matched pressure boundary conditions and dedicated 1D stenosis and bifurcation loss terms, so their result supports an FFR-specific matched-contract comparison rather than field-level or tier-level equivalence [9]. Together with CT-FFR methodology work, this shows that the scalar is not a raw geometry measurement; it is an output of an acquisition, modeling, boundary, and pressure-observation chain [33, 34]. Thus a cross-model discrepancy is uninterpretable until the observed quantity is declared and matched.

## 6.3 Uncertainty in Boundaries, Walls, Rheology, and Geometry

Boundary-condition uncertainty is one of the most persistent barriers to interpretable hemodynamics. Inlet waveforms, outlet impedances, terminal resistances, and coupled-domain histories can change the local solution as much as the internal discretization. In a multiscale setting, these quantities are not auxiliary inputs; they are the mechanism by which upstream and downstream vascular beds communicate with the modeled region [4, 18]. A reduced stenosis result that does not declare its boundary approximation therefore cannot be cleanly compared with a resolved result, even if both models share a nominal geometry. The source contrast is direct: multiscale papers treat the boundary as an interface model, whereas single-domain diagnostic studies can hide the same choice behind an inlet or outlet label.

Wall and rheology uncertainty create a second closure layer. Wall stiffness and thickness determine wave speed and area response; rheology and velocity-profile assumptions determine dissipative behavior and momentum transport. In 1D models, these effects enter through pressure–area laws, friction terms, effective viscosity choices, and momentum-flux correction factors [5]. In FSI models, they enter through wall material laws, structural boundary conditions, and the coupling algorithm. Both settings face the same interpretive problem: when the wall or rheology is not independently constrained, a discrepancy can be a physics signal, a closure signal, or a compensation between uncertain choices.

These uncertainties are not of one kind. Aleatory variability concerns physiological variation across cycles or populations; epistemic or model-form uncertainty concerns missing mechanisms; parameter uncertainty concerns unknown values in a selected model; numerical error concerns the computed realization; and data-matching uncertainty concerns whether two models, measurements, or files describe the same state. Liu and coauthors show why the category must be tied to an observable: rheology choice was negligible for a pressure ratio in their intracranial stenosis setting but visible in low-WSS regions [14].

Geometry is similarly double-edged. Patient-specific geometry is attractive because it carries clinically meaningful morphology, but segmentation, registration, current-versus-reference configuration, and wall-motion state can dominate the modeling specification. Idealized geometry removes many of those ambiguities and exposes controlled severity and refinement parameters, but it cannot by itself answer patient-specific questions. The useful distinction is not realistic versus artificial. It is whether the geometry serves the claim: patient-specific geometry for patient-specific interpretation, and idealized geometry for isolating numerical, closure, and comparison behavior. Gambaruto and coauthors provide a bounded example: in one aneurysm case, reconstruction variation within imaging resolution changed WSS enough that geometry sensitivity was greater than, but comparable to, rheology sensitivity [31]. That is not a universal uncertainty range; it is evidence that reconstruction can be a claim-level model input.

## 6.4 Validation Status, Reproducibility, and Evidence Limits

Verification, validation, uncertainty quantification, and reproducibility should not be collapsed. ASME VVUQ terminology is useful because it separates conformity to a mathematical description, representation of a real application, and propagation of uncertain numerical or physical quantities [41]. Manufactured solutions and grid studies ask whether an implementation behaves as expected for a declared mathematical problem. Benchmarks and open solvers support reproducibility and comparison across workflows [6, 7]. The benchmark study contributes shared cases and scheme comparison; OpenBF contributes an inspectable open finite-volume workflow. Resolved-flow platforms and model repositories extend that reproducibility lesson to 3D workflows by making geometry, meshing, solver, and repository materials easier to inspect [42, 43]. None of these materials is automatically a validation claim unless the model, data, observable, and accepted reference match the target question.

Equilibrium preservation is a particularly sharp consistency test. A continuous stenotic balance law may admit a rest or steady state through cancellation between flux gradients, geometry terms, wall response, friction, and boundary data. A discrete method that does not preserve that state can generate motion from the numerical representation itself. Well-balanced methods for 1D blood flow make this point directly and raise the standard for stenosis computations: manufactured solutions and grid studies are useful, but an equilibrium problem needs an equilibrium-preservation check [40]. The 2023 Pimentel-Garcia et al. claim concerns all stationary solutions for discontinuous geometrical and mechanical properties, so its evidence role is equilibrium preservation rather than generic high-order behavior. The broader verification and validation gap is that many studies report plausible hemodynamic outputs without enough evidence to separate implementation error, discretization error, closure uncertainty, cross-model mismatch, and external validation against an accepted target.

### 6.5 Reproducibility and Emerging Learned Methods

Reproducibility is the practical bridge between modeling ambition and reviewable evidence. Benchmark studies help by fixing common cases and comparison quantities, but stenosis studies also need persisted geometry definitions, boundary histories, closure parameters, solver settings, diagnostics, and output maps [6, 7]. The lesson from the benchmark/open-solver pair is not that every stenosis study must use the same code. It is that disagreement between two models is hard to assign to physics, parameters, numerics, or data handling unless the comparison design is inspectable.

Learned methods do not remove this requirement. Physics-informed neural networks offer a framework for solving and inverting PDE-constrained problems [24], and hemodynamics reviews now treat PINNs as an active direction for vascular modeling [10]. At the same time, gradient-pathology analyses show that training dynamics can be a limiting numerical issue rather than a transparent solver detail [45]. Operator-learning methods such as DeepONet and Fourier neural operators shift the ambition from one solution to a map between function spaces [25, 26]. Recent hemodynamics-specific neural differential and stenosed-vessel operator-learning work shows why such models are appealing for fast parameterized flow prediction. Csala and coauthors report lower error than a specified 1D FEM baseline and sub-1.2% relative error against averaged 3D data within their declared train/test setting [27]. Velikorodny and coauthors learn pressure and flow maps and then derive FFR in a simplified stenosis setting with 201 training and 68 test cases [28]. These are conditional dataset-level claims, not evidence of patient-specific or out-of-distribution clinical validity.

Their open challenge is the same synthesis problem in a new form: the training distribution, geometry family, boundary inputs, wall and rheology assumptions, loss normalization, and validation observables must be declared before speed or fit can be interpreted as hemodynamic evidence.

### 6.6 Role of the Idealized Stenosis Case Study

The idealized stenosis case study is useful because it applies four review propositions in one controlled setting: model contract, equilibrium preservation, common observation operator, and bounded cross-model comparison. It fixes a geometry family, closure contract, boundary approximation, discretization, and output map tightly enough that the consequences can be inspected. It removes segmentation and clinical-matching ambiguity so that closure dependence, boundary sensitivity, equilibrium preservation, verification evidence, and cross-dimensional observability can be examined in a controlled setting. Patient-specific interpretation remains the domain of studies with matched patient geometry, boundary data, wall state, and external validation data.

# 7 Idealized Stenosis Case Study

This chapter uses one controlled stenosis computation to show how the review framework is applied in practice. It fixes a smooth idealized geometry, states the one-dimensional area–flow model contract, declares the closure and boundary assumptions, identifies the principal finite-volume realization, checks the evidence categories introduced in Section 5, and compares one-dimensional output with available resolved three-dimensional velocity data through a declared observation operator. The chapter is a worked example of model interpretation, not a standalone solver-development report.

The result is deliberately bounded. The manufactured-solution run gives positive implementation-verification evidence for the forced operator; the geometry-rest run shows that the current MUSCL/Rusanov realization is not exactly well balanced for the stenotic rest state; and the final-time comparison localizes section-mean velocity differences for the 23% and 40% stenosis cases. These findings support the review framework: they show how a model contract, a numerical evidence category, and an observation map determine what a stenosis comparison can mean. Read the chapter in that order: the methodology defines the mathematical model and observation contract, the verification subsection separates the positive MMS evidence from the rest-state limitation, and the comparison subsection interprets the 23% and 40% final-time velocity discrepancies under the declared section map.

The discretization views in the following figure orient the case study only: they preview the stationary-Stokes finite-element sampling geometry and the one-dimensional finite-volume grid before the model contract and evidence categories are stated.

## 7.1 Case-Study Role and Model Contract

This section applies the review framework to one controlled stenosis example. The model tier, retained variables, closure choices, boundary approximation, finite-volume operator, observation map, and matching limits are stated before any result is interpreted. Dense operator formulas, secondary numerical surfaces, and reproducibility commands remain in the appendices unless they are needed to understand a main-body claim. The paragraphs below therefore name mathematical roles first; implementation labels appear only as traceability tags for reproducing the selected runs.

**Solver-coordinate convention.** In the comparison outputs, the stored solver state `A` is $a = R^2$, not physical circular area. The physical map is Definition 7.3: $A_{\text{phys}} = \pi a$, $Q_{\text{phys}} = \pi q$, and $\bar{u} = q/a = Q_{\text{phys}}/A_{\text{phys}}$.

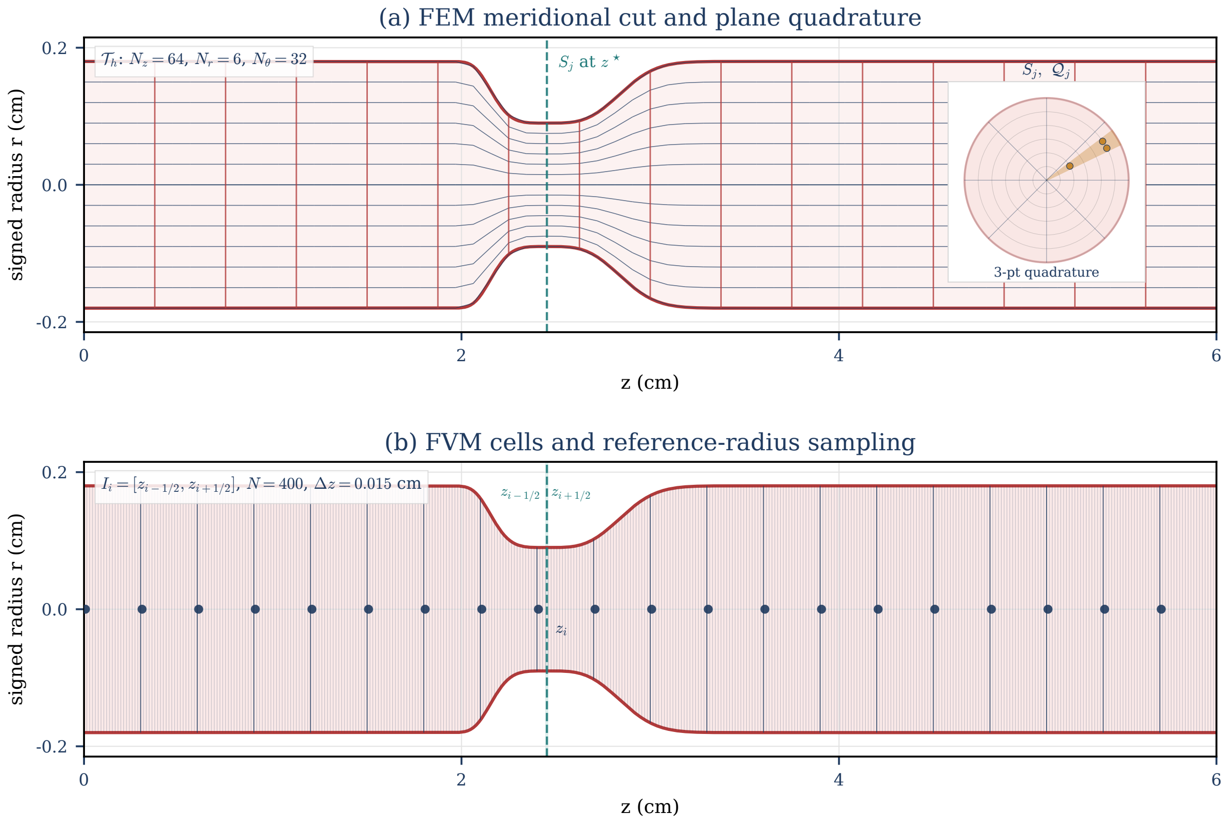


Figure 2: Finite-element and finite-volume discretization views of the same $C^\infty$ reference stenosis geometry defined in Definition 7.4. Panel (a) shows a meridional projection of the stationary-Stokes tetrahedral mesh $\mathcal{T}_h$ against the wall curves $r = \pm R_0(z)$; the inset sketches a section $S_j = \Omega_{3D} \cap \{z = z_j\}$, the centroid-fan cut triangles $\mathcal{Q}_j$, and the three barycentric quadrature points used by the declared cross-section quadrature operator. Panel (b) shows the 1D finite-volume partition $I_i = [z_{i-1/2}, z_{i+1/2}]$ with centers $z_i = (i - \frac{1}{2})\Delta z$, sampled along $R_0(z)$.

### 7.1.1 Model Contract and Native Variables

The computational problem is a single idealized stenotic vessel represented by a selected one-dimensional area–flow model. The model assumes a distinguished axial direction and cross-sectionally averaged pressure and velocity, retaining distributed pulse-wave and pressure–flow structure while discarding secondary flow, separation, recirculation, and resolved wall traction. The reduction follows standard compliant-artery and dimension-reduced modeling arguments [3, Sec. 2, pp. 253–257] [11, Ch. 3, Secs. 3.2 and 3.6] [5, Sec. 3, pp. 24–26].

**Assumption 7.1** (Straight single-vessel geometry)**.** The selected 1D model assumes a vessel of length $L$ aligned with the $z$-axis and a circular cross-section with radius $R(z,t)$. For $0 < z < L$, $S(z,t)$ is the interior cross-section used for averaging, not a boundary component of the open lumen.

**Definition 7.2** (Section-averaged fields and observed pressure)**.** The retained reduced kinematic variables

Table 8: Case-study contract for the retained 23% and 40% stenosis-severity numerical evidence. The table states how the worked example should be read before the numerical results are interpreted.

| **Contract item** | **Main-body statement** | **Interpretation boundary** |
|---|---|---|
| Case-study purpose | A controlled idealized stenosis example applies the review framework to a selected 1D area–flow model and two resolved velocity fields. | The chapter illustrates verification, reproducibility, and diagnostic cross-model velocity comparison; it is not a pressure-ratio study or patient-specific interpretation. |
| Model and closures | The main comparison uses the $R_{\max}$-normalized Canic-derived extended 1D member with Newtonian $\nu = 0.04$ cm$^2$/s, parabolic profile closure, $N = 400$ MUSCL finite-volume cells, Rusanov flux, and SSPRK3 time integration. | Other variants are secondary sensitivity context rather than the principal two-case comparison method. |
| Observation operator | The 3D velocity field is reduced by the declared cross-section quadrature operator; the 1D state is observed as physical flow $Q_{1D} = \pi q$ and mean velocity $\bar{u}_{1D} = q/a$ on 200 axial sections. A second retained comparison applies the resolved displacement field before forming the same plane cuts. | The comparison supports a claim only for these common section observables. Radial-profile outputs are not promoted because their area-closure gate fails. |
| Numerical evidence | The main text reports manufactured-solution behavior, output-operator verification, geometry-rest drift, subcritical boundary diagnostics, CFL, positivity, and finite-volume balance. | These checks support run admissibility and numerical interpretation; they do not identify a physical prediction target. |
| Matching limits | The analytic reference profile is shared and the static cut-area check is reported for both reference and deformed coordinates. A quasi-static membrane-FSI comparator is also retained for the radial wall displacement solve. | Velocity differences cannot be attributed uniquely to a specific closure, wall, boundary, or geometry mechanism because the resolved data available here do not carry matched boundary histories or full transient wall history. |

and observed pressure representative are

$$
\begin{aligned}
A_{\text{phys}}(z,t) &:= \int_{S(z,t)} 1\, dS, && \text{physical cross-sectional area,}\\
Q_{\text{phys}}(z,t) &:= \int_{S(z,t)} u_z\, dS, && \text{physical volumetric flow rate,}\\
\bar{u}(z,t) &:= \frac{Q_{\text{phys}}(z,t)}{A_{\text{phys}}(z,t)}, && \text{mean axial velocity,}\\
\bar{p}_{3D,\text{g}}(z,t) &:= \frac{1}{A_{\text{phys}}(z,t)} \int_{S(z,t)} p_{3D,\text{g}}\, dS, && \text{observed 3D cross-sectional mean gauge pressure.}
\end{aligned}
$$

Here $p_{3D,\text{g}}$ is continuum pressure in the wall-law gauge frame. The observed representative $\bar{p}_{3D,\text{g}}$ is distinct from the reduced wall-law pressure variable introduced below.

**Definition 7.3** (Physical-to-solver map for the implemented 1D state)**.** The continuum 1D model uses physical area $A_{\text{phys}}$ and physical flow $Q_{\text{phys}}$. The implemented radius-squared solver instead stores

$$
a = R^2, \qquad q = \frac{Q_{\text{phys}}}{\pi}.
$$

Thus

$$
A_{\text{phys}} = \pi a, \qquad Q_{\text{phys}} = \pi q, \qquad \bar{u} = \frac{Q_{\text{phys}}}{A_{\text{phys}}} = \frac{q}{a}.
$$

With $K = Eh/(1-\sigma^2)$ and the Canic conservative-form convention $R_0^* = R_{\max}$, the selected evolution wall law gives the solver elastic flux contribution defined in Appendix G.8:

$$\Psi_h(a) = \frac{K}{3\rho R_{\max}^2} a^{3/2}, \qquad \partial_a \Psi_h(a) = \frac{K}{2\rho R_{\max}^2} \sqrt{a}.$$

This map is the convention used by the Section 7.3 velocity and flow comparisons.

### 7.1.2 Closure Choices: Geometry and Wall Law

**Definition 7.4** (Stenosis reference geometry)**.** A stenosis is encoded through the reference radius or reference area. Let $R_{\text{base}}(z)$ be the nominal healthy radius and $s(z) \in [0, 1)$ a fractional radius-reduction profile. Then

$$R_0(z) = R_{\text{base}}(z)\big(1 - s(z)\big), \qquad A_0(z) = \pi R_0(z)^2. \tag{7.1}$$

The baseline idealized stenotic-vessel profile is the smooth asymmetric kernel

$$g(z) = z - z_s + \kappa \exp\left(-\frac{(z - z_a)^2}{2\sigma_a^2}\right), \qquad \psi(z) = \exp\left(-\lambda g(z)^4\right), \tag{7.2}$$

$$s(z) = s_{\max}\psi(z), \qquad 0 \le s_{\max} < 1. \tag{7.3}$$

In the default centimeter-scale simulation setting, $z_a = 2.5\,\text{cm}$, $z_s = 3.4\,\text{cm}$, $\sigma_a = 1\,\text{cm}$, $\kappa = 0.95\,\text{cm}$, and $\lambda = 50\,\text{cm}^{-4}$. This geometry is the baseline idealized simulation geometry for the worked example.

*Remark* 7.5 (Stenosis-profile and variable-radius boundary)*.* The profile in Eq. 7.3 is $C^\infty$ and rapidly localized rather than compactly supported. This smooth idealized geometry is the baseline for the reduced simulations in this report and is also useful for fixed-wall 3D studies: the same analytic radius profile supplies controlled derivatives, wall normals, severity variation, and repeatable mesh-refinement families without introducing segmentation noise. Canic et al. show that standard 1D variable-geometry models may miss stenotic pressure and velocity trends unless additional variable-radius terms are included [16, Abstract, Secs. 1–2]. Those corrections are included by the Canic-derived extended member (implementation label `canic-extended-1d`); the classical parabolic-profile baseline (implementation label `classical-1d-no-slip`) retains the same selected wall law but omits the Canic variable-radius corrections.

**Definition 7.6** (Classical parabolic-profile 1D baseline)**.** The classical baseline used as a model-family member is a parabolic-profile reduced model. A no-slip wall trace is compatible with a parabolic profile for fully developed Newtonian flow in a straight circular tube, but no-slip alone does not imply that profile in an unsteady stenotic vessel. The parabolic profile is therefore an independent cross-sectional closure with $\alpha_{\mathcal{P}} = 4/3$ and $g_{\mathcal{P}} = 4$. In the mathematical discussion this model is the classical parabolic-profile baseline; the implementation label is `classical-1d-no-slip`. It keeps the same reduced state variables and selected Canic-derived pressure–area wall law as the extended 1D framework, but omits the Canic effective-alpha variable-radius correction. The $R_{\max}$-normalized Canic-derived extended member is labeled `canic-extended-1d` in implementation manifests.

**Definition 7.7** (Selected Canic–Koiter pressure–area wall law)**.** The selected elastic wall closure is the Canic–Koiter thin-wall pressure–area law for the evolution wall-law gauge pressure $p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}$. In conventional area notation it may be written

$$p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}(A,z,t) = p_{\mathrm{ext}}(z,t) + \frac{\beta(z)}{A_0(z)}\left(\sqrt{A(z,t)} - \sqrt{A_0(z)}\right). \tag{7.4}$$

Here $A_0(z)$ is the reference area, $p_{\mathrm{ext}}$ is the external pressure, and $\beta(z)$ is a wall-stiffness parameter. In the radius-squared coordinate $a = R^2$ used by the implemented finite-volume evolution, the selected continuous source-balanced law uses the Canic conservative-form reference radius $R_0^* = R_{\max}$:

$$p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}(a,z,t) = p_{\mathrm{ext}}(z,t) + \frac{K}{R_{\max}^2}\left(\sqrt{a} - R_0(z)\right), \qquad K = \frac{Eh}{1-\sigma^2}.$$

With $A = \pi a$ and $A_0 = \pi R_0^2$, the conventional law in Eq. 7.4 reduces to this selected evolution law only when

$$\beta(z) = \sqrt{\pi}\, K\, \frac{R_0(z)^2}{R_{\max}^2}.$$

Thus the implemented evolution uses a constant reference-radius denominator, not the local $R_0^{-2}$ denominator. In this report $R_{\max}$ is the healthy reference-radius parameter used by the solver; for the default stenosis profile it equals $\max_z R_0(z)$. The selected comparison runs use the cgs parameter values $\rho = 1.055\,\mathrm{g/cm^3}$, $E = 5.02 \times 10^6\,\mathrm{dyn/cm^2}$, $h = 0.06\,\mathrm{cm}$, $\sigma = 0.5$, $K = 4.016 \times 10^5\,\mathrm{dyn/cm}$, $R_{\max} = 0.18\,\mathrm{cm}$, and $p_{\mathrm{ext}} \equiv 0$. This is the wall law paired with the implemented elastic potential and wall/geometry source below. Legacy pressure-output helpers in the implementation still report a local-denominator pressure diagnostic $p^{\mathrm{diag}}_{\mathrm{loc,g}}$; benchmark pressure extrema and pressure-diagnostic bars refer to that legacy diagnostic, not to $p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}$. The 1D model identifies $p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}$ with a cross-sectional pressure representative as a closure assumption and does not resolve radial pressure variation.

### 7.1.3 Reduced Balance Law and Rest-State Contract

The selected 1D model is an area–flow balance law closed by the profile, rheology, wall, geometry, and boundary choices stated above. The important mathematical point is that the retained state is only axial area and flow; transverse velocity, wall response, friction, rheology, and variable-radius effects enter through closures. Appendix G.8 is the operator authority in solver coordinates. The main text uses it only to state the contract needed to interpret the evidence.

The selected continuous law has a rest state that matters for the evidence standard. Under constant external pressure and compatible zero-flow boundary data, the reference geometry $a(z) = R_0(z)^2$ with $q(z) = 0$ should remain at rest in the continuous source-balanced model. This property is the target of the geometry-rest verification test in Section 7.2.3.

**Proposition 7.8** (Rest-state compatibility of the selected wall source)**.** *Suppose $p_{\text{ext}}$ is constant in the gauge convention and the boundary data are compatible with $q = 0$. Then $a(z) = R_0(z)^2$ and $q(z) = 0$ are an exact steady state of the selected continuous source-balanced evolution law.*

*Proof.* Using the operator definitions in Appendix G.8, the mass equation gives $\partial_z q = 0$. In the momentum equation, all terms proportional to $q$ vanish, so $S_{\text{fric}} = S_\alpha = S_{p_2} = 0$. At $a = R_0^2$,

$$\partial_z \Psi_h(R_0^2) = \partial_z \left( \frac{K}{3\rho R_{\max}^2} R_0^3 \right) = \frac{K}{\rho R_{\max}^2} R_0^2 R_0' = S_{\text{wall}}(R_0^2, z).$$

Thus the elastic flux derivative is exactly canceled by the wall/geometry source, and $\partial_t q = 0$. □

Continuous compatibility does not imply discrete preservation: the current MUSCL/Rusanov method exhibits the reported non-well-balanced geometry-rest drift.

#### 7.1.4 Source-to-Implementation Differences

The selected realization is a documented Canic-derived extended model, not a literal reproduction of every source-paper parameter and normalization in [16]. The report therefore uses the phrase $R_{\max}$*-normalized Canic-derived extended model* when source fidelity matters. The main differences are the $R_{\max}^{-2}$ elastic normalization, the parabolic principal velocity-profile closure, the fixed-area characteristic boundary approximation, and the locally frozen-viscosity treatment used for the diagnostic $p_2$ derivative. Appendix G.2 contains the detailed source-to-realization table. The diagnostic extended pressure correction is

$$p_{2,\text{g}}(a, q, z, t) = g_{\mathcal{P}} \rho \nu_{\text{eff}}^{\mathcal{R}} \frac{q}{a} \frac{R_0'(z)}{R_0(z)}.$$

This $p_2$ term is a variable-radius viscous correction, not a replacement wall law. In generalized-Newtonian closure runs, the implemented analytic $p_2$ derivative uses the local value of $\nu_{\text{eff}}^{\mathcal{R}}$ and does not differentiate that effective-viscosity factor; it is therefore a locally frozen-viscosity computational closure for this term.

#### 7.1.5 Boundary Approximation and Subcritical Condition

The comparison runs start from the geometry-rest state and then impose the finite-volume boundary approximation stated below.

**Definition 7.9** (Single-vessel boundary data)**.** Generic single-vessel models may prescribe inlet flow or velocity and an outlet pressure or terminal relation; these alternatives are model-family context rather than the reported boundary rule. The realized finite-volume run prescribes an inlet solver flow derived from a nominal reference-area mean velocity,

$$q_{\mathrm{in}} = R_0(0)^2 \bar{u}_{\mathrm{in}},$$

solves the inlet ghost area from the approximate outgoing characteristic invariant, fixes the outlet ghost coordinate $a_{\mathrm{out}} = R_0(L)^2$, and obtains the outlet ghost flow from the corresponding approximate invariant. Hence the imposed inlet condition is a prescribed reference-area flow, not a fixed realized mean velocity $q_{\mathrm{in}}/a_{\mathrm{in}}$, and the outlet condition fixes the elastic reference pressure only for the selected wall component.

**Definition 7.10** (Characteristic speeds for the reduced elastic subsystem)**.** Let $\lambda_{\pm}$ denote the two characteristic speeds of the positive-area elastic 1D subsystem obtained from the homogeneous $(A, Q)$ balance law. Their formula and assumptions are given in Appendix G.7.

**Assumption 7.11** (Subcritical boundary regime)**.** The baseline boundary treatment is used only in the subcritical regime $\lambda_{-} < 0 < \lambda_{+}$, where one characteristic enters through the inlet and one characteristic enters through the outlet.

Under this regime, a finite-volume scheme constrains the incoming information at each boundary and leaves the outgoing state determined by the interior solve. The finite-volume boundary treatment in Appendix G.10 uses an $\alpha = 1$, fixed-area characteristic approximation as the boundary invariant for the variable-$\alpha_{\mathrm{eff}}$ solver. It is therefore an implemented boundary approximation, not an exact invariant derivation for the full variable-radius balance law.

### 7.1.6 Principal MUSCL/Rusanov/SSPRK3 Method

Only the principal comparison method belongs in the main narrative. The reported two-case severity comparison advances the radius-squared state $(a, q)$ on a one-dimensional finite-volume grid using the MUSCL/Rusanov/SSPRK3 operator specified in Appendix G.8. The principal numerical comparison uses $N = 400$ cells on $0 \leq z \leq 6$ cm, a timestep cap of $10^{-5}$ s, and a CFL cap of 0.45. The optional output-sensitivity check reruns the same comparison protocol at $N = 200, 400, 800$ cells and is interpreted only as a production-grid sensitivity check. The boundary-state approximation is the fixed-area characteristic contract defined in Appendix G.10. Other finite-volume, modal-DG, time-stepper, and ODE-integration surfaces are secondary sensitivity context summarized in Table 25; they are not the principal two-case comparison method.

### 7.1.7 Plane–Tetrahedron Observation Operator

The 1D state directly supplies a section mean, $\bar{u}_{1D}(z,t) = q(z,t)/a(z,t)$, and physical flow $Q_{1D}(z,t) = \pi q(z,t)$. The implementation can also emit radial-profile rows using the same cut triangles, and the package audit now classifies same-cut area and flow closure. Those rows are treated as secondary output, and no radial-profile figure or localization claim is retained until regenerated report evidence is reviewed and accepted.

The resolved-data loader also accepts pressure and displacement XDMF/HDF5 companions for a velocity field. When the comparison is explicitly run in deformed-coordinate mode, the observation operator applies the node-centered displacement before forming the plane cuts. Both the reference-coordinate and deformed-coordinate section comparisons are retained in Section 7.3.4.

For a target plane $S_j = \Omega_{\mathrm{3D}} \cap \{z = z_j\}$, the default cross-section quadrature operator intersects every tetrahedron with $S_j$, linearly interpolates axial velocity on the cut edges, sorts the cut polygon vertices in the plane, and triangulates the polygon by a centroid fan. Let $\mathcal{Q}_j$ denote the resulting cut triangles. The reported 3D area, flow, and mean velocity are

$$A_{3D,j} = \sum_{\Delta \in \mathcal{Q}_j} |\Delta|, \qquad Q_{3D,j} = \sum_{\Delta \in \mathcal{Q}_j} |\Delta|\, \bar{u}_{z,\Delta}, \qquad \bar{u}_{3D,j} = Q_{3D,j}/A_{3D,j}.$$

For a cut triangle $\Delta$ whose vertices carry reconstructed axial velocities $u_{z,1}$, $u_{z,2}$, and $u_{z,3}$,

$$\bar{u}_{z,\Delta} = \frac{u_{z,1} + u_{z,2} + u_{z,3}}{3}.$$

The corresponding 1D quantities use physical flow, not the solver flow coordinate:

$$Q_{1D,j} = \pi q(z_j, t), \qquad \bar{u}_{1D,j} = q(z_j, t)/a(z_j, t).$$

Radial-profile rows are not part of the retained comparison metrics.

### 7.1.8 Comparison Design and Discrepancy Metrics

The resolved-velocity comparison evaluates the selected 1D solution against two available resolved 3D velocity comparison datasets at the matched final-time sample $T = 0.9995$ s. The retained cases are described by their 23% and 40% stenosis severities. The comparison is a diagnostic cross-model velocity comparison under the declared observation operator. Source identifiers, input policy, and static asset ownership are kept in Appendix H.

Velocity and flow differences are reported as sample discrepancy measures over valid section targets. With $d_{u,j} = \bar{u}_{1D,j} - \bar{u}_{3D,j}$ and $d_{Q,j} = Q_{1D,j} - Q_{3D,j}$, the retained section metrics are the signed mean bias, mean absolute discrepancy, RMS discrepancy, maximum absolute discrepancy, and relative RMS discrepancy,

$$\bar{d}_u = \frac{1}{n}\sum_j d_{u,j}, \qquad D_{u,1} = \frac{1}{n}\sum_j |d_{u,j}|, \qquad D_{u,2} = \left(\frac{1}{n}\sum_j d_{u,j}^2\right)^{1/2},$$

$$D_{u,\infty} = \max_j |d_{u,j}|, \qquad D_{u,2}^{\mathrm{rel}} = \frac{\left(\sum_j d_{u,j}^2\right)^{1/2}}{\left(\sum_j \bar{u}_{3D,j}^2\right)^{1/2}},$$

$$\bar{d}_Q = \frac{1}{n}\sum_j d_{Q,j}, \qquad D_{Q,1} = \frac{1}{n}\sum_j |d_{Q,j}|, \qquad D_{Q,2} = \left(\frac{1}{n}\sum_j d_{Q,j}^2\right)^{1/2}.$$

## 7.2 Verification Evidence and Rest-State Limitation

This section applies the evidence standards from Section 5 to the worked example. The retained main-body evidence has three parts: manufactured-solution verification supports the forced finite-volume operator, the geometry-rest test exposes the non-well-balanced limitation of the current method, and the admissibility diagnostics show that the accepted comparison runs stayed within the declared boundary, CFL, positivity, and conservation regime. Secondary checks for nonprincipal numerical surfaces remain in Appendix G.

### 7.2.1 Evidence Hierarchy and Scope

The hierarchy used below is explicit. The strongest positive evidence is the manufactured-solution test, because it compares the implemented forced operator with a declared smooth exact solution and an independently assembled forcing consistency check. The principal limitation is the geometry-rest test: the selected comparison runs start from that rest state, but the current MUSCL/Rusanov realization does not exactly preserve it. Boundary, CFL, positivity, and conservation diagnostics then state whether accepted runs remained numerically admissible. Self-convergence, stationary-Stokes projection, broad rheology/profile sweeps, resolved-velocity benchmark stages, and nonselected DG rows are secondary checks kept in Appendix G.

The observation operator has its own check because the retained 1D–3D comparison is only meaningful after the resolved field is mapped to section area, physical flow, and mean axial velocity. The synthetic check in Table 9 verifies the declared plane–tetrahedron quadrature implementation on a single tetrahedron with constant and affine node-centered axial fields. It checks that constant fields return $\bar{u} = c$ and $Q = cA$, that affine triangle means integrate to the centroid value up to roundoff, and that shifted planes produce the expected area and mean-velocity changes. This is output-operator verification, not a matched-data validation of the available 3D cases.

Standard finite-volume context is supplied by conservative hyperbolic discretization with consistent numerical fluxes and sources, CFL-controlled time stepping, limiter and Rusanov dissipation choices, and SSP

Table 9: Synthetic verification check for the declared cross-section quadrature operator. Differences are measured against analytic constant or affine references on the synthetic tetrahedron; the plane-shift rows report changes relative to the $z = 0.5$ affine cut.

| case | z | shift | area | flow err. | mean err. | shift $\Delta A$ | shift $\Delta \bar{u}$ | status |
|---|---|---|---|---|---|---|---|---|
| constant | 0.25 | 0.000e+00 | 0.2812 | 0.000e+00 | 0.000e+00 | – | – | pass |
| constant | 0.5 | 0.000e+00 | 0.125 | 0.000e+00 | 0.000e+00 | – | – | pass |
| constant | 0.75 | 0.000e+00 | 0.03125 | 0.000e+00 | 0.000e+00 | – | – | pass |
| affine | 0.25 | 0.000e+00 | 0.2812 | 0.000e+00 | 0.000e+00 | – | – | pass |
| affine | 0.5 | 0.000e+00 | 0.125 | 0.000e+00 | 0.000e+00 | – | – | pass |
| affine | 0.75 | 0.000e+00 | 0.03125 | 2.776e-17 | 8.882e-16 | – | – | pass |
| plane_shift | 0.45 | -0.05 | 0.1513 | 0.000e+00 | 0.000e+00 | 0.02625 | -0.2167 | pass |
| plane_shift | 0.5 | 0.000e+00 | 0.125 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 0.000e+00 | pass |
| plane_shift | 0.55 | 0.05 | 0.1012 | 1.110e-16 | 8.882e-16 | -0.02375 | 0.2167 | pass |

Runge–Kutta time integration [36]. In 1D blood-flow practice, MUSCL/Rusanov and Lax–Friedrichs-type discretizations are commonly assessed through exact-solution, benchmark, grid-refinement, and equilibrium tests [6, 7, 37, 38, 39]. Here those checks are empirical implementation evidence, not a formal convergence theorem for the full nonlinear stenosis solver.

#### 7.2.2 Manufactured-Solution Evidence

For verification only, the selected finite-volume operator is paired with a manufactured forcing term. Production runs use no manufactured forcing. The verification run supplies a smooth exact state $(a^\star(z,t), q^\star(z,t))$, exact initial data, exact boundary traces, and forcing residuals for the selected operator. In solver coordinates, let

$$s(z) = \sin\left(\frac{\pi z}{L}\right), \qquad a_0(z) = R_0(z)^2, \qquad \omega = \frac{2\pi}{T_{\mathrm{mms}}}.$$

The default manufactured state uses

$$\epsilon_a = 0.02, \qquad U_{\mathrm{mms}} = 2.0\,\mathrm{cm/s}, \qquad T_{\mathrm{mms}} = 0.02\,\mathrm{s},$$

and defines

$$a^\star(z,t) = a_0(z)\left[1 + \epsilon_a s(z)\cos(\omega t)\right],$$
$$q^\star(z,t) = a_0(z) U_{\mathrm{mms}} s(z)^2 \sin(\omega t).$$

The tabulated run uses $0 \le z \le L = 6\,\text{cm}$ and $0 \le t \le 2 \times 10^{-3}\,\text{s}$, equal to 10% of the manufactured period. The forcing terms are the residuals required by the selected flux and source realization,

$$
\begin{aligned}
f_a(z,t) &= \partial_t a^\star + \partial_z F_a(a^\star, q^\star; z),\\
f_q(z,t) &= \partial_t q^\star + \partial_z F_q(a^\star, q^\star; z) - S_q(a^\star, q^\star; z),
\end{aligned}
$$

where $F_a, F_q$ and $S_q$ are the selected point flux and source functions. The forcing consistency check is independent enough to test the declared finite-volume operator rather than only replaying a stored output; the table reports its maximum absolute mismatch. The run therefore checks the operator against a declared manufactured state. It remains implementation-verification evidence, not a physical stenosis experiment.

The verification table reports cell-center discrete errors

$$
\|e\|_1 = \frac{1}{N}\sum_i |e_i|, \qquad \|e\|_2 = \left(\frac{1}{N}\sum_i e_i^2\right)^{1/2}, \qquad \|e\|_\infty = \max_i |e_i|,
$$

with observed orders computed from the discrete $L_2$ rows as

$$
p_{\text{obs}} = \frac{\log(E_{\text{coarse}}/E_{\text{fine}})}{\log(\Delta_{\text{coarse}}/\Delta_{\text{fine}})}.
$$

These are discrete $L_2$ norm values for the stated error samples. For spatial rows $\Delta$ is the grid spacing. For requested-timestep rows the native solver also reports realized CFL diagnostics because the accepted timestep is $\min(\Delta t_{\text{request}}, \Delta t_{\text{CFL}})$. The timestep-insensitivity rows in the generated MMS table are therefore read through accepted timestep and realized CFL, not as a clean temporal-order study.

Table 10: Manufactured-solution spatial verification errors. The order columns use adjacent-grid $L_2$ errors; the final column checks the inserted forcing against an independently assembled residual.

| $N$ | $\Delta t_{\min}$ | $\text{CFL}_{\max}$ | $\|e_a\|_1$ | $\|e_a\|_2$ | $\|e_a\|_\infty$ | $p_a$ | $\|e_q\|_1$ | $\|e_q\|_2$ | $\|e_q\|_\infty$ | $p_q$ | forcing check |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 50 | 2.500e-06 | 0.02154 | 7.412e-07 | 1.018e-06 | 2.377e-06 | 1.752 | 0.001163 | 0.003447 | 0.017 | 1.497 | 0.000e+00 |
| 100 | 2.500e-06 | 0.04309 | 1.895e-07 | 3.022e-07 | 9.223e-07 | 1.719 | 3.318e-04 | 0.001221 | 0.008493 | 1.502 | 0.000e+00 |
| 200 | 2.500e-06 | 0.08617 | 4.866e-08 | 9.177e-08 | 3.661e-07 | 1.696 | 8.996e-05 | 4.310e-04 | 0.004241 | 1.503 | 0.000e+00 |
| 400 | 2.500e-06 | 0.1723 | 1.245e-08 | 2.833e-08 | 1.457e-07 | – | 2.369e-05 | 1.520e-04 | 0.002119 | – | 0.000e+00 |

Table 11: Manufactured-solution timestep-insensitivity rows on the finest MMS grid. These rows report accepted timestep and realized CFL rather than temporal order.

| $N$ | requested $\Delta t$ | accepted $\Delta t_{\min}$ | accepted $\Delta t_{\max}$ | $\text{CFL}_{\max}$ | $\|e_a\|_2$ | $\|e_q\|_2$ | forcing check |
|---|---|---|---|---|---|---|---|
| 400 | 2.000e-05 | 2.168e-06 | 6.532e-06 | 0.45 | 2.833e-08 | 1.520e-04 | 0.000e+00 |
| 400 | 1.000e-05 | 2.168e-06 | 6.532e-06 | 0.45 | 2.833e-08 | 1.520e-04 | 0.000e+00 |
| 400 | 5.000e-06 | 5.000e-06 | 5.000e-06 | 0.3447 | 2.833e-08 | 1.520e-04 | 0.000e+00 |
| 400 | 2.500e-06 | 2.500e-06 | 2.500e-06 | 0.1723 | 2.833e-08 | 1.520e-04 | 0.000e+00 |

The spatial rows provide bounded, positive implementation-verification evidence for the declared forced operator. The observed $L_2$ rates are approximately 1.70 for area and 1.50 for flow over this grid range, below the formal smooth-region MUSCL/SSPRK3 expectation, so this table is not reported as a formal-order result for every model variant. The $L^1$ history tracks integrated error behavior, while $L^\infty$ highlights localized boundary, limiter, or source-discretization effects.

#### 7.2.3 Geometry-Rest Preservation Test

The continuous selected source-balanced law admits the geometry-rest state $a_i = R_0(z_i)^2$, $q_i = 0$ under compatible zero-flow boundary data (Proposition 7.8). The current principal implementation is not exactly well balanced for that state. The Rusanov/MUSCL operator includes numerical viscosity in the mass flux, and for spatially varying $R_0$ this produces a nonzero discrete residual unless an equilibrium reconstruction or path-conservative flux is added.

Three levels of this statement must be kept separate. Continuous rest-state compatibility is the cancellation property of the source-balanced differential model. The instantaneous discrete residual at $t = 0$ is the semidiscrete right-hand side obtained after sampling that state and applying the selected reconstruction, Rusanov flux, source, and boundary rules. The time-evolved rest drift below is the accumulated response of the time integrator to that residual. Table 12 reports the initial component magnitudes before the drift history is interpreted.

Table 12: Initial geometry-rest residual component magnitudes at $t = 0$. The area row is the Rusanov mass-flux residual. The momentum rows split the elastic-flux difference and wall/geometry source before summing them into the total flow residual. Values are maxima over cells in solver coordinates.

| Case | $N$ | $\max\lvert R_a^{\mathrm{Rus}}\rvert$ | $z_a$ | $\max\lvert R_q^{\mathrm{el}}\rvert$ | $\max\lvert S_q^{\mathrm{wall}}\rvert$ | $\max\lvert R_q^{\mathrm{tot}}\rvert$ | $z_{q,\mathrm{tot}}$ |
|---|---|---|---|---|---|---|---|
| 23% | 400 | 0.69 | 2.122 | 6.908e+04 | 6.845e+04 | 630.3 | 2.137 |
| 23% | 800 | 0.1892 | 2.134 | 6.849e+04 | 6.839e+04 | 175.7 | 2.141 |
| 40% | 400 | 1.234 | 2.122 | 1.043e+05 | 1.040e+05 | 1437.0 | 2.137 |
| 40% | 800 | 0.3225 | 2.126 | 1.043e+05 | 1.042e+05 | 236.7 | 2.134 |

The zero-forcing rest test starts the selected 23% and 40% stenosis geometries from the rest state with a zero-velocity inlet and reports

$$\max_i |q_i|, \qquad \max_i |a_i - R_0(z_i)^2|, \qquad \left|\sum_i a_i \Delta z - \sum_i a_i(0)\Delta z\right|$$

as functions of grid size and elapsed time. This test is required context for any comparison run initialized from the same geometry-rest state.

**Principal numerical limitation.** The zero-forcing geometry-rest runs retain nonzero localized artificial solver flow despite zero requested and applied inlet flow. The method is not exactly well balanced for this geometry-rest state; the defect decreases strongly under refinement but remains non-negligible on the production grid.

Table 13: Zero-forcing, zero-inlet geometry-rest drift summary. The table reports the requested/applied inlet flow, the largest cell flow over positive elapsed times, and the signed finite-volume balance at the final reported time.

| Severity | $N$ | time of peak $\max_i \lvert q_i \rvert$ | requested $q_{\mathrm{in}}$ | applied $q_{\mathrm{in}}$ | peak $\max \lvert q_i \rvert$ | $z_{\max \lvert q \rvert}$ | final $\max \lvert q_i \rvert$ | final $\Delta \int a\,dz$ | final flux integral | final balance residual |
|---|---|---|---|---|---|---|---|---|---|---|
| 23 | 400 | 0.001 | 0.000e+00 | 0.000e+00 | 0.04387 | 2.078 | 0.03839 | 4.744e-07 | -4.744e-07 | -5.654e-11 |
| 23 | 800 | 0.001 | 0.000e+00 | 0.000e+00 | 0.01116 | 2.074 | 0.00973 | 1.841e-07 | -1.841e-07 | -6.173e-13 |
| 40 | 400 | 0.001 | 0.000e+00 | 0.000e+00 | 0.07053 | 2.078 | 0.06339 | 4.047e-06 | -4.047e-06 | -5.310e-11 |
| 40 | 800 | 0.001 | 0.000e+00 | 0.000e+00 | 0.01782 | 2.074 | 0.01605 | 1.254e-06 | -1.254e-06 | -2.340e-12 |

The corrected rest-state run has zero requested and applied inlet flow, so the previously reported $q_{\mathrm{in}}t$ volume signature is not retained. The finite-volume balance $\Delta \int a\,dz + \int (\widehat{F}^a_{\mathrm{out}} - \widehat{F}^a_{\mathrm{in}})\,dt$ closes to near roundoff in the produced CSV. The production comparison-flow scale remains $q_{\mathrm{comp}} \approx 2.288/\pi = 0.7283\,\mathrm{cm}^3/\mathrm{s}$ in the solver coordinate, corresponding to $Q_{\mathrm{comp}} \approx 2.288\,\mathrm{cm}^3/\mathrm{s}$ in physical circular flow, as used in the axial-flow comparison (Figure 5). On the $N = 400$ production comparison grid, the final zero-inlet rest-flow values are 0.03839 and $0.06339\,\mathrm{cm}^3/\mathrm{s}$ in solver $q$ for the 23% and 40% stenosis cases, equal to 5.3% and 8.7% of $q_{\mathrm{comp}}$. The corresponding peak production-grid values are 0.04387 and $0.07053\,\mathrm{cm}^3/\mathrm{s}$. On the finer $N = 800$ check grid, the $t = 1$ s values fall to 0.00973 and $0.01605\,\mathrm{cm}^3/\mathrm{s}$; the observed final-flow refinement rates from $N = 400$ to $N = 800$ are 1.98 for both severities. The defect is therefore strongly localized and strongly grid-decreasing, but it remains non-negligible on the production comparison grid.

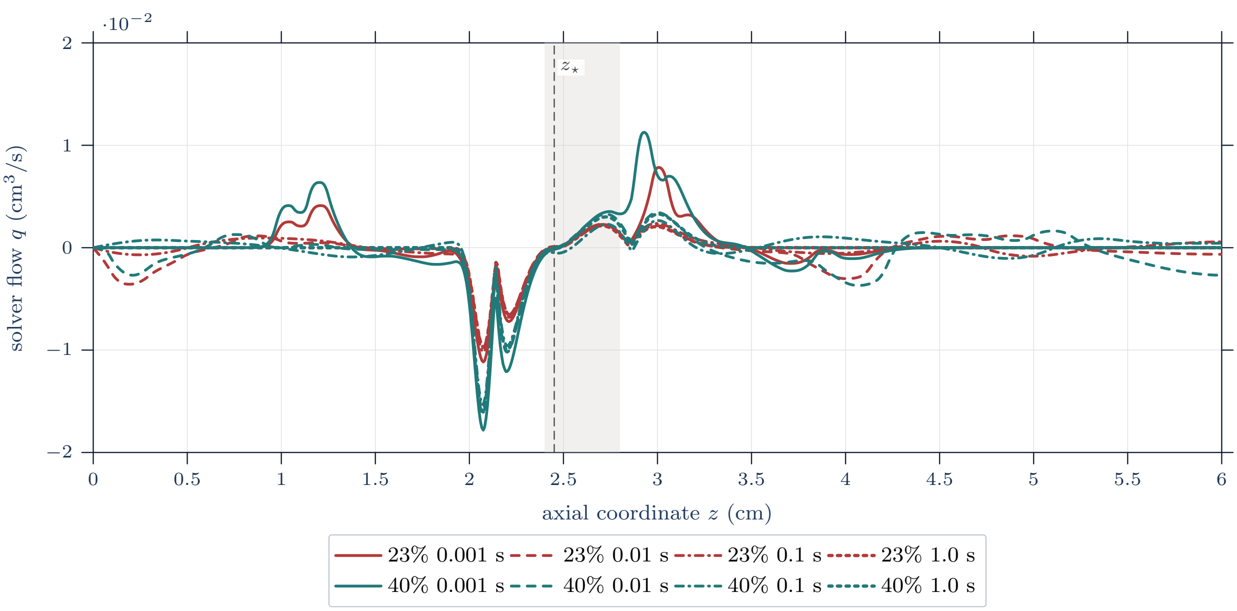


Figure 3: Zero-inlet rest-state solver-flow profiles on the $N = 800$ grid. The nonzero rest flow is localized near the stenosis throat rather than a spatially coherent imposed through-flow.

This geometry-rest result is the principal negative numerical finding of the chapter. The selected 23% and 40% stenosis comparison runs are initialized from the same geometry-rest state and use the same operator. The corrected zero-inlet magnitude no longer supports a large through-flow interpretation, but the nonzero localized residual means the production comparison is read as a diagnostic section-mean velocity comparison, not as a clean production-grid discretization-accuracy study.

### 7.2.4 Boundary, CFL, Positivity, and Conservation Diagnostics

The accepted runs report timestep, realized CFL, characteristic-speed, solver-volume, positivity, and finite-volume balance diagnostics. The fixed-area characteristic boundary rule is used under the subcritical sign condition $\lambda_- < 0 < \lambda_+$. A positive characteristic radicand alone is not the boundary-regime evidence; the persisted sign extrema and subcriticality margin are the relevant diagnostics.

In the rest-state table, the finest-grid 23% and 40% stenosis $t = 1$ s rows retain positive subcritical margins, zero requested and applied inlet flow, signed solver-volume defects, boundary-flux integrals, and near-roundoff finite-volume balance residuals. The rest CSV reports zero positivity projection counts for these rest tests. The production comparison diagnostics in Table 14 provide the corresponding characteristic-speed evidence for the runs later used in the velocity-output comparison. These diagnostics are admissibility and finite-volume balance checks. They do not remove the non-well-balanced geometry-rest drift reported above.

Table 14: Subcritical-boundary diagnostics for the same two 1D runs used in the resolved-velocity comparison. The characteristic-radicand values are in solver units of $\mathrm{cm}^2/\mathrm{s}^2$; the signs of $\lambda_-$ and $\lambda_+$ support the boundary-regime gate in Assumption 7.11.

| Case | $\min \alpha_{\mathrm{eff}}$ | $\max \alpha_{\mathrm{eff}}$ | $\min \chi^2$ | $\min \lambda_-$ | $\max \lambda_-$ | $\min \lambda_+$ | $\max \lambda_+$ | $\min\{\lambda_+, -\lambda_-\}$ |
|---|---|---|---|---|---|---|---|---|
| 23% stenosis | 1.33058 | 1.33333 | $8.15 \times 10^5$ | -998.85 | -852.17 | 953.33 | 1058.76 | 852.17 |
| 40% stenosis | 1.32500 | 1.33333 | $6.36 \times 10^5$ | -999.08 | -713.79 | 880.69 | 1058.94 | 713.79 |

Table 15: Production diagnostics from the 23% and 40% severity 1D comparison runs completed at $T = 0.9995$ s. These rows summarize timestep, CFL, positivity, area, and finite-volume balance for the 1D runs used in the resolved-comparator evidence.

| Stenosis | $T_{\mathrm{target}}$ | $T_{\mathrm{done}}$ | $\lvert\Delta t_{1\mathrm{D}--3\mathrm{D}}\rvert$ | $\Delta t_{\min}$ | $\Delta t_{\max}$ | $\mathrm{CFL}_{\max}$ | $\min \pi a$ | $\Delta \int a\,dz$ | pos. count | area-flux balance | $\max \lvert\dot{a}\rvert$ | $\max \lvert\dot{q}\rvert$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 23% | 0.9995 | 0.9995 | $5.47 \cdot 10^{-14}$ | $3.12 \cdot 10^{-7}$ | $6.56 \cdot 10^{-6}$ | 0.45 | 0.06037 | $1.32 \cdot 10^{-4}$ | 0 | $2.12 \cdot 10^{-5}$ | $9.01 \cdot 10^{-6}$ | 0.02446 |
| 40% | 0.9995 | 0.9995 | $5.47 \cdot 10^{-14}$ | $6.16 \cdot 10^{-7}$ | $6.56 \cdot 10^{-6}$ | 0.45 | 0.03659 | $1.73 \cdot 10^{-4}$ | 0 | $-9.19 \cdot 10^{-6}$ | $1.11 \cdot 10^{-5}$ | 0.01965 |

### 7.2.5 Appendix-Held Secondary Checks

The fourth evidence tier is kept out of the main argument. Appendix G retains package-level checks, sampled self-convergence, nonselected DG p/h behavior, resolved-velocity benchmark context, rheology/profile sensitivity, and fixed-wall stationary-Stokes refinement. Those secondary checks can reveal runnability, boundedness, solver-policy, or sensitivity issues. They do not replace the MMS evidence, and they do not change the geometry-rest conclusion.

## 7.3 Resolved Comparator Evidence

### 7.3.1 Available Resolved Data and Matching Limits

This section reports final-time comparisons between the selected 1D area–flow solution and two resolved stenotic-vessel datasets. Under the reference-coordinate section observation map, the 23% case has mean absolute section-velocity discrepancy 0.505 cm/s and relative RMS discrepancy 2.77%; the 40% case has 0.966 cm/s and 6.88%. The larger discrepancies occur near the throat, and the available matching data do not isolate one closure, wall, boundary, or geometry cause. Both cases are sampled at $T = 0.9995$ s; the measured 1D–3D final-time offset is $5.47 \times 10^{-14}$ s in each case. This timestamp agreement is a final-time alignment check, not a complete transient-history match.

The evidence below separates three resolved-comparator questions. First, the reference-coordinate velocity comparison maps the 1D and 3D fields to common section-mean observables. Second, the deformed-coordinate comparison applies the available node-centered displacement field before forming plane cuts. Third, the quasi-static membrane-FSI solve supplies a resolved wall-deformation comparator based on the membrane relation used by Canic et al. for their wall model [16, Eq. (35)]. The third item is a Canic-style resolved comparator for this report, not a reproduction of the full transient FSI calculation in that paper. The metadata table identifies the resolved files used for these three questions; the scientific comparison is reported by stenosis severity.

Table 16: Resolved-data metadata used by the velocity and displacement-aware comparisons. Dataset identifiers are provenance labels; the case descriptions in the text use stenosis severity.

| Case | Dataset | Nodes | Tetra. | $t_{3D}$ (s) | $\lvert t_{3D} - T\rvert$ (s) | Velocity | Pressure | Displacement |
|---|---|---|---|---|---|---|---|---|
| 23% stenosis | 77 | 22691 | 110420 | 0.9995 | $5.47 \times 10^{-14}$ | node-centered | node-centered | applied in deformed-coordinate cuts |
| 40% stenosis | 60 | 21975 | 106547 | 0.9995 | $5.47 \times 10^{-14}$ | node-centered | node-centered | applied in deformed-coordinate cuts |

The 40% case is therefore both larger in relative RMS discrepancy and more localized near the throat. That localization is tied to the declared section observation map, but the available data do not assign the difference to one closure, wall, boundary, or geometry mechanism.

### 7.3.2 Observation Operator and Deformed Coordinates

The comparison uses the declared cross-section quadrature operator. For a target plane $S_j = \Omega_{\mathrm{3D}} \cap \{z = z_j\}$, the operator intersects the tetrahedral mesh with the plane, triangulates the cut polygon, and computes

$$A_{3D,j} = \sum_{\Delta \in \mathcal{Q}_j} |\Delta|, \qquad Q_{3D,j} = \sum_{\Delta \in \mathcal{Q}_j} |\Delta| \, \bar{u}_{z,\Delta}, \qquad \bar{u}_{3D,j} = Q_{3D,j}/A_{3D,j}.$$

The 1D quantities use the solver-to-physical map $Q_{1D,j} = \pi q(z_j, t)$ and $\bar{u}_{1D,j} = q(z_j, t)/a(z_j, t)$. In deformed-coordinate mode the plane cut is formed after replacing each mesh coordinate $x$ by $x + d(x)$, where $d$ is the

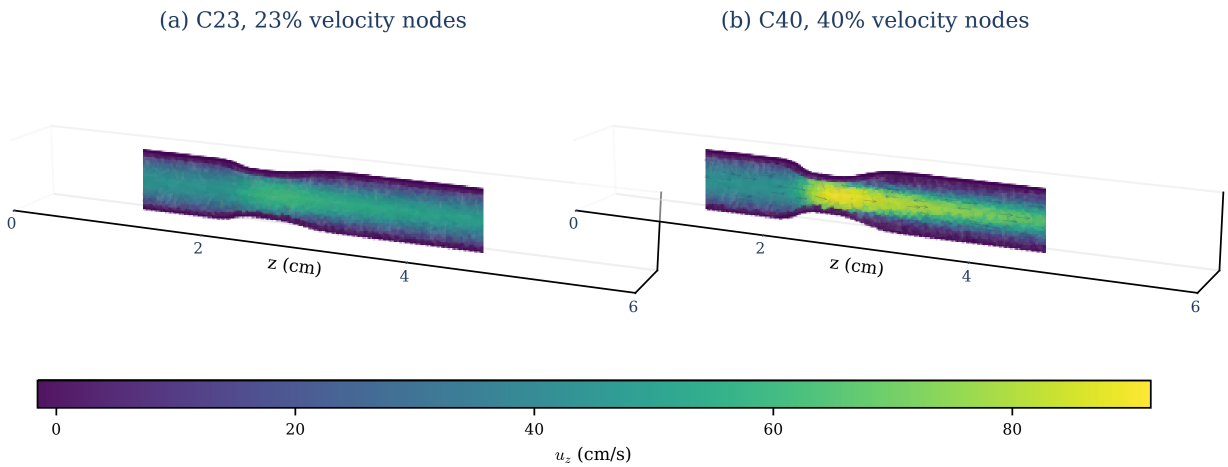


Figure 4: Final-time resolved velocity samples used by the 23% and 40% stenosis comparisons. The panels show node-centered $u_z$ values; the quantitative comparison enters only after applying the declared cross-sectional observation operator.

node-centered resolved displacement field.

The stenosis throat is the minimizer $z^\star = \operatorname{argmin}_z R_0(z)$ of the smooth reference-radius profile in Eq. 7.3. For the comparison geometry this gives $z^\star \approx 2.451\,\mathrm{cm}$, with $R_{\min} = 0.1386\,\mathrm{cm}$ for the 23% stenosis case and $R_{\min} = 0.1080\,\mathrm{cm}$ for the 40% stenosis case. The area check below shows that the section cuts track the analytic reference geometry closely enough for the velocity comparison to be interpreted as an observation-map comparison rather than a gross cut-geometry mismatch.

Table 17: Cut-area check against the static reference $A_{\mathrm{ref}}(z) = \pi R_0(z)^2$ for both coordinate choices. Discrepancies are percentages over the 200 valid section targets.

| **Case** | **Coordinates** | **min $\epsilon_A$** | **median $\epsilon_A$** | **mean $\epsilon_A$** | **max $\epsilon_A$** |
|---|---|---|---|---|---|
| 23% stenosis | reference | 0.045 | 0.281 | 0.389 | 2.94 |
| 23% stenosis | deformed | 0.001 | 0.254 | 0.336 | 3.03 |
| 40% stenosis | reference | 0.005 | 0.291 | 0.397 | 4.46 |
| 40% stenosis | deformed | 0.003 | 0.286 | 0.417 | 4.60 |

### 7.3.3 Reference-Coordinate Velocity Comparison

The reference-coordinate comparison is retained as the baseline final-time velocity comparison. The section means are $\langle Q_{1D} \rangle = 2.288\,\mathrm{cm}^3/\mathrm{s}$ and $\langle Q_{3D} \rangle = 2.241\,\mathrm{cm}^3/\mathrm{s}$ for the 23% stenosis case, and $\langle Q_{1D} \rangle = 2.287\,\mathrm{cm}^3/\mathrm{s}$ and $\langle Q_{3D} \rangle = 2.219\,\mathrm{cm}^3/\mathrm{s}$ for the 40% stenosis case. The discrepancy summaries below show that the 40% case has the larger section-mean velocity gap, concentrated near the stenosis throat.

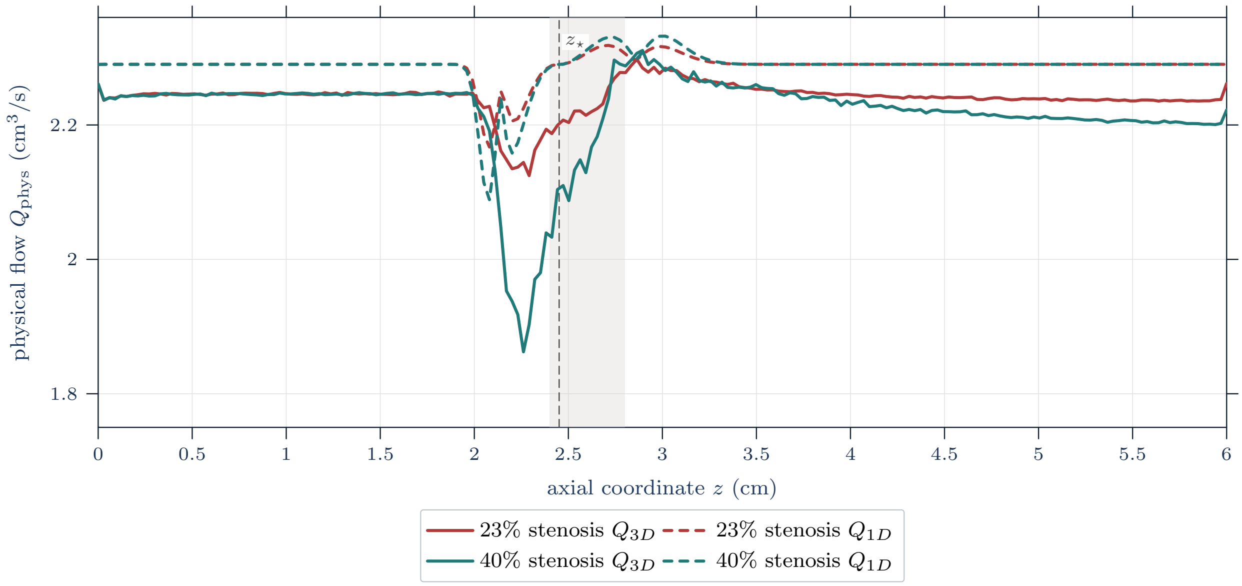


Figure 5: Axial physical-flow comparison after applying the reference-coordinate plane-quadrature observation operator. Dashed curves are $Q_{1D}(z) = \pi q(z)$ and solid curves are $Q_{3D}(z)$ from the resolved velocity fields; the figure compares observed section flow, not native resolved velocity fields or pressure-ratio output.

The algebraic split used for flow and area effects is

$$\frac{Q_{1D}}{A_{1D}} - \frac{Q_{3D}}{A_{3D}} = \frac{Q_{1D} - Q_{3D}}{A_{1D}} + Q_{3D}\left(\frac{1}{A_{1D}} - \frac{1}{A_{3D}}\right), \qquad A_{1D} = \pi a.$$

This decomposition separates sampled-section terms algebraically; it is not causal attribution.

Table 18: Section-wise decomposition of the reference-coordinate mean-velocity discrepancy into physical-flow and area-denominator contributions. Values are in cm/s. This algebraic split is tied to the declared section observation operator and is not a causal error attribution.

| **Case** | $\langle \lvert (Q_{1D} - Q_{3D})/A_{1D} \rvert \rangle$ | $\langle \lvert Q_{3D}(A_{1D}^{-1} - A_{3D}^{-1}) \rvert \rangle$ | $z_{\max \lvert d_u \rvert}$ | $\lvert d_u \rvert_{\max}$ | **Larger term at max** |
|---|---|---|---|---|---|
| 23% stenosis | 0.529 | 0.112 | 2.291 | 2.35 | Flow term |
| 40% stenosis | 1.012 | 0.133 | 2.261 | 9.92 | Flow term |

Table 19: Reference-coordinate section-mean velocity discrepancy summary at the matched final-time sample after applying the common section observation map. Values are in cm/s except for the axial location column in cm; the table reports diagnostic cross-model discrepancy, not clinical validation.

| **Case** | **Sections** | $\langle \bar{u}_{1D} \rangle$ | $\langle \bar{u}_{3D} \rangle$ | $\overline{d}_u$ | $D_{u,1}$ | $D_{u,2}$ | **rel.** $D_{u,2}$ | **max** $\lvert d_u \rvert$ | $z_{\max \lvert d_u \rvert}$ |
|---|---|---|---|---|---|---|---|---|---|
| 23% stenosis | 200 | 24.11 | 23.63 | 0.480 | 0.505 | 0.664 | 0.0277 | 2.35 | 2.291 |
| 40% stenosis | 200 | 26.43 | 25.52 | 0.909 | 0.966 | 1.87 | 0.0688 | 9.92 | 2.261 |

The larger 40% discrepancy is therefore throat-localized and depends on the declared section observation map. It cannot be assigned uniquely to one closure, wall, boundary, geometry, or numerical mechanism from the available matched data.

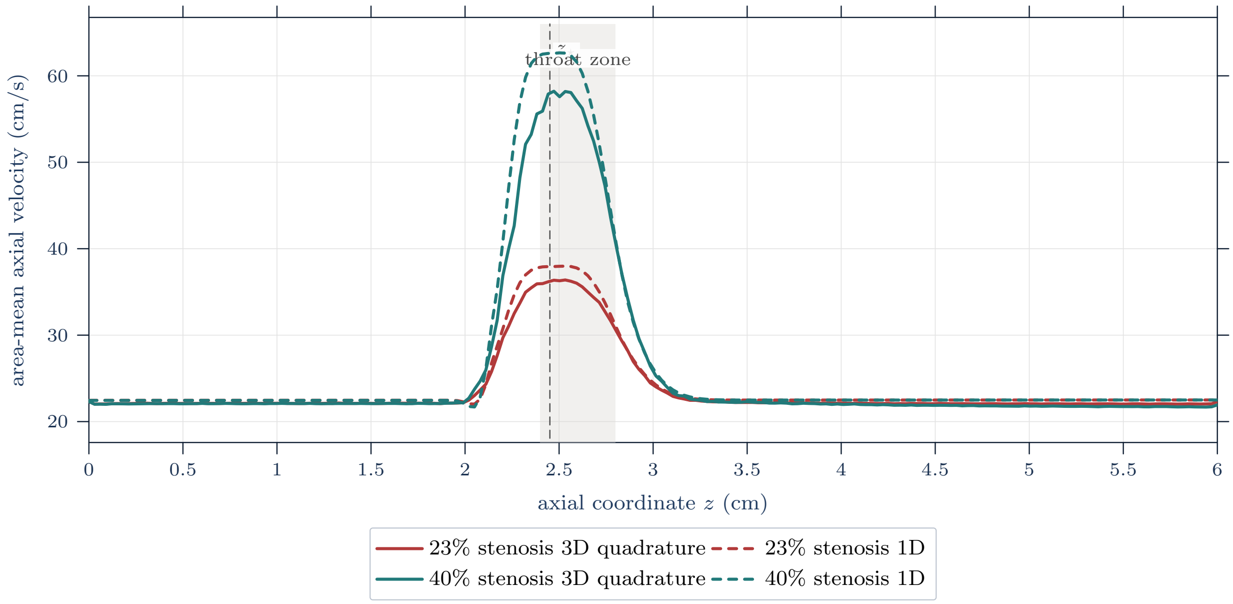


Figure 6: Area-mean axial velocity comparison at $t = 0.9995$ s under the reference-coordinate section observation map. Solid curves are tetrahedral cross-section quadrature means from the resolved 3D velocity field; dashed curves are interpolated 1D means $\bar{u} = q/a = Q_{\text{phys}}/A_{\text{phys}}$.

Table 20: Normalized scale comparison between the production-grid geometry-rest limitation and final-time 1D–3D velocity-discrepancy measures. These ratios place the diagnostic comparison on a common scale; they do not convert the comparison into an external validation result.

| Case | rest max $\lvert q\rvert$/prod. $q$ | $D_{Q,1}/\langle Q_{1D}\rangle$ | $D_{u,1}$ (cm/s) | rel. $D_{u,2}$ |
|---|---|---|---|---|
| 23% stenosis | 5.27% | 2.11% | 0.505 | 2.77% |
| 40% stenosis | 8.71% | 3.10% | 0.966 | 6.88% |

Appendix G.4 retains the optional production-grid sensitivity rows and the five largest section-wise velocity discrepancies. Those rows support the stated localization and numerical-sensitivity reading without changing the main comparison claim.

### 7.3.4 Deformed-Coordinate Comparison

The deformed-coordinate comparison repeats the same section operator after moving each resolved node by its displacement value. Table 21 shows that applying displacement changes the final-time discrepancy measures only modestly for these two datasets. This is still not a full wall-motion match: the available comparison data contain the final displacement field, not the resolved wall history or boundary histories.

Table 21: Reference-coordinate versus deformed-coordinate discrepancy measures for the same section observation map. $D_{u,1}$ and $D_{u,2}$ are the mean absolute and RMS section-mean velocity discrepancies in cm/s; $D_{Q,1}$ is the mean absolute physical-flow discrepancy in cm$^3$/s.

| Case | Coordinates | $D_{u,1}$ | $D_{u,2}$ | $\max\lvert d_u\rvert$ | $D_{Q,1}$ |
|---|---|---|---|---|---|
| 23% stenosis | reference | 0.5048 | 0.6636 | 2.349 | 0.04821 |
| 23% stenosis | deformed | 0.5117 | 0.6684 | 2.357 | 0.04750 |
| 40% stenosis | reference | 0.9660 | 1.868 | 9.923 | 0.07087 |
| 40% stenosis | deformed | 0.9741 | 1.869 | 9.935 | 0.07003 |

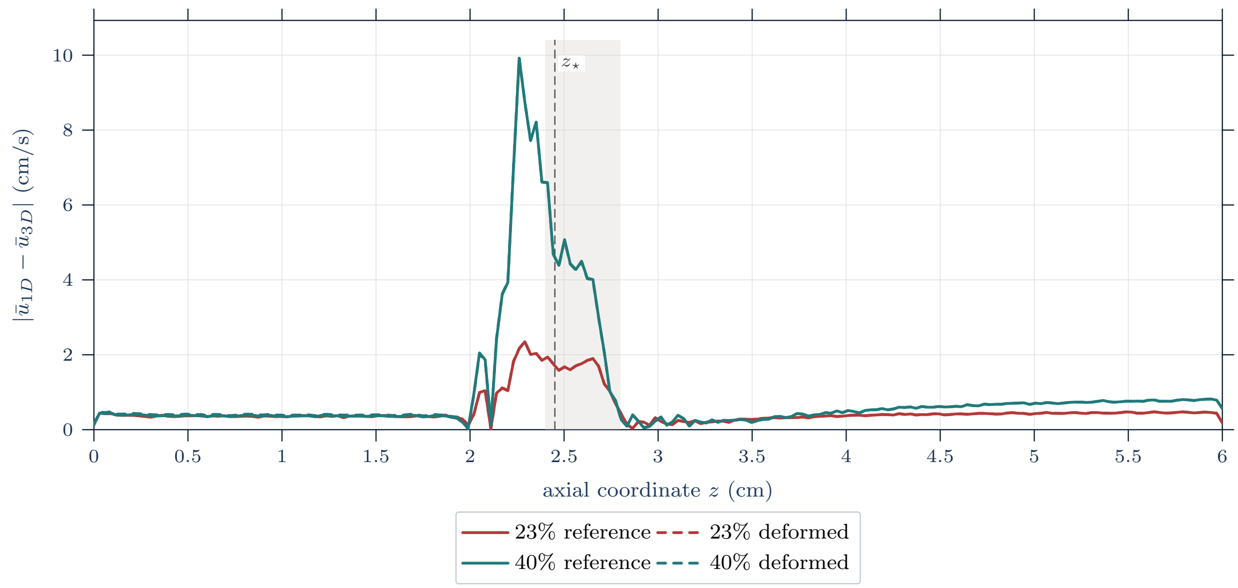


Figure 7: Effect of the displacement-aware observation map on the section-wise velocity discrepancy. Solid curves use reference mesh coordinates, while dashed curves form the same plane cuts after the nodal map $x \mapsto x + d(x)$; the figure changes the coordinate mode, not the 1D model or the resolved velocity field.

### 7.3.5 Quasi-static Membrane-FSI Comparator

The resolved wall comparator solves a quasi-static fixed point between the Stokes lumen solve and the radial membrane update. In the notation of Appendix G.6, the wall equation is the inertia-free form

$$C_0\eta_r(z) = f_r(z), \qquad \eta_r(0) = \eta_r(L) = 0, \qquad C_0 = \frac{Eh}{(1-\sigma^2)(R_0^*)^2}.$$

This is the Canic-style membrane relation used for the resolved comparator [16, Eq. (35)]. It is separate from the selected 1D pressure–area closure used in the reduced model. The retained comparator is a quasi-static radial membrane update coupled to repeated Stokes solves; it informs wall-motion scale, not full transient moving-boundary FSI validation.

Table 22: Quasi-static membrane-FSI fixed-point results on the retained finer mesh. The reported displacement is the radial wall displacement $\eta_r$; the minimum radius is the deformed lumen radius $R_0 + \eta_r$. This comparator informs wall-motion scale only and does not supply evidence for a full transient moving-boundary FSI model.

| Case | Mesh | Iter. | Residual (cm) | $\max \eta$ (cm) | $\min R$ (cm) | $\langle Q \rangle$ (cm$^3$/s) |
|---|---|---|---|---|---|---|
| 23% stenosis | $16 \times 4 \times 16$ | 9 | 5.972e-8 | 3.052e-5 | 0.1402 | 0.5472 |
| 40% stenosis | $16 \times 4 \times 16$ | 9 | 6.051e-8 | 3.092e-5 | 0.1108 | 0.416 |

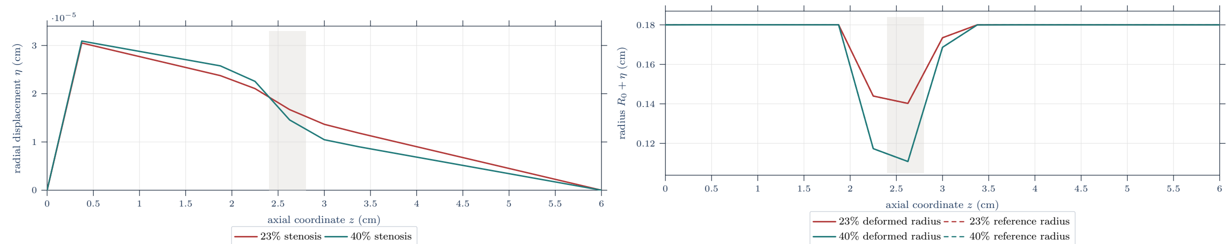


Figure 8: Quasi-static wall deformation from the membrane-FSI comparator. The upper panel shows the radial displacement $\eta_r(z)$ solving the fixed-end membrane equation; the lower panel shows the resulting deformed radius $R_0(z) + \eta_r(z)$ alongside the reference radius.

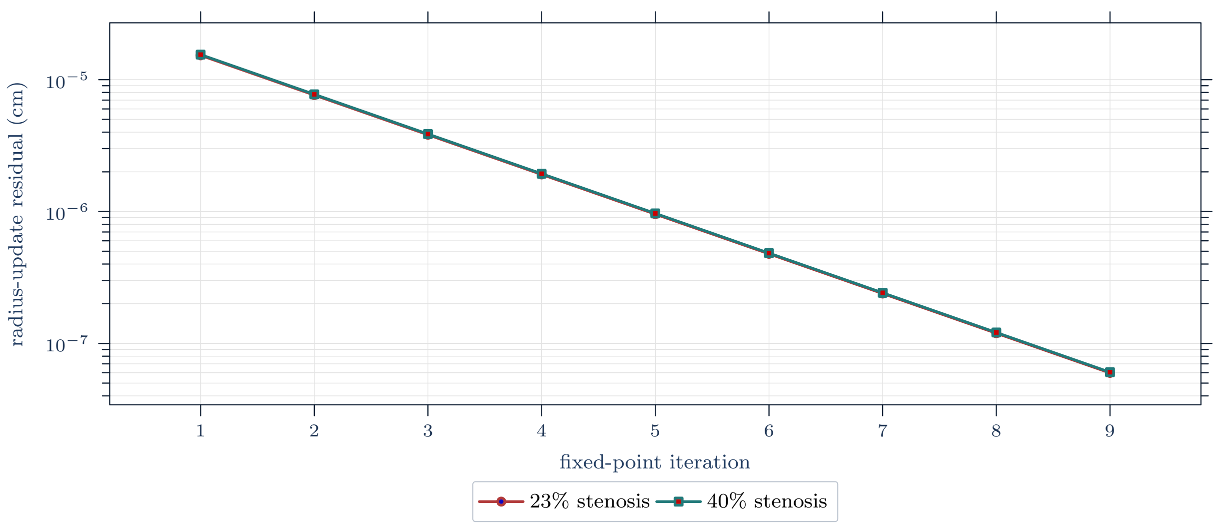


Figure 9: Fixed-point convergence for the quasi-static membrane-FSI comparator. The residual is the maximum radial-radius update between successive coupled wall solves, reported in centimeters; it checks the comparator solve, not the 1D–3D velocity observation error.

#### 7.3.6 Radial-Profile Limitation

Radial distributions of axial velocity are not promoted because the retained report evidence has not yet accepted regenerated same-cut radial-profile rows. The package audit now classifies section and radial-bin closure on identical cuts, but the manuscript retains only the section-mean axial-velocity comparison until those regenerated report assets are reviewed and accepted.

#### 7.3.7 Interpretation Limits

This chapter reports a controlled diagnostic cross-model comparison against declared resolved comparators: a reference-coordinate velocity comparison, a displacement-aware observation comparison, and a quasi-static membrane-wall comparator. The evidence supports the implemented observation and comparison workflow and localizes final-time velocity discrepancies under the stated operators. It does not provide externally validated clinical prediction evidence, pressure-ratio evidence, or equivalence to a full transient FSI computation with matched boundary and wall-history data.

# 8 Integrated Discussion

## 8.1 RQ1: Model Hierarchy and Question Matching

The first research question asked how blood-flow model classes are organized and what distinguishes them. The answer is that the hierarchy is a set of retained-state, closure, coupling, and observation choices, not a ranking of accuracy. Formaggia, Lamponi, and Quarteroni show that section-averaging and wall simplification define different 1D operators; Shi, Lawford, and Hose distinguish lumped global pressure–flow interactions from distributed 1D pulse propagation; and Quarteroni–Veneziani multiscale coupling shows that boundary data can be a reduced model of omitted circulation [3, 4, 5]. Thus 0D, 1D, and 2D formulations should not be described as weaker versions of 3D CFD; they retain different states and require different observation maps.

The case study demonstrates this answer by using a distributed 1D area–flow state only for quantities it can natively or reconstructively support: area, flow, section-mean velocity, wall-law pressure, and a declared velocity observation map. It does not ask that state to produce native WSS, secondary flow, wall stress, or clinical pressure-ratio evidence.

## 8.2 RQ2: Observables and Closure Choices

The second research question asked how rheology, wall mechanics, geometry, boundary data, numerical discretization, evidence standards, and observation operators affect interpretation. The answer is that these choices define the quantity being reported. Blanco and coauthors support this point for FFR-specific 1D–3D comparison under matched pressure boundary conditions and dedicated loss closures; Liu and coauthors show observable-dependent rheology sensitivity in an intracranial stenosis setting; John and coauthors show that WSS-vector convention changes AWSS/OSI; and Gambaruto and coauthors show that geometry reconstruction can be a material sensitivity source in a patient aneurysm case [9, 14, 17, 31].

The case study therefore presents its closure list as mathematical information, not as implementation detail: Newtonian principal rheology, parabolic profile closure, an elastic pressure–area wall law, analytic reference geometry, fixed-area characteristic boundary approximation, and a plane–tetrahedron velocity observation operator. The resulting comparison is a velocity-observation comparison under those assumptions, not a WSS, wall-displacement, or FFR evidence claim. The membrane discussion sharpens the same boundary: Newtonian or generalized-Newtonian terminology belongs to the fluid stress law, while the membrane model is a structural wall closure and the quasi-static comparator is not evidence for a full transient moving-boundary FSI model.

### 8.3 RQ3: Numerical Evidence and Cross-Dimensional Interpretation

The third research question asked what limitations remain when resolved scale, cost, identifiability, validation status, reproducibility, and uncertainty are compared. The answer is that each evidence type supports a different claim. ASME VVUQ terminology separates verification, validation, and uncertainty quantification; Boileau and coauthors provide benchmark comparison evidence; Pimentel-Garcia and coauthors show that a well-balanced claim must name the preserved stationary family; and Colebank shows that practical identifiability depends on measured outputs, boundary-condition class, and design assumptions [6, 40, 41, 44].

The worked stenosis example demonstrates the distinction. Its manufactured solution supports code-verification claims for a forced operator. Its geometry-rest test supports a separate statement about equilibrium preservation and exposes the current non-well-balanced limitation. Its 1D–3D comparison is interpretable only after physical flow and area-mean velocity are extracted through a common observation operator. These checks support a bounded diagnostic cross-model comparison, not an external validation result. The positive MMS result, negative rest-state result, and final-time comparator are useful precisely because they cannot be collapsed into one validation grade.

### 8.4 Contribution and Limitations

The contribution is a review framework for reading mathematical hemodynamics across model tiers. It supplies a contract that connects retained state, closures, boundary data, discretization, observation operator, and evidence type. It applies that contract once in an idealized stenosis computation.

The limitations are correspondingly bounded. The review is narrative and scoping, not systematic. Some sources remain preprints, software artifacts, or metadata entries with roles narrower than peer-reviewed empirical evidence. The case geometry, wall law, rheology, and boundary approximation are idealized. The numerical evidence distinguishes verification, equilibrium preservation, and diagnostic comparison, but it does not provide externally validated clinical prediction evidence or a fully matched transient FSI target.

## 9 Conclusion

This report's main conclusion is that hemodynamic simulation claims support a claim only when the equations, retained state, closures, boundary data, discretization, evidence standard, and observation operator are declared together. Model dimension alone does not determine interpretation, and neither a benchmark, a grid study, a cross-model comparison, nor a learned surrogate result substitutes for validation against an external target for a specified quantity and context. Likewise, a membrane-wall update, a pressure–area law, and a resolved FSI calculation answer different structural questions unless their wall state, loads, and observation operators are matched.

The report does not claim externally validated clinical prediction for the idealized stenosis case or universal superiority of any model tier. Future work should strengthen the same chain where the present evidence is intentionally limited: construct or adopt a well-balanced discretization for the geometry-rest state, obtain matched wall and boundary metadata for resolved comparisons, report uncertainty separately from model discrepancy, and specify validation observables before predictive or clinical claims are made.

# A Acronyms and Abbreviations

**General abbreviations.**

| | | | |
|---|---|---|---|
| e.g. | *exempli gratia* (for example) | i.e. | *id est* (that is) |

**Report acronyms.**

| | | | |
|---|---|---|---|
| CFD | computational fluid dynamics | CFL | Courant–Friedrichs–Lewy |
| DG | discontinuous Galerkin | FFR | fractional flow reserve |
| FSI | fluid–structure interaction | HDF5 | Hierarchical Data Format version 5 |
| MMS | method of manufactured solutions | ODE | ordinary differential equation |
| PDE | partial differential equation | SSPRK | strong-stability-preserving Runge–Kutta |
| WSS | wall shear stress | XDMF | eXtensible Data Model and Format |
| 0D | zero-dimensional | 1D | one-dimensional |
| 2D | two-dimensional | 3D | three-dimensional |

# B Mathematical Notation

**Notation B.1** (Report mathematical notation)**.** This appendix records the shared mathematical shorthand that remains active in this report after the main text has separated the review notation from the implemented 1D solver, verification evidence, and diagnostic velocity comparison. Elementary function-space definitions and compactness background are not repeated here.

| Symbol | Meaning |
|---|---|
| $:=$ | defines |
| $\equiv$ | notational equivalence or identified-with relation |
| $\mathbb{R}$, $\mathbb{R}^+$, $\mathbb{N}$, $\mathbb{N}_0$ | real numbers, positive real numbers, natural numbers, and nonnegative integers |
| $[n]$ | index set $\{1, 2, \dots, n\}$ |
| $I_T$ | open time interval $(0, T_{\text{final}})$ for transient interior-time statements |
| $\overline{I}_T$ | closed time interval $[0, T_{\text{final}}]$ for initial data and sampled outputs |
| $K$ | compact output rectangle, typically $[0, L] \times [0, T_{\text{final}}]$, when a sampled norm is locally declared |
| $\Omega$, $\overline{\Omega}$, $\partial\Omega$ | spatial domain, closure, and boundary |
| $\Omega_B(t)$ | spatial blood domain at time $t$ when a continuum tier is being referenced |
| $\mathcal{Q}_T$ | space-time domain $\{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}$ |
| $d\boldsymbol{x}$, $dS_{\boldsymbol{x}}$, $dV$, $dS$ | volume and surface measure elements in the locally stated dimension |
| $\nabla$, div, **div**, $\Delta$ | gradient, vector divergence, row-wise tensor divergence, and scalar Laplacian |
| $\partial_t$, $\partial_z$, $\partial_A$, $\partial_r$, $\partial_\theta$ | compact partial-derivative notation with respect to the named coordinate |
| $\partial_{\hat{\boldsymbol{n}}}\phi$ | normal derivative $\langle \nabla\phi,\ \hat{\boldsymbol{n}} \rangle$ on a boundary with unit normal $\hat{\boldsymbol{n}}$ |
| $\boldsymbol{v}_i$ | $i$th component of a vector $\boldsymbol{v}$ |
| $\boldsymbol{S} : \boldsymbol{T}$ | Frobenius contraction of second-order tensors |
| $\langle \cdot,\ \cdot \rangle_X$ | inner product on a stated vector space $X$ |
| $\lVert\cdot\rVert_X$ | norm on a stated normed space $X$; a bare norm is local shorthand only |
| $\lVert\cdot\rVert_{\mathrm{F}}$ | Frobenius norm of a matrix or second-order tensor |
| $(r, \theta, z)$, $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$ | cylindrical coordinates and basis, used only when a retained continuum reference requires them |

## C Domain-Specific Notation

**Definition C.1** (SI unit convention)**.** Unless otherwise stated, physical continuum quantities use SI units: length in meters, time in seconds, density in $\mathrm{kg/m^3}$, pressure in Pa, dynamic viscosity in Pa·s, and kinematic viscosity in $\mathrm{m^2/s}$.

**Convention C.2** (Centimeter–second numerical-record convention)**.** The Julia solver and the generated report tables use the centimeter–gram–second numerical-record convention unless a table states otherwise. Lengths are in cm, time in s, density in $\mathrm{g/cm^3}$, velocity in cm/s, pressure and stress in $\mathrm{dyn/cm^2}$, dynamic viscosity in g/(cm s), and kinematic viscosity in $\mathrm{cm^2/s}$.

The implemented 1D solver stores a radius-squared coordinate $a = R^2$ and a solver flow $q$. Physical circular area and physical flow are

$$A_{\mathrm{phys}} = \pi a, \qquad Q_{\mathrm{phys}} = \pi q, \qquad \bar{u} = \frac{q}{a} = \frac{Q_{\mathrm{phys}}}{A_{\mathrm{phys}}}.$$

Thus legacy CSV headers such as `A_cm2` and `Q_cm3_s` are read in this report as the solver-coordinate fields $a$ and $q$ unless a physical quantity is explicitly labelled `Aphys_cm2` or `Qphys_cm3_s`. A Pa-valued command-line pressure is converted before storage by $1\,\mathrm{Pa} = 10\,\mathrm{dyn/cm^2}$.

**Convention C.3** (Pressure-output convention)**.** Pressure-drop and pressure-ratio symbols are retained only as reduced-output vocabulary. A pressure-drop trace requires declared proximal and distal scalar pressure observations. A pressure-ratio output additionally requires a declared observation tuple

$$\pi_{\mathrm{PR}} := (\mathcal{O}_{\mathrm{prox}}, \mathcal{O}_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}}),$$

with a finite averaging window and a positive denominator admissibility bound. For observed gauge-pressure traces $P_{\mathrm{prox}}$ and $P_{\mathrm{dist}}$,

$$\begin{aligned} \Delta p(t) &:= P_{\mathrm{prox}}(t) - P_{\mathrm{dist}}(t), \\ \mathrm{PR}_{1D}(\pi_{\mathrm{PR}}) &:= \frac{\langle p_{\mathrm{ref}} + P_{\mathrm{dist}} \rangle_{\mathcal{T}_{\mathrm{HF}}}}{\langle p_{\mathrm{ref}} + P_{\mathrm{prox}} \rangle_{\mathcal{T}_{\mathrm{HF}}}}. \end{aligned}$$

The subscript HF is only a declared averaging-window label in this report, and no pressure-ratio result is reported in the diagnostic velocity comparison. The shorthand is not a clinical measurement model [46, Ch. 5, Sec. 5.5, pp. 72–73] [47, p. 1703].

**Convention C.4** (Shear-output convention)**.** Shear terminology is model-tier dependent. A resolved wall-shear-stress output requires a resolved stress tensor, wall normal, wall location, output operator, and, for a signed component, a tangent direction [17, Abstract, Sec. 2, pp. 8–10]. A reduced 1D shear proxy requires its closure, profile, location, and sign or window convention. Reduced shear proxies are closure diagnostics, not resolved traction outputs.

| **Symbol** | **Meaning** | **Units** |
|---|---|---|
| $R$, $d_v = 2R$ | vessel radius and diameter | length |
| $R_0(z)$ | reference stenosed radius used by the geometry and wall source | length |
| $R_{\max}$ | healthy reference-radius parameter used by the implemented $R_{\max}$-normalized wall law | length |
| $a = R^2$ | solver radius-squared area coordinate stored as `A` in legacy output | length$^2$ |
| $q$ | solver flow coordinate stored as `Q` in legacy output | length$^3$/time |
| $A_{\text{phys}} = \pi a$ | physical circular cross-sectional area | length$^2$ |
| $Q_{\text{phys}} = \pi q$ | physical circular-area flow | length$^3$/time |
| $\bar{u} = q/a$ | section-mean velocity used by the 1D–3D observation comparison | length/time |
| $\alpha_{\mathcal{P}}$ | momentum-flux correction for the declared velocity profile; the parabolic baseline uses 4/3 | – |
| $g_{\mathcal{P}}$ | profile shear-rate factor used in the reduced wall-friction and viscosity proxy | – |
| $\mathcal{R}$ | rheology closure family used by the reduced closure | mixed |
| $\mathcal{W}$ | wall closure family; the selected report model uses the Canic–Koiter law | mixed |
| $K = Eh/(1-\sigma^2)$ | wall stiffness coefficient used by the implemented wall law | force/length |
| $\beta(z)$ | wall-law stiffness parameter; for the selected evolution law $\beta = \sqrt{\pi} K R_0^2/R_{\max}^2$ | pressure $\cdot$ length |
| $\Psi$ | elastic flux potential satisfying $\partial_A \Psi = (A/\rho)\partial_A p^{\text{evol}}_{\mathcal{W},\text{g}}$ in the physical-area notation | length$^4$/time$^2$ |
| $\mathbf{U} = (a, q)^\top$ | implemented finite-volume state in solver coordinates | mixed |
| $\widehat{\mathbf{F}}_{i+1/2}$ | numerical flux at a finite-volume interface | mixed |
| $\widehat{S}_i$ | cell-centered source term used by the implemented operator | mixed |
| $D_z^h$ | recorded finite-difference operator on cell-centered quantities | 1/length |
| $\dot{\gamma}_{1D} = g_{\mathcal{P}}\lvert q\rvert/(a\sqrt{a})$ | reduced shear-rate proxy used for generalized-Newtonian computed closure-family cases | 1/time |
| $\nu_{\text{eff}}^{\mathcal{R}}$ | density-scaled effective kinematic viscosity used in reduced source terms | length$^2$/time |
| $c$ or $c_{\text{wave}}$ | 1D pulse-wave speed for the declared wall law | length/time |
| $p^{\text{evol}}_{\mathcal{W},\text{g}}$ | reduced gauge pressure produced by the selected evolution wall law | pressure |
| $p^{\text{diag}}_{\text{loc,g}}$ | legacy local-denominator pressure diagnostic from the implementation helper | pressure |
| $\tau^{1D}_{w,z}$ | signed 1D axial shear proxy after declaring closure and output convention | pressure |
| $\mathcal{G}^{r_h,e}_{1D,h}$ | computed fixed-grid 1D map indexed by numerical and closure records | mixed |

*Remark* C.5 (Viscosity notation). This report writes $\eta$ for dynamic viscosity and

$$\nu := \frac{\eta}{\rho}$$

for kinematic viscosity. Generalized-Newtonian closure families use a stress-level effective dynamic viscosity $\eta_{\text{eff}}(\dot{\gamma})$ and the corresponding density-scaled coefficient $\nu_{\text{eff}}^{\mathcal{R}} = \eta_{\text{eff}}^{\mathcal{R}}/\rho$ in reduced source terms. This convention avoids overloading $\nu$ with the dynamic coefficient used in continuum stress laws [11, Sec. 3.1, p. 35, Sec. 3.6, p. 215] [17, Sec. 1.1, pp. 6–7, Eqs. (1.2)–(1.4), Fig. 4.4, p. 18].

# D Mathematical Convention Appendix

This appendix records only the analytic conventions needed by the continuum derivations and coordinate appendices. Elementary compactness, Banach-space, and function-space background is not repeated here.

**Convention D.1** (Standing regularity convention)**.** Unless a section states a stronger or weaker hypothesis, displayed classical identities are used only under the regularity needed for their terms to be defined. Fields have the indicated classical derivatives on the stated domains; boundary traces are assumed when boundary terms are written; and integration-by-parts or adjoint identities are invoked only when the required boundary terms are meaningful. When a weak formulation is discussed, the relevant Sobolev space, trace theorem, and boundary regularity assumptions are stated locally rather than inferred from this convention. Boundaries are at least piecewise $C^1$ for classical boundary integrals, and outward unit normals are interpreted on the regular boundary pieces.

**Definition D.2** (Domain, time, and field notation)**.** For a bounded open connected domain $\Omega \subset \mathbb{R}^n$ with the locally stated regularity, $\overline{\Omega}$ denotes its closure and $\partial\Omega$ its boundary. The report uses

$$I_T := (0, T_{\mathrm{final}}), \qquad \overline{I}_T := [0, T_{\mathrm{final}}], \qquad K := [0, \ell] \times [0, T_{\mathrm{final}}]$$

unless a section declares a local final time or length. Scalar, vector, and second-order tensor fields are written

$$\phi : \Omega \to \mathbb{R}, \qquad \boldsymbol{f} : \Omega \to \mathbb{R}^n, \qquad \boldsymbol{T} : \Omega \to \mathbb{R}^{n \times n}.$$

Multi-index notation is used in the standard way: $D^\alpha \phi$ denotes the classical derivative of order $|\alpha| = \alpha_1 + \cdots + \alpha_n$ when it exists. Spaces such as $C^k(\overline{\Omega})$ and $L^p(\Omega)$ are used with their standard meanings; the norm subscript always identifies the intended space when a norm is needed.

**Definition D.3** (Differential-operator conventions)**.** Named-coordinate partial derivatives use compact notation, for example

$$\partial_t \phi = \frac{\partial \phi}{\partial t}, \qquad \partial_z f = \frac{\partial f}{\partial z}, \qquad \partial_A p = \frac{\partial p}{\partial A}.$$

For scalar fields, $\nabla \phi$ is the column gradient and $\Delta \phi = \nabla \cdot \nabla \phi$. For vector fields, $\nabla \boldsymbol{f}$ is the Jacobian with entries $(\nabla \boldsymbol{f})_{ij} = \partial_{x_j} f_i$ and

$$\operatorname{div} \boldsymbol{f} = \nabla \cdot \boldsymbol{f} = \sum_{i \in [n]} \partial_{x_i} f_i.$$

For tensor fields, the report uses the row-divergence convention

$$(\mathbf{div}\boldsymbol{T})_i = \sum_{j\in[n]} \partial_{x_j} T_{ij}.$$

This is the convention used in the continuum momentum balance and in the Navier–Stokes coordinate appendix. Componentwise integrals and Laplacians are used for vector- and tensor-valued fields whenever the relevant components are integrable or differentiable.

**Definition D.4** (Norm conventions)**.** The report uses the usual $L^p$ and $C^k$ norms when they are explicitly named. For vector-valued $L^p$ quantities, the pointwise range norm is Euclidean. For matrix or second-order tensor values, the pointwise range norm is the Frobenius norm,

$$\|\boldsymbol{T}\|_{\mathrm{F}} \ := \ (\boldsymbol{T} : \boldsymbol{T})^{1/2} = \left( \sum_{i,j\in[n]} T_{ij}^2 \right)^{1/2}.$$

Bare norms are local shorthand only; global notation tables state the intended space or range norm where ambiguity would otherwise arise.

# E Continuum Derivation Details

This appendix supplements the continuum mechanics details that support the main-body assumptions without interrupting the reduced-order forward-model narrative within this report. It is appendix background for notation and model hierarchy, not additional evidence for the numerical study. All identities below are classical identities under the regularity needed for the displayed derivatives, traces, and changes of variables. In particular, **div** is the row-divergence of a tensor field, and boundary forms assume the trace and normal hypotheses needed for the displayed integrals. The final subsection gives compact integral and variational bridge forms used to distinguish finite-volume, finite-element, and moving-domain discretization contracts.

## E.1 Material Derivative and Trajectory Details

For a scalar field $\phi \in C^1(\mathcal{Q}_T)$ and a sufficiently regular flow map $\mathcal{L}_t$, define the Lagrangian pullback $\widehat{\phi}(t, \boldsymbol{\xi}) := \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$. Differentiating with respect to time along a trajectory and using $\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$ gives

$$\frac{d}{dt}\phi(t, \mathcal{L}_t(\boldsymbol{\xi})) = \partial_t \phi(t, \boldsymbol{x}) + \nabla\phi(t, \boldsymbol{x}) \cdot \boldsymbol{u}(t, \boldsymbol{x}), \qquad \boldsymbol{x} = \mathcal{L}_t(\boldsymbol{\xi}).$$

The identity

$$\operatorname{div}(\phi \boldsymbol{u}) = \nabla\phi \cdot \boldsymbol{u} + \phi \operatorname{div}\boldsymbol{u}$$

also gives

$$D_t\phi = \partial_t\phi + \operatorname{div}(\phi\boldsymbol{u}) - \phi\operatorname{div}\boldsymbol{u}.$$

## E.2 Jacobian Evolution and Reynolds Transport

Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ and $\widehat{J}_t = \det \widehat{\boldsymbol{F}}_t$. Here $\mathcal{L}_t$ is assumed to be an orientation-preserving $C^1$ change of variables on the material volume being considered, with the additional time regularity needed to differentiate $\widehat{\boldsymbol{F}}_t$ and $\widehat{J}_t$. The standard Jacobi identity gives

$$\partial_t \widehat{J}_t = \widehat{J}_t \operatorname{tr}\Big(\partial_t \widehat{\boldsymbol{F}}_t \, \widehat{\boldsymbol{F}}_t^{-1}\Big) = \widehat{J}_t \operatorname{div}\boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi})),$$

which is the Lagrangian form of $D_t J_t = J_t \operatorname{div}\boldsymbol{u}$ [12, Part I, Ch. 1, Sec. 1.3].

For $V(t) = \mathcal{L}_t(V(0))$, change variables in $I(t) = \int_{V(t)} \phi(t, \boldsymbol{x})\, d\boldsymbol{x}$:

$$I(t) = \int_{V(0)} \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))\widehat{J}_t(\boldsymbol{\xi})\, d\boldsymbol{\xi}.$$

Differentiating under the integral sign and substituting the Jacobian-evolution identity gives

$$I'(t) = \int_{V(t)} (D_t\phi + \phi\mathrm{div}\boldsymbol{u})\ d\boldsymbol{x} = \int_{V(t)} (\partial_t\phi + \mathrm{div}(\phi\boldsymbol{u}))\ d\boldsymbol{x}.$$

If $V(t)$ has Lipschitz boundary and the trace of $\phi\boldsymbol{u}$ is integrable on $\partial V(t)$, the divergence theorem gives the boundary form

$$I'(t) = \int_{V(t)} \partial_t\phi\ d\boldsymbol{x} + \int_{\partial V(t)} \phi\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dS.$$

## E.3 Linear Momentum Balance Details

Starting from the material-volume balance

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS + \int_{V(t)} \rho\boldsymbol{f_b}\, dV,$$

Reynolds transport gives

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{V(t)} (D_t(\rho\boldsymbol{u}) + \rho\boldsymbol{u}\,\mathrm{div}\boldsymbol{u})\, dV.$$

Using mass conservation, $D_t\rho + \rho\mathrm{div}\boldsymbol{u} = 0$, this reduces to

$$\int_{V(t)} \rho D_t\boldsymbol{u}\, dV = \int_{V(t)} \rho\,(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u})\, dV.$$

The surface traction term satisfies

$$\int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS = \int_{V(t)} \mathbf{div}(\boldsymbol{T})\, dV.$$

This uses the tensor row-divergence convention under the regularity needed for the stress trace $\boldsymbol{T}\hat{\boldsymbol{n}}$ and the volume integral of $\mathbf{div}\boldsymbol{T}$. Since $V(t)$ is arbitrary within the localization class, the differential form is

$$\rho\,(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}.$$

The conservative Eulerian form

$$\partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}$$

is equivalent whenever the continuity equation holds, because

$$\partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) + \boldsymbol{u}\left(\partial_t\rho + \mathrm{div}(\rho\boldsymbol{u})\right).$$

## E.4 Integral and variational bridge forms

For a fixed control volume $K$, the finite-volume bridge is the integral balance

$$\frac{d}{dt}\int_K U\,d\boldsymbol{x} + \int_{\partial K} F(U)\cdot\hat{\boldsymbol{n}}\,dS = \int_K S(U)\,d\boldsymbol{x},$$

with the numerical flux supplying the discrete approximation to the boundary integral. This is distinct from a mixed finite-element incompressible weak form. On a fixed fluid domain $\Omega_B$ with boundary parts $\Gamma_D$ and $\Gamma_N$, the Newtonian schematic weak form is: find $(\boldsymbol{u}, p)$ satisfying the essential velocity data such that

$$\int_{\Omega_B}\rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right)\cdot\boldsymbol{v}\,d\boldsymbol{x} + \int_{\Omega_B} 2\eta\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{v})\,d\boldsymbol{x} - \int_{\Omega_B} p\,\mathrm{div}\boldsymbol{v}\,d\boldsymbol{x} = \int_{\Omega_B}\rho\boldsymbol{f_b}\cdot\boldsymbol{v}\,d\boldsymbol{x} + \int_{\Gamma_N}\boldsymbol{t}_N\cdot\boldsymbol{v}\,dS,$$

$$\int_{\Omega_B} r\,\mathrm{div}\boldsymbol{u}\,d\boldsymbol{x} = 0,$$

for admissible velocity tests $\boldsymbol{v}$ and pressure tests $r$. The pressure space must specify a gauge or quotient convention when the boundary conditions leave pressure determined only up to a constant. In an ALE or moving-domain FSI variant, the convective derivative is written with relative velocity $\boldsymbol{u} - \boldsymbol{w}$, and the fluid–structure interface contributes the kinematic and traction conditions

$$\boldsymbol{u}_f = \partial_t\boldsymbol{d}_s, \qquad \boldsymbol{T}_f\hat{\boldsymbol{n}}_f + \boldsymbol{T}_s\hat{\boldsymbol{n}}_s = \mathbf{0}.$$

These bridge forms identify the mathematical contracts later numerical methods may discretize; they do not assert that every retained computational workflow implements the full moving-domain FSI form.

# F Navier–Stokes Coordinate and Energy Details

This appendix collects mathematical orientation material for the 3D Navier–Stokes model. It is background for the continuum hierarchy; it is not evidence for the 1D study specification or any downstream validation claim. All formulas are read under the standing regularity and differential-operator conventions in Appendix D. In particular, $\mathbf{div}$ is row-divergence for tensor fields, $\Delta \boldsymbol{u}$ is the componentwise vector Laplacian, tensor norms are Frobenius norms when written, and $\eta$ is dynamic viscosity while $\nu = \eta/\rho$ is kinematic viscosity as in Remark C.5.

## F.1 Viscous-Term Simplification and Energy Context

For a Newtonian fluid with constant dynamic viscosity $\eta$ and sufficiently smooth velocity field,

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \mathbf{div}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top) = \eta \Delta \boldsymbol{u} + \eta \nabla(\mathrm{div} \boldsymbol{u}).$$

For incompressible flow, $\mathrm{div} \boldsymbol{u} = 0$, so

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \Delta \boldsymbol{u}, \qquad \rho^{-1} \mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \nu \Delta \boldsymbol{u}, \qquad \nu = \eta/\rho.$$

This term is the momentum-diffusion contribution; it does not disappear in the incompressible Newtonian limit.

Classical local well-posedness results for Navier–Stokes depend on the domain, boundary conditions, forcing, initial-data space, and solution class. Under sufficiently smooth compatible data one obtains a unique strong solution on a maximal time interval. If the maximal interval is finite, continuation fails through blow-up of the norm controlled by the chosen well-posedness theory.

## F.2 Cylindrical Coordinate Form

For $(r, \theta, z)$ with orthonormal basis $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$,

$$x = r \cos \theta, \qquad y = r \sin \theta, \qquad \boldsymbol{u} = u_r \boldsymbol{e}_r + u_\theta \boldsymbol{e}_\theta + u_z \boldsymbol{e}_z.$$

The component formulas are local cylindrical-coordinate formulas on patches where $r > 0$. Axis regularity, wall traces, and boundary conditions require separate hypotheses and are not supplied by the coordinate notation alone. The differential operators used in cylindrical divided-by-density Navier–Stokes forms are

$$\begin{aligned}
(\boldsymbol{u}\cdot\nabla) &= u_r\partial_r + \frac{u_\theta}{r}\partial_\theta + u_z\partial_z,\\
\nabla p &= \partial_r p\,\boldsymbol{e}_r + \frac{1}{r}\partial_\theta p\,\boldsymbol{e}_\theta + \partial_z p\,\boldsymbol{e}_z,\\
\mathrm{div}\boldsymbol{u} &= \frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z.
\end{aligned}$$

The rate-of-deformation components in the cylindrical orthonormal basis are

$$\begin{aligned}
D_{rr} &= \partial_r u_r, & D_{\theta\theta} &= \frac{1}{r}\partial_\theta u_\theta + \frac{u_r}{r}, & D_{zz} &= \partial_z u_z,\\
D_{r\theta} &= \frac{1}{2}\left(\partial_r u_\theta - \frac{u_\theta}{r} + \frac{1}{r}\partial_\theta u_r\right), & D_{rz} &= \frac{1}{2}(\partial_r u_z + \partial_z u_r), & D_{\theta z} &= \frac{1}{2}\left(\frac{1}{r}\partial_\theta u_z + \partial_z u_\theta\right).
\end{aligned}$$

Use the scalar shear-rate convention $\dot\gamma = \sqrt{2\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{u})}$. Then

$$\tau_{ij} = 2\eta_{\text{eff}}(\dot\gamma)D_{ij}, \qquad i,j\in\{r,\theta,z\}.$$

When a source writes the law as a function of $D:D$ or $|D|_{\mathrm{F}}^2$, this report translates it through this convention, consistent with Remark C.5. Writing $\boldsymbol{f} = f_r\boldsymbol{e}_r + f_\theta\boldsymbol{e}_\theta + f_z\boldsymbol{e}_z$, the momentum components are

$$\begin{aligned}
\text{(radial)}\quad & \partial_t u_r + u_r\partial_r u_r + \frac{u_\theta}{r}\partial_\theta u_r + u_z\partial_z u_r - \frac{u_\theta^2}{r}\\
&\quad = -\frac{1}{\rho}\partial_r p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rr}) + \frac{1}{r}\partial_\theta\tau_{r\theta} + \partial_z\tau_{rz} - \frac{1}{r}\tau_{\theta\theta}\right] + f_r,\\
\text{(azimuthal)}\quad & \partial_t u_\theta + u_r\partial_r u_\theta + \frac{u_\theta}{r}\partial_\theta u_\theta + u_z\partial_z u_\theta + \frac{u_r u_\theta}{r}\\
&\quad = -\frac{1}{\rho r}\partial_\theta p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{r\theta}) + \frac{1}{r}\partial_\theta\tau_{\theta\theta} + \partial_z\tau_{\theta z} + \frac{1}{r}\tau_{r\theta}\right] + f_\theta,\\
\text{(axial)}\quad & \partial_t u_z + u_r\partial_r u_z + \frac{u_\theta}{r}\partial_\theta u_z + u_z\partial_z u_z\\
&\quad = -\frac{1}{\rho}\partial_z p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rz}) + \frac{1}{r}\partial_\theta\tau_{\theta z} + \partial_z\tau_{zz}\right] + f_z.
\end{aligned}$$

The incompressibility condition is

$$\frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z = 0.$$

# G Numerical Methods Details

This appendix gives the numerical-method details needed to interpret the implemented solver contract, the verification evidence, and the diagnostic 1D–3D velocity comparison. Secondary solver surfaces are included only where they support retained secondary numerical diagnostics; the principal comparison method remains MUSCL finite volume with minmod limiting, Rusanov flux, source splitting, and native SSPRK3.

**Convention G.1** (Computed realization record)**.** A computed realization is interpreted only through the model, closure, discretization, grid, timestep policy, boundary-state rule, output operator, and diagnostic metadata recorded for that realization. Symbols such as $D_t$, $D_z$, numerical fluxes, residual arrays, and sampled norms denote recorded discrete objects unless a section explicitly switches back to a continuum identity.

## G.1 Case-study rheology closure details

The reduced model evaluates a declared rheology closure family $\mathcal{R}$ through an effective dynamic-viscosity law $\eta_{\mathrm{eff}}^{\mathcal{R}}(\dot\gamma)$ and then uses $\nu_{\mathrm{eff}}^{\mathcal{R}} = \eta_{\mathrm{eff}}^{\mathcal{R}}/\rho$ in the source terms. For computed non-Newtonian closure-family cases, singular or degenerate zero-shear behavior is regularized by

$$\dot\gamma_\epsilon \;:=\; \max\{|\dot\gamma|, \epsilon_\gamma\}, \qquad \epsilon_\gamma > 0,$$

and optional declared dynamic-viscosity bounds are applied through

$$\Pi_{[\eta_{\min},\eta_{\max}]}(\xi) \;:=\; \min\{\eta_{\max}, \max\{\eta_{\min}, \xi\}\}, \qquad 0 \le \eta_{\min} \le \eta_{\max} \le \infty.$$

Without declared bounds, this projection is the identity. The implemented catalog laws are

$$\begin{aligned}
\eta_{\mathrm{eff}}^{\mathrm{newt}} &\equiv \eta, \\
\eta_{\mathrm{eff}}^{\mathrm{carreau}}(\dot\gamma) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\mathrm{C}}\dot\gamma_\epsilon)^2\right)^{(n_{\mathrm{C}}-1)/2}, \\
\eta_{\mathrm{eff}}^{\mathrm{CY}}(\dot\gamma) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\mathrm{CY}}\dot\gamma_\epsilon)^{a_{\mathrm{Y}}}\right)^{(n_{\mathrm{CY}}-1)/a_{\mathrm{Y}}}, \\
\eta_{\mathrm{eff}}^{\mathrm{casson}}(\dot\gamma) &= \left(\sqrt{\tau_y/\dot\gamma_\epsilon} + \sqrt{\eta_p}\right)^2, \\
\eta_{\mathrm{eff}}^{\mathrm{power}}(\dot\gamma) &= K_{\mathrm{pl}}\dot\gamma_\epsilon^{\,n_{\mathrm{pl}}-1}.
\end{aligned}$$

The Carreau and Carreau–Yasuda parameters satisfy $\eta_0 > 0$, $\eta_\infty \ge 0$, $\eta_0 \ge \eta_\infty$, $\lambda_{\mathrm{C}}, \lambda_{\mathrm{CY}} \ge 0$, $n_{\mathrm{C}}, n_{\mathrm{CY}} > 0$, and $a_{\mathrm{Y}} > 0$; the Casson parameters satisfy $\tau_y \ge 0$ and $\eta_p > 0$; and the power-law parameters satisfy $K_{\mathrm{pl}} > 0$ and $n_{\mathrm{pl}} > 0$. The dynamic-viscosity parameters have viscosity units, $\tau_y$ has stress units, and $K_{\mathrm{pl}}$ has units $\mathrm{Pa}\cdot\mathrm{s}^{n_{\mathrm{pl}}}$ in SI.

## G.2 Relationship to the extended 1D model of Canic et al.

Section 4.1 of Canic et al. is the source comparison target for this case-study family, but the retained report implementation is a documented specialization rather than a literal reproduction of every paper parameter, wall normalization, and resolved-FSI setting. The source model is an $(A, Q)$ area–flow system with a variable-radius correction $\alpha_c$, a viscous correction $p_2$, a pressure–area wall law, and a velocity-profile closure. The selected report realization uses the solver coordinates $(a, q)$ with $A_{\mathrm{phys}} = \pi a$ and $Q_{\mathrm{phys}} = \pi q$, the parabolic-profile constants $\alpha_{\mathcal{P}} = 4/3$ and $g_{\mathcal{P}} = 4$, the $R_{\max}^{-2}$ wall-law normalization, and a locally frozen effective-viscosity treatment in the $p_2$ derivative.

Table 23: Source-to-implementation map for the selected 1D model.

| Feature | Canic et al. source form | Implemented report form |
|---|---|---|
| Pressure denominator | Elastic pressure written with a local $R_0(z)^{-2}$ denominator. | Evolution wall potential and source use the constant $R_{\max}^{-2}$ normalization specified in Appendix G.8. |
| Profile parameters | Source simulations state $\gamma = 9$ and $\alpha = 1.1$. | Principal runs use the parabolic profile $\alpha_{\mathcal{P}} = 4/3$, $g_{\mathcal{P}} = 4$; power-profile runs are sensitivity/diagnostic records. |
| Variable-radius momentum correction | The extended 1D model includes the correction $\alpha_c$ tied to reference-radius variation. | The extended member uses the retained $\alpha_c$ switch in the finite-volume operator; the classical baseline sets that switch to zero. |
| Friction coefficient | Profile-dependent wall/friction closure in the reduced model. | Implemented through the declared velocity-profile shear factor and rheology closure family. |
| $p_2$ differentiation | Variable-radius pressure correction differentiated in the model. | Locally frozen-viscosity treatment for the $p_2$ derivative. |
| External pressure | External pressure term may be included in the formulation. | $p_{\mathrm{ext}} \equiv 0$ in the reported runs. |
| Coordinates | Physical area and flow notation in the source paper. | Solver state is $(a, q)$ with $A_{\mathrm{phys}} = \pi a$, $Q_{\mathrm{phys}} = \pi q$. |
| Boundary rule | Boundary conditions are model/problem dependent. | Fixed-area characteristic approximation applied with persisted subcritical-regime diagnostics. |
| Radial velocity reconstruction | Canic-style formulas may reconstruct a radial velocity component from the reduced fields. | The current axial-profile helper reconstructs axial velocity distributions only; it is not a Canic-style $u_r$ reconstruction. |
| Discrete well balancing | Continuous rest equilibrium is a model-level property. | Rusanov/MUSCL is not exactly well balanced for spatially varying $R_0$; Section 7.2 reports zero-forcing rest-state drift. |

## G.3 Secondary numerical diagnostics

The package benchmark suite reports secondary numerical diagnostics for the Julia solver: closure-catalog health, sampled self-convergence, native/integrator discrepancy, stationary-Stokes projection rows, rheology/profile sensitivity, OpenBF-style boundary handling, and resolved-velocity output diagnostics. These diagnostics keep documented solver surfaces runnable and generated assets schema-consistent. They do not supersede the MMS or geometry-rest evidence in Section 7.2, and they are not the principal two-case comparison method. When a secondary diagnostic lacks a declared reference solution, hardware-normalized runtime target, or admitted threshold, it is interpreted only as numerical-diagnostic context.

Table 24: Package benchmark stage summary.

| Stage | Rows | OK | Skipped/error |
|---|---|---|---|
| case results | 18 | 18 | 0 |
| refinement | 128 | 128 | 0 |
| time-integrator comparison | 18 | 18 | 0 |
| stokes ic | 12 | 12 | 0 |
| rheology profile | 60 | 60 | 0 |
| boundary openbf | 3 | 3 | 0 |
| resolved3d | 4 | 4 | 0 |

Table 25: Implemented discretization surfaces retained as secondary appendix context. The principal 23% and 40% stenosis comparison uses the MUSCL finite-volume row with native SSPRK3; other rows are diagnostic, sensitivity, or integrator-context surfaces.

| **Surface** | **Implemented option** | **Report role** |
|---|---|---|
| Spatial | First-order finite-volume Rusanov | Low-order cell-average baseline and DG $p = 0$ bridge. |
| Spatial | MUSCL finite volume, minmod limiter, Rusanov flux | Principal 23% and 40% stenosis comparison method and main verification target. |
| Spatial | Characteristic WENO3 and Richtmyer/Lax–Wendroff finite volume | Native sensitivity and diagnostic paths, not principal comparison methods. |
| Spatial | Modal Legendre DG degrees $p = 0, \ldots, 4$ | Secondary native-DG surface; selected benchmark rows may use only subsets such as $p = 0, 1, 2$. |
| Time | Forward Euler, SSPRK2, SSPRK3, SSPRK54, and selected integrator wrappers | Native SSPRK3 is selected for the principal comparison; other integrators are diagnostic context. |

**Nonselected DG p/h demonstration.** The manufactured-solution p/h demonstration is a secondary smooth-solution implementation check for the native modal-DG surface. The implementation supports modal Legendre DG degrees $p = 0, \ldots, 4$; selected closure-catalog or package-benchmark rows may exercise only subsets such as $p = 0, 1, 2$, while the p-refinement verification workflow runs through $p = 4$. The fixed-grid p-refinement rows plateau after $p = 1$ in the generated record, so they are not used here as evidence for a p-convergence claim.

Table 26: Manufactured-solution p- and h-refinement demonstration. The h-refinement rows report observed order from adjacent grid spacings; the p-refinement rows report error reduction relative to the previous polynomial degree.

| Sweep | $p$ | $N$ | DOFs | $\|e_a\|_2$ | h-order | p-reduction | $\|e_q\|_2$ | h-order | p-reduction |
|---|---|---|---|---|---|---|---|---|---|
| h-refinement | 2 | 20 | 120 | 9.797e-06 | 1.457 | – | 0.009629 | 1.052 | – |
| h-refinement | 2 | 40 | 240 | 3.569e-06 | 1.488 | – | 0.004643 | 1.209 | – |
| h-refinement | 2 | 80 | 480 | 1.272e-06 | 1.51 | – | 0.002009 | 1.424 | – |
| h-refinement | 2 | 160 | 960 | 4.467e-07 | – | – | 7.489e-04 | – | – |
| p-refinement | 0 | 40 | 80 | 1.078e-05 | – | – | 0.002585 | – | – |
| p-refinement | 1 | 40 | 160 | 3.569e-06 | – | 3.022 | 0.004645 | – | 0.5565 |
| p-refinement | 2 | 40 | 240 | 3.569e-06 | – | 1 | 0.004643 | – | 1 |
| p-refinement | 3 | 40 | 320 | 3.569e-06 | – | 0.9999 | 0.004643 | – | 1 |
| p-refinement | 4 | 40 | 400 | 3.569e-06 | – | 1 | 0.004642 | – | 1 |

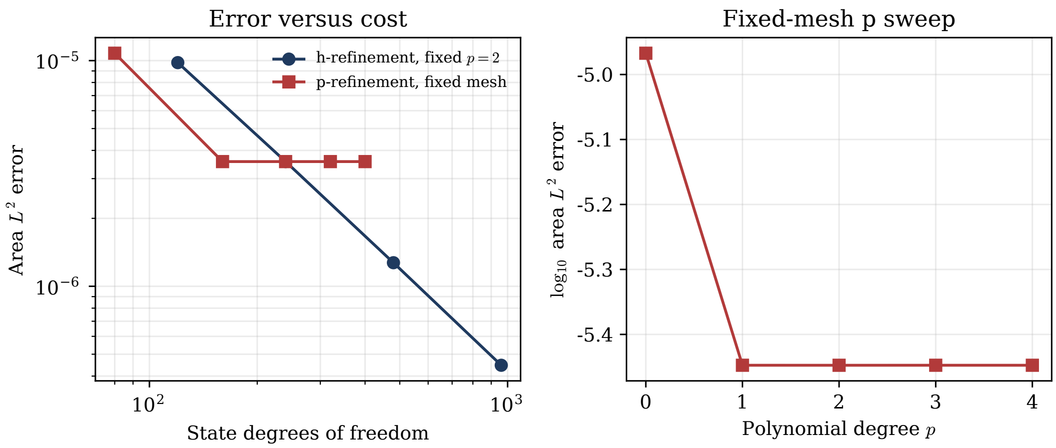


Figure 10: Manufactured-solution p- and h-refinement demonstration for the secondary modal-DG surface. This is smooth-case implementation evidence; the fixed-grid p-sweep plateau is not interpreted as a p-convergence result.

**Self-convergence, integrator comparison, and closure sensitivity.** The remaining package-benchmark figures are retained as compact secondary numerical diagnostics. They can expose grid sensitivity, integration-policy discrepancies, or closure-family wiring problems, but they do not establish a convergence theorem for the nonlinear operator. The benchmark pressure traces are legacy local-denominator diagnostics $p_{\mathrm{loc,g}}^{\mathrm{diag}}$ from the implementation helpers, not selected evolution wall-law pressure observations.

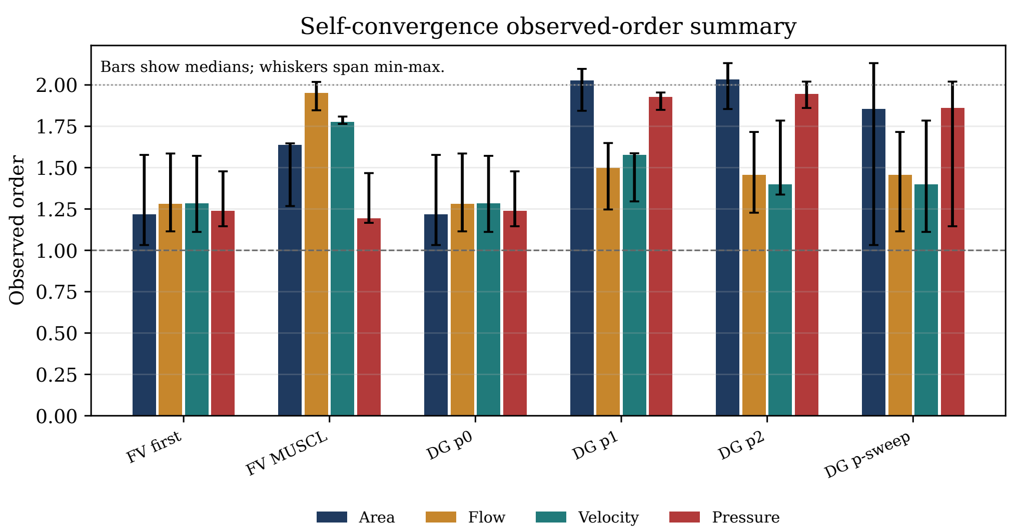


Figure 11: Secondary self-convergence summary reported by the benchmark renderer. Pressure bars, where present, are legacy local-denominator diagnostics $p_{\mathrm{loc,g}}^{\mathrm{diag}}$ rather than selected evolution wall-law pressure observations.

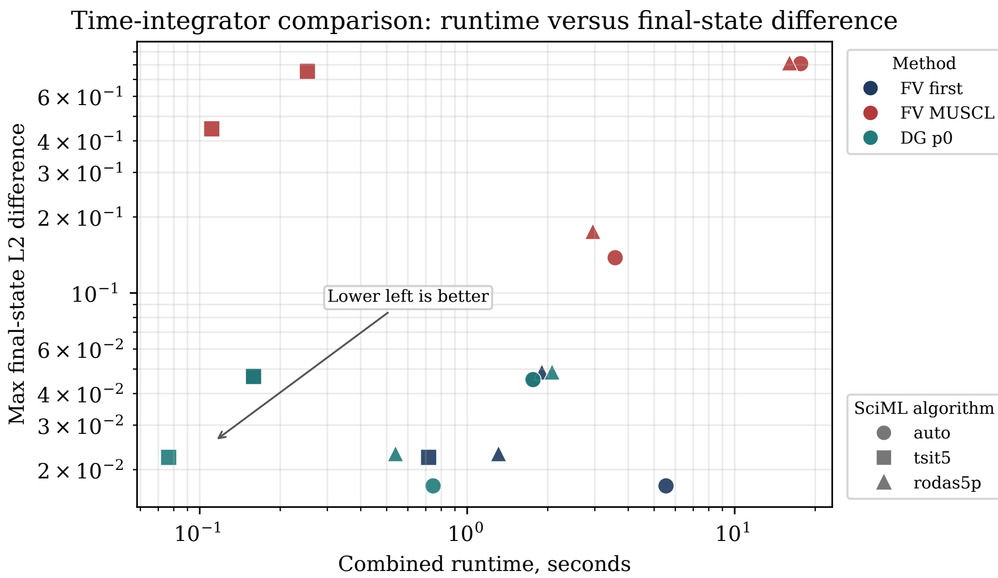


Figure 12: Secondary native/integrator discrepancy for compatible methods. This is an integrator-comparison check without an admitted reference solution, hardware-normalized runtime specification, or pass/fail threshold.

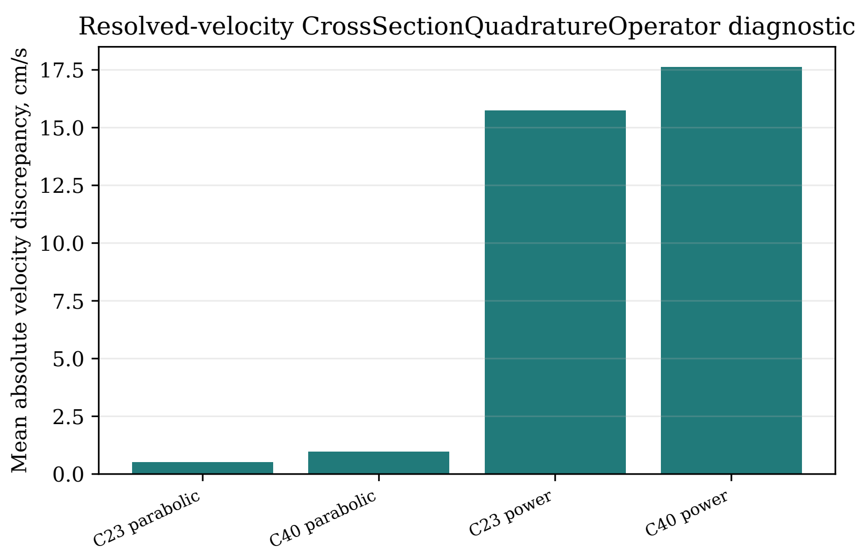


Figure 13: Secondary resolved-velocity output-operator diagnostic for available local cases.

## G.4 Additional resolved-comparator diagnostics

Section 7.3 reports the main final-time velocity comparison. The optional rows below retain the production-grid sensitivity summary and the largest section-wise discrepancies so that the main text can state the comparison result without turning the chapter into a diagnostic listing. The retained cross-dimensional observation is section-mean axial velocity on declared plane cuts. Radial distributions of axial velocity remain secondary output until regenerated same-cut radial-profile evidence is reviewed and accepted for the report.

Table 27: Optional production-grid sensitivity for the reference-coordinate resolved-velocity observation map. Flow discrepancies are in $\mathrm{cm}^3/\mathrm{s}$ and velocity discrepancies are in cm/s; the rows test numerical sensitivity of the comparison output, not patient-specific interpretation.

| Case | $N$ | $D_{Q,1}$ | $D_{Q,2}$ | $D_{u,1}$ | $D_{u,2}$ | rel. $D_{u,2}$ | max $\lvert d_u \rvert$ | adj. $D_{u,2}$ |
|---|---|---|---|---|---|---|---|---|
| 23% stenosis | 200 | 0.05965 | 0.07358 | 0.6182 | 0.8996 | 0.03757 | 4.533 | – |
| 23% stenosis | 400 | 0.04821 | 0.05085 | 0.5048 | 0.6636 | 0.02771 | 2.349 | 0.678 |
| 23% stenosis | 800 | 0.04874 | 0.05322 | 0.5049 | 0.7063 | 0.0295 | 2.94 | 0.1785 |
| 40% stenosis | 200 | 0.08395 | 0.1132 | 1.082 | 1.894 | 0.06975 | 7.93 | – |
| 40% stenosis | 400 | 0.07087 | 0.08927 | 0.966 | 1.868 | 0.06881 | 9.923 | 1.328 |
| 40% stenosis | 800 | 0.07108 | 0.09539 | 0.9843 | 2.019 | 0.07435 | 11.57 | 0.3499 |

Table 28: Five largest section-wise velocity discrepancies in the reference-coordinate comparison. Velocities and velocity discrepancies are in cm/s; physical flows and flow discrepancies are in $\mathrm{cm}^3/\mathrm{s}$. The ranking identifies where the declared observation map differs most strongly between tiers.

| **Rank** | **Case** | $z$ **(cm)** | $\bar{u}_{1D}$ | $\bar{u}_{3D}$ | $\lvert d_u \rvert$ | $Q_{1D}$ | $Q_{3D}$ | $\lvert d_Q \rvert$ | **Tri.** |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 40% stenosis | 2.261 | 52.58 | 42.66 | 9.92 | 2.201 | 1.862 | 0.339 | 759 |
| 2 | 40% stenosis | 2.291 | 56.94 | 48.20 | 8.74 | 2.230 | 1.903 | 0.327 | 749 |
| 3 | 40% stenosis | 2.352 | 61.43 | 53.22 | 8.21 | 2.270 | 1.980 | 0.290 | 743 |
| 4 | 40% stenosis | 2.322 | 59.81 | 52.09 | 7.72 | 2.253 | 1.970 | 0.283 | 771 |
| 5 | 40% stenosis | 2.231 | 46.95 | 39.94 | 7.01 | 2.174 | 1.918 | 0.256 | 833 |

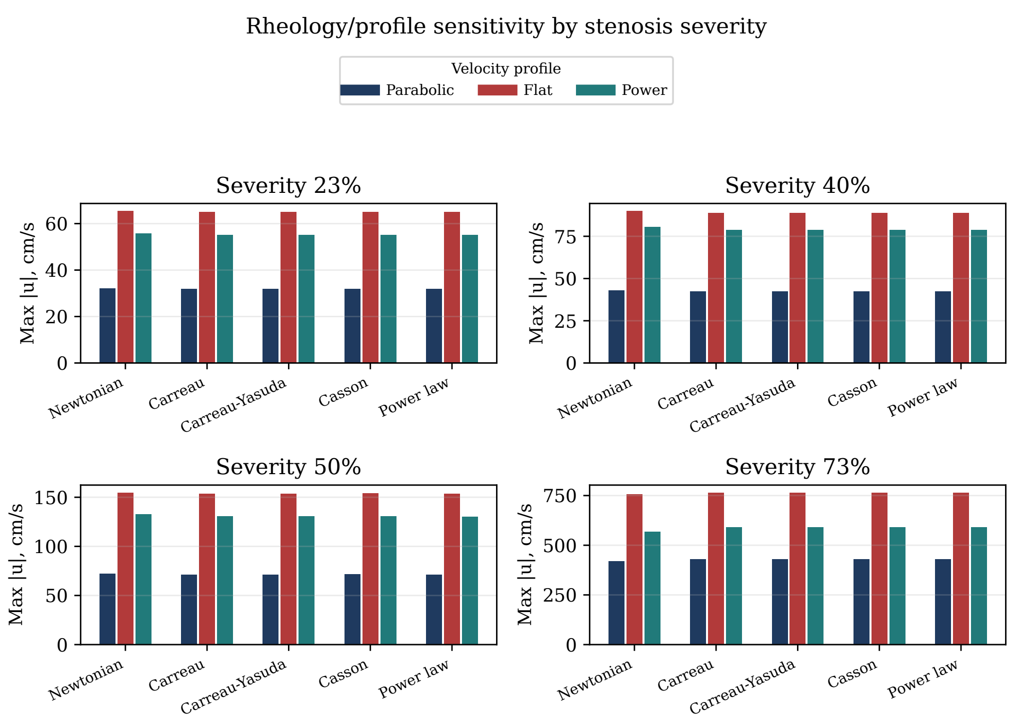


Figure 14: Secondary rheology/profile closure-family wiring stress test for declared combinations, not a calibrated physiological sensitivity study.

## G.5 Fixed-wall stationary-Stokes refinement check

The local three-dimensional refinement study is limited to a fixed-wall stationary-Stokes problem on the smooth idealized stenosis geometry. It is an initializer and projection diagnostic, not a replacement for the 1D initial-condition contract and not a resolved FSI record. The rows document successful projection and coarse-to-fine discrepancies relative to the largest completed mesh; no asymptotic convergence claim is made from this short refinement set.

Table 29: Stationary-Stokes projection and mesh-refinement diagnostics. The finest-mesh relative discrepancy is zero on the finest row by construction; it is included to show the refinement reference used.

| Case | Nodes | Cells | $\langle Q\rangle_{\mathrm{FE}}$ | rel. diff.$(u_{\mathrm{FE}}, u_{\mathrm{proj}})$ | rel. diff.$(u_{\mathrm{FE}}, u_{\mathrm{finest}})$ | max $\lvert\tau_w\rvert$ |
|---|---|---|---|---|---|---|
| stokes-s23-8x2x8 | 153 | 576 | 0.1743 | 0.2012 | 0.3643 | 5.644 |
| stokes-s23-16x4x16 | 1105 | 5376 | 0.1743 | 0.1928 | 0.000e+00 | 8.446 |
| stokes-s40-8x2x8 | 153 | 576 | 0.1286 | 0.378 | 0.5212 | 6.034 |
| stokes-s40-16x4x16 | 1105 | 5376 | 0.1286 | 0.2201 | 0.000e+00 | 11.06 |

## G.6 Quasi-static membrane-FSI comparator workflow

The Julia package now contains a separate membrane-FSI comparator workflow for Canic-style resolved-wall comparison experiments. It is not the fixed-wall stationary-Stokes refinement record above. The steady mode solves a fixed-point problem on the current lumen geometry: a Stokes solve supplies the wall pressure load, the radial membrane equation updates the wall displacement, and the tetrahedral mesh is regenerated on the updated radius profile until the wall update reaches the declared tolerance.

For the quasi-static mode the wall equation is the inertia-free specialization of the Canic membrane relation [16, Eq. (35)],

$$C_0\eta_r(z) = f_r(z), \qquad \eta_r(0) = \eta_r(L) = 0, \qquad C_0 = \frac{Eh}{(1-\sigma^2)(R_0^*)^2}.$$

Here $\eta_r$ is the radial displacement, $f_r$ is the physical radial fluid load sampled on the wall, and $R_0^*$ is the constant reference-radius convention recorded in the workflow manifest. Outlet-gauge normalization is a diagnostic and export convention rather than the forcing convention for the partitioned membrane update. The implementation also accepts a named positive, sufficiently smooth reference-radius callback, so the membrane-FSI solve is not tied to the default Canic stenosis profile. The retained report comparison promotes the quasi-static summary table, wall-profile curves, and fixed-point history into Section 7.3.5; the scratch output directory and manifest remain provenance records for regenerating the report-owned assets.

The native resolved-FSI exact-boundary path is separate package-development diagnostic work, not a promoted report-evidence surface. Focused package tests now check the density-consistent transient and convection terms, symmetric-gradient Cauchy viscous form, boundary-aware pressure-space policy, physical wall-pressure forcing convention, exact Canic geometry, and same-cut radial-profile closure classifications. That retires the Lane 11 P0 mathematical-contract blocker only at focused package-test scope. It does not establish the severity-23 preproduction execution gate, production parity, imported parity, moving-wall/ALE fidelity, durable restart or resume support, regenerated report tables, promoted radial-profile evidence, or manuscript-grade reproduction of Section 4.1 of Canic et al. Current Gridap-based resolved-flow language is therefore focused contract evidence rather than a backend-replacement or production-claim basis.

The corresponding report-generation entry point is:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics fsi validate \
  --wall-mode quasi-static \
  --severities 23,40 \
  --meshes 8x2x8,16x4x16 \
  --output-dir tmp/simulations/output/membrane_fsi_validation \
  --publish-report-assets \
  --report-assets-dir report/assets \
  --overwrite
```

The workflow writes a summary CSV, a TEX table fragment, per-case wall profiles, per-case coupling histories, and a JSON manifest under the selected output directory, with the retained report assets copied to `report/assets/data/membrane-fsi/` and `report/assets/tables/membrane-fsi/`. Dynamic wall mode is a reduced radial membrane time integrator coupled to repeated quasi-steady Stokes solves, not a transient moving-boundary fluid solve, and its outputs remain diagnostic output unless a later artifact-refresh lane explicitly promotes them.

## G.7 Elastic-potential balance-law form

The reduced state is $\mathbf{U} = (A, Q)^\top$. For a law in the $\mathcal{W}_{\mathrm{elastic1D}}$ balance-law contract with prescribed momentum-flux correction $\alpha$, the reference elastic-potential form is

$$\begin{aligned} \partial_t A + \partial_z Q &= 0, \\ \partial_t Q + \partial_z \left( \alpha \frac{Q^2}{A} + \Psi(A,z,t) \right) &= S_{\mathrm{geom}}(A,z,t) - C_f(z,A,Q)\frac{Q}{A}. \end{aligned}$$

The elastic potential satisfies

$$\partial_A \Psi(A,z,t) = \frac{A}{\rho} \partial_A p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}(A,z,t),$$

and the geometry source is

$$S_{\mathrm{geom}}(A,z,t) = \partial_z \Psi(A,z,t)\big|_A - \frac{A}{\rho}\, \partial_z p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}(A,z,t)\big|_A .$$

For the selected Canic–Koiter wall law, assuming $p_{\mathrm{ext}}$ is independent of $A$,

$$c^2(A,z) = \frac{A}{\rho}\partial_A p^{\mathrm{evol}}_{\mathcal{W},\mathrm{g}}(A,z,t) = \frac{\beta(z)\sqrt{A}}{2\rho A_0(z)}.$$

Under the selected solver-coordinate map $A = \pi a$ and $A_0 = \pi R_0^2$, the implemented law corresponds to $\beta(z) = \sqrt{\pi} K R_0(z)^2 / R_{\max}^2$. Therefore

$$c^2(a,z) = \frac{K}{2\rho R_{\max}^2}\sqrt{a}$$

for the evolution wall law used in the reported finite-volume runs. Freezing $z$ and the wall-closure data gives homogeneous elastic-subsystem speeds

$$\lambda_{\pm}(A,Q;z) = \alpha\frac{Q}{A} \pm \sqrt{c^2(A,z) + \alpha(\alpha-1)\left(\frac{Q}{A}\right)^2}.$$

The subcritical boundary case is $\lambda_- < 0 < \lambda_+$.

## G.8 Implemented semi-discrete solver operator

The implemented solver stores the radius-squared coordinate in the variable `A`; here write that coordinate as $a = R^2$. The physical circular area is $A_{\mathrm{phys}} = \pi a$, the physical flow is $Q_{\mathrm{phys}} = \pi q$, and the mean velocity is $q/a$.

For $N$ finite-volume cells, let

$$z_i = \left(i - \frac{1}{2}\right)\Delta z, \qquad \Delta z = \frac{L}{N}, \qquad u_h(t) = \left(a_1(t), \ldots, a_N(t), q_1(t), \ldots, q_N(t)\right)^{\top}.$$

The semi-discrete system exposed by `semidiscretize` and `rhs!` is

$$\dot{u}_h = \mathcal{L}_h(u_h, t; \theta),$$

where $\theta$ denotes the recorded physical, geometry, grid, and solver parameters. Componentwise,

$$\begin{aligned}
\dot{a}_i &= -\frac{\widehat{F}^a_{i+1/2} - \widehat{F}^a_{i-1/2}}{\Delta z}, \\
\dot{q}_i &= -\frac{\widehat{F}^q_{i+1/2} - \widehat{F}^q_{i-1/2}}{\Delta z} + \widehat{S}_i.
\end{aligned}$$

All divisions by $a$ use the positivity-regularized radius-squared coordinate. With $K = Eh/(1-\sigma^2)$, Canic conservative-form reference radius $R_0^* = R_{\max}$, velocity-profile momentum factor $\alpha_{\mathcal{P}}$, and $\alpha_c(R_0') =$

$-2(R_0')^2/35$, the model-family switch is

$$\iota_{\mathcal{M}} = \begin{cases} 1, & \text{extended 1D model member,} \\ 0, & \text{classical parabolic-profile baseline.} \end{cases}$$

The semi-discrete full-state flux is

$$\mathbf{F}(a,q,z) = \left(q,\, \left(\alpha_{\mathcal{P}} + \iota_{\mathcal{M}}\alpha_c(R_0'(z))\right)\frac{q^2}{a} + \frac{K}{3\rho R_{\max}^2}a^{3/2}\right)^{\top}.$$

At each interface,

$$\widehat{\mathbf{F}}_{i+1/2} = \frac{1}{2}\left[\mathbf{F}(\mathbf{U}^-_{i+1/2}, z_{i+1/2}) + \mathbf{F}(\mathbf{U}^+_{i+1/2}, z_{i+1/2})\right] - \frac{1}{2}s_{i+1/2}\left(\mathbf{U}^+_{i+1/2} - \mathbf{U}^-_{i+1/2}\right),$$

where $\mathbf{U} = (a,q)^{\top}$. MUSCL reconstruction limits the conservative solver variables componentwise:

$$\delta_i^a = \operatorname{minmod}(a_i - a_{i-1},\, a_{i+1} - a_i), \qquad \delta_i^q = \operatorname{minmod}(q_i - q_{i-1},\, q_{i+1} - q_i),$$

$$\mathbf{U}^-_{i+1/2} = (a_i + \tfrac{1}{2}\delta_i^a,\, q_i + \tfrac{1}{2}\delta_i^q)^{\top}, \qquad \mathbf{U}^+_{i+1/2} = (a_{i+1} - \tfrac{1}{2}\delta_{i+1}^a,\, q_{i+1} - \tfrac{1}{2}\delta_{i+1}^q)^{\top}.$$

Boundary-adjacent slopes are set to zero and paired with the ghost states in Appendix G.10. Positivity limiting is applied to reconstructed $a$ states before flux evaluation. In the stencil language of Section 5, a first-order two-point flux would give the familiar three-cell interior dependency for residual $i$, but the MUSCL slopes above expand the interior residual dependency to the additional neighbors needed to reconstruct the two adjacent interface states. Boundary ghost states replace part of that dependency near the inlet and outlet. The local numerical speed for the Rusanov interface flux is

$$s_{i+1/2} = \max\left\{s_h(\mathbf{U}^-_{i+1/2}, z_{i+1/2}), s_h(\mathbf{U}^+_{i+1/2}, z_{i+1/2})\right\},$$

with

$$\begin{aligned} s_h(a,q,z) &= \max\left\{\left|\alpha_{\mathrm{eff}}u - \chi\right|, \left|\alpha_{\mathrm{eff}}u + \chi\right|\right\}, \\ \alpha_{\mathrm{eff}} &= \alpha_{\mathcal{P}} + \iota_{\mathcal{M}}\alpha_c(R_0'(z)), \qquad u = \frac{q}{a}, \\ \chi^2 &= (\alpha_{\mathrm{eff}}u)^2 - \alpha_{\mathrm{eff}}u^2 + \frac{K}{2\rho R_{\max}^2}\sqrt{a}. \end{aligned}$$

This operator is not an exactly well-balanced discretization of the geometry-rest state. At $q_i = 0$ with spatially varying $a_i = R_0(z_i)^2$, the Rusanov mass-flux viscosity generally contributes a nonzero discrete residual unless an equilibrium reconstruction or path-conservative flux is added.

**Rest-state compatibility detail.** Proposition 7.8 is a continuous source-balance statement. At the geometry-rest state $a = R_0^2$ and $q = 0$, the mass equation gives $\partial_z q = 0$. The profile, friction, and pressure-correction terms that contain $q$, $\partial_z q$, or $q^2$ vanish:

$$S_{\text{fric}} = 0, \qquad S_\alpha = 0, \qquad S_{p_2} = 0.$$

The remaining momentum balance is the cancellation between the elastic flux gradient and the wall/geometry source. Since

$$\Psi_h(R_0^2) = \frac{K}{3\rho R_{\max}^2} R_0^3, \qquad \partial_z \Psi_h(R_0^2) = \frac{K}{\rho R_{\max}^2} R_0^2 R_0',$$

and

$$S_{\text{wall}}(R_0^2, z) = \frac{K}{\rho R_{\max}^2} R_0^2 R_0',$$

the continuous source-balanced equation gives $\partial_t q = 0$. The current discrete operator fails to inherit this cancellation because the interface Rusanov flux is applied to reconstructed conservative states rather than to an equilibrium variable that makes the flux/source pair cancel at the discrete interfaces. A well-balanced reconstruction would need to make the discrete elastic flux difference and the discrete wall source cancel for the target equilibrium; the present implementation records the resulting drift instead of suppressing it.

The source term uses the specified centered interior and one-sided boundary finite difference $D_z^h$ on cell averages. Let

$$\dot\gamma_{1D,i} = g_{\mathcal{P}} \frac{|q_i|}{a_i \sqrt{a_i}}, \qquad \nu_{\text{eff},i}^{\mathcal{R}} = \frac{\eta_{\text{eff}}^{\mathcal{R}}(\dot\gamma_{1D,i})}{\rho},$$

and $\alpha_c'(z) = -4R_0'(z)R_0''(z)/35$. The implemented source is

$$\begin{aligned}
\widehat{S}_i &= -2\nu_{\text{eff},i}^{\mathcal{R}} g_{\mathcal{P}} \frac{q_i}{a_i} + \frac{K}{\rho R_{\max}^2} a_i R_0'(z_i) - \iota_{\mathcal{M}} a_i \widehat{\partial_z p_2}_i + \iota_{\mathcal{M}} \frac{q_i^2}{a_i} \alpha_c'(z_i), \\
\widehat{\partial_z p_2}_i &= \nu_{\text{eff},i}^{\mathcal{R}} g_{\mathcal{P}} \left[ \frac{q_i R_0''(z_i)}{a_i R_0(z_i)} - \frac{q_i (R_0'(z_i))^2}{a_i R_0(z_i)^2} + \frac{(D_z^h q)_i R_0'(z_i)}{a_i R_0(z_i)} - \frac{q_i (D_z^h a)_i R_0'(z_i)}{a_i^2 R_0(z_i)} \right].
\end{aligned}$$

The $\widehat{\partial_z p_2}_i$ term is the density-scaled discrete derivative of the Canic variable-radius viscous pressure correction and is inactive when $\iota_{\mathcal{M}} = 0$.

Other implemented surfaces include lower-order finite volume, WENO3 finite volume, a Lax–Wendroff variant, modal Legendre DG with $p = 0, \ldots, 4$, and selected integrator wrappers. They remain numerical-diagnostic context unless a retained table or figure explicitly invokes them.

## G.9 Time-integration realization

The reported comparison advances the semi-discrete operator with native SSPRK3:

$$u_h^{n+1} = \Phi_{\Delta t_n}^{\mathrm{SSPRK3}} \left( u_h^n; \mathcal{L}_h \right).$$

Using $\Pi$ for the recorded positive-area projection on the $a$ components,

$$\begin{aligned}
u^{(1)} &= \Pi \left( u^n + \Delta t_n \mathcal{L}_h(u^n, t^n) \right), \\
u^{(2)} &= \Pi \left( \frac{3}{4} u^n + \frac{1}{4} \left[ u^{(1)} + \Delta t_n \mathcal{L}_h(u^{(1)}, t^n + \Delta t_n) \right] \right), \\
u^{n+1} &= \Pi \left( \frac{1}{3} u^n + \frac{2}{3} \left[ u^{(2)} + \Delta t_n \mathcal{L}_h(u^{(2)}, t^n + \tfrac{1}{2} \Delta t_n) \right] \right).
\end{aligned}$$

The native step size is

$$\Delta t_n = \min \left\{ \Delta t_{\max}, \frac{\mathrm{CFL}\, \Delta z}{\max_i s_h(a_i^n, q_i^n, z_i)}, T_{\mathrm{final}} - t^n \right\}.$$

## G.10 Boundary-state realization

The finite-volume boundary realization imposes one incoming boundary condition at each boundary in the subcritical regime using an $\alpha = 1$, fixed-area characteristic approximation. This is the implemented boundary-state rule for the variable-$\alpha_{\mathrm{eff}}$ solver, not an exact invariant derivation for the full variable-radius balance law. Let

$$c_0 = \left( \frac{K}{2\rho (R_0^*)^2} \right)^{1/2} = \left( \frac{K}{2\rho R_{\max}^2} \right)^{1/2}.$$

At the inlet, the incoming flow is $q_{\mathrm{in}} = R_0(0)^2 \bar{u}_{\mathrm{in}}$. The outgoing invariant from the first interior cell is

$$w_- = \frac{q_1}{a_1} - 4 c_0 a_1^{1/4},$$

and the ghost coordinate $a_{\mathrm{in}}$ solves

$$\frac{q_{\mathrm{in}}}{a_{\mathrm{in}}} - 4 c_0 a_{\mathrm{in}}^{1/4} = w_-.$$

At the outlet, the radius-squared coordinate is fixed to $a_{\mathrm{out}} = R_0(L)^2$. The outgoing invariant from the last interior cell is

$$w_+ = \frac{q_N}{a_N} + 4 c_0 a_N^{1/4},$$

and the ghost flow is

$$q_{\rm out} = a_{\rm out} w_+ - 4c_0 a_{\rm out}^{5/4}.$$

Boundary-quality plots and reflected-wave sidecars are diagnostics until explicit thresholds are admitted.

This fixed-area characteristic rule is intentionally narrower than common physiological outflow models. A Windkessel or impedance boundary would replace the fixed outlet area by a terminal pressure–flow relation with additional lumped parameters and time history. A prescribed pressure, traction, or reflection-coefficient boundary would impose a different incoming characteristic or terminal relation. Those alternatives are appropriate when downstream vascular compliance, resistance, or measured pressure data are part of the question; they are not implemented as principal comparison conditions in this report.

## G.11 Residual diagnostics

Stored residuals, CFL summaries, positivity records, conservation summaries, and boundary-quality sidecars are post-processing diagnostics on sampled fields. They are interpreted through the recorded discrete operators and grids, not as standalone continuum residual proofs.

## G.12 Full rest-state drift grid

Section 7.2 reports a compressed summary of the zero-forcing, zero-inlet geometry-rest drift test. The full generated grid and elapsed-time table is retained here so that the main-text interpretation can be checked against the underlying rows.

Table 30: Full zero-forcing, zero-inlet geometry-rest drift diagnostics. The volume-defect column is the signed solver-coordinate integral $\Delta\int a\,dz$; the balance residual is $\Delta\int a\,dz + \int(\widehat{F}^a_{\mathrm{out}} - \widehat{F}^a_{\mathrm{in}})\,dt$.

| Severity | $N$ | $t$ | requested $q_{\mathrm{in}}$ | applied $q_{\mathrm{in}}$ | $\widehat{F}^a_{\mathrm{in}}$ | $\widehat{F}^a_{\mathrm{out}}$ | $\max\lvert q_i\rvert$ | $z_{\max\lvert q\rvert}$ | $\Delta\int a\,dz$ | balance residual |
|---:|---:|---:|---:|---:|---:|---:|---:|---:|---:|---:|
| 23 | 100 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.03 | 0.000e+00 | 0.000e+00 |
| 23 | 100 | 0.001 | 0.000e+00 | 0.000e+00 | -4.560e-11 | 0.000e+00 | 0.5146 | 2.13 | 5.851e-14 | -5.976e-15 |
| 23 | 100 | 0.01 | 0.000e+00 | 0.000e+00 | 6.518e-07 | -0.0347 | 0.4579 | 2.13 | -5.088e-05 | 2.391e-10 |
| 23 | 100 | 0.1 | 0.000e+00 | 0.000e+00 | -6.403e-09 | 0.008553 | 0.4609 | 2.13 | -2.712e-05 | 1.720e-10 |
| 23 | 100 | 1.0 | 0.000e+00 | 0.000e+00 | 1.259e-10 | -1.346e-05 | 0.4647 | 2.13 | -2.531e-05 | 1.524e-10 |
| 23 | 200 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.015 | 0.000e+00 | 0.000e+00 |
| 23 | 200 | 0.001 | 0.000e+00 | 0.000e+00 | -1.432e-10 | 0.000e+00 | 0.1662 | 2.085 | 6.753e-14 | 2.102e-14 |
| 23 | 200 | 0.01 | 0.000e+00 | 0.000e+00 | 2.167e-07 | -0.009325 | 0.1467 | 2.085 | 1.587e-06 | -5.402e-11 |
| 23 | 200 | 0.1 | 0.000e+00 | 0.000e+00 | 5.150e-11 | 0.004369 | 0.1502 | 2.085 | 5.042e-07 | -1.587e-10 |
| 23 | 200 | 1.0 | 0.000e+00 | 0.000e+00 | -6.622e-11 | -2.304e-05 | 0.1473 | 2.085 | 7.243e-07 | -1.358e-10 |
| 23 | 400 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.0075 | 0.000e+00 | 0.000e+00 |
| 23 | 400 | 0.001 | 0.000e+00 | 0.000e+00 | 1.293e-10 | 0.000e+00 | 0.04387 | 2.078 | 5.970e-14 | 2.669e-14 |
| 23 | 400 | 0.01 | 0.000e+00 | 0.000e+00 | 1.087e-08 | -0.002511 | 0.0383 | 2.078 | 1.025e-06 | -5.474e-11 |
| 23 | 400 | 0.1 | 0.000e+00 | 0.000e+00 | 8.434e-11 | 0.001825 | 0.03992 | 2.078 | 4.646e-07 | -5.642e-11 |
| 23 | 400 | 1.0 | 0.000e+00 | 0.000e+00 | -3.518e-11 | -8.001e-06 | 0.03839 | 2.078 | 4.744e-07 | -5.654e-11 |
| 23 | 800 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.00375 | 0.000e+00 | 0.000e+00 |
| 23 | 800 | 0.001 | 0.000e+00 | 0.000e+00 | 1.468e-10 | 0.000e+00 | 0.01116 | 2.074 | 5.976e-14 | 2.669e-14 |
| 23 | 800 | 0.01 | 0.000e+00 | 0.000e+00 | 3.438e-10 | -6.460e-04 | 0.009717 | 2.074 | 3.900e-07 | -7.549e-13 |
| 23 | 800 | 0.1 | 0.000e+00 | 0.000e+00 | -3.879e-11 | 5.684e-04 | 0.01021 | 2.074 | 2.013e-07 | -5.507e-13 |
| 23 | 800 | 1.0 | 0.000e+00 | 0.000e+00 | -1.119e-10 | -3.876e-06 | 0.00973 | 2.074 | 1.841e-07 | -6.173e-13 |
| 40 | 100 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.03 | 0.000e+00 | 0.000e+00 |
| 40 | 100 | 0.001 | 0.000e+00 | 0.000e+00 | -1.464e-10 | 0.000e+00 | 0.8534 | 2.07 | 5.618e-14 | -1.243e-14 |
| 40 | 100 | 0.01 | 0.000e+00 | 0.000e+00 | -1.357e-10 | -0.1178 | 0.7525 | 2.07 | -1.940e-04 | -2.172e-11 |
| 40 | 100 | 0.1 | 0.000e+00 | 0.000e+00 | -5.977e-10 | 0.01485 | 0.7664 | 2.07 | -9.340e-05 | -3.742e-10 |
| 40 | 100 | 1.0 | 0.000e+00 | 0.000e+00 | 4.146e-11 | 2.450e-05 | 0.7684 | 2.07 | -8.853e-05 | -7.961e-11 |
| 40 | 200 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.015 | 0.000e+00 | 0.000e+00 |
| 40 | 200 | 0.001 | 0.000e+00 | 0.000e+00 | -1.432e-10 | 0.000e+00 | 0.2704 | 2.085 | 6.747e-14 | 2.097e-14 |
| 40 | 200 | 0.01 | 0.000e+00 | 0.000e+00 | 1.564e-08 | -0.03815 | 0.2433 | 2.085 | -1.184e-06 | 5.695e-12 |
| 40 | 200 | 0.1 | 0.000e+00 | 0.000e+00 | -7.420e-10 | 0.007515 | 0.2421 | 2.085 | -1.204e-05 | -4.058e-10 |
| 40 | 200 | 1.0 | 0.000e+00 | 0.000e+00 | 8.087e-11 | 2.623e-05 | 0.2435 | 2.085 | 1.043e-06 | -1.523e-10 |
| 40 | 400 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.0075 | 0.000e+00 | 0.000e+00 |
| 40 | 400 | 0.001 | 0.000e+00 | 0.000e+00 | 1.301e-10 | 0.000e+00 | 0.07053 | 2.078 | 5.967e-14 | 2.663e-14 |
| 40 | 400 | 0.01 | 0.000e+00 | 0.000e+00 | 1.054e-09 | -0.01045 | 0.06402 | 2.078 | 7.734e-06 | -1.454e-11 |
| 40 | 400 | 0.1 | 0.000e+00 | 0.000e+00 | -3.933e-11 | 0.001859 | 0.06172 | 2.078 | -2.559e-06 | -1.077e-10 |
| 40 | 400 | 1.0 | 0.000e+00 | 0.000e+00 | 1.411e-10 | 1.050e-05 | 0.06339 | 2.078 | 4.047e-06 | -5.310e-11 |
| 40 | 800 | 0.000e+00 | 0.000e+00 | 0.000e+00 | 1.106e-10 | 0.000e+00 | 0.000e+00 | 0.00375 | 0.000e+00 | 0.000e+00 |
| 40 | 800 | 0.001 | 0.000e+00 | 0.000e+00 | 1.470e-10 | 0.000e+00 | 0.01782 | 2.074 | 5.979e-14 | 2.671e-14 |
| 40 | 800 | 0.01 | 0.000e+00 | 0.000e+00 | -8.786e-11 | -0.002687 | 0.01625 | 2.074 | 2.455e-06 | -2.782e-12 |
| 40 | 800 | 0.1 | 0.000e+00 | 0.000e+00 | 1.410e-11 | 4.312e-04 | 0.01535 | 2.074 | -6.424e-07 | -1.825e-12 |
| 40 | 800 | 1.0 | 0.000e+00 | 0.000e+00 | -2.531e-11 | 3.360e-06 | 0.01605 | 2.074 | 1.254e-06 | -2.340e-12 |

# H Software, Reproducibility, and AI-Assisted Work Record

This report is built from the LaTeX sources rooted at `report/final-report.tex`. The solver package is the root Julia package `StenoticHemodynamics`, with package identity in `packages/stenotic-hemodynamics/Project.toml`, resolved Julia dependencies in `packages/stenotic-hemodynamics/Manifest.toml`, and the package entry point at `packages/stenotic-hemodynamics/src/StenoticHemodynamics.jl`. The command-line surface used by the report is `packages/stenotic-hemodynamics/bin/stenotic-hemodynamics`, which dispatches through `packages/stenotic-hemodynamics/bin/stenotic-hemodynamics.jl` and the Julia project under `packages/stenotic-hemodynamics/`.

**Source-control and archive record.** This manuscript refresh was prepared from the local `master` branch. The exact source tree for a submitted PDF should be identified by the git commit or release tag that contains this appendix and the corresponding `public/final-report.pdf`. In this checkout, the local branch and retained release tag are:

```
branch: master
local_tag_available: masters-report-final-2026-06-19 -> 1
    a15a52d6e986f08d745c9e40ecbcd04d4475104
```

The local release manifest remains the concrete source-access record rather than a final public archive record:

```
public/reproducibility/release-manifest.json
```

This appendix therefore records local reproducibility rather than claiming a public archive. No public repository URL, immutable archive URL, or DOI is asserted by this local thesis copy. Unless a later public release adds one of those records, access to `StenoticHemodynamics` is governed by the repository owner's source-control policy.

**AI-assisted drafting and code work.** Substantive AI assistance, including OpenAI ChatGPT and Codex, was used during preparation of this report and repository. That assistance included outlining and reorganizing prose, drafting and polishing manuscript passages, checking consistency between the report and local code, suggesting LaTeX and Julia/Python edits, and helping inspect build, test, and validation output. AI-generated text or code was treated as draft material, not as an independent source of evidence or scientific authority. I reviewed, edited, and accept responsibility for the final manuscript and repository state, including the mathematical statements, citations, code descriptions, figure and table descriptions, numerical claims, and conclusions. No AI tool is listed as an author, credited as a contributor capable of final approval or accountability, or cited as a primary source.

**Computational environment.** The checked Julia environment and local hardware record were:

```
manifest_julia_version: 1.12.6
launcher_julia_version: julia version 1.12.6
operating_system: macOS 26.5.1, Darwin 25.5.0, arm64
hardware: Apple M1 Max, 10 logical CPUs, 32 GiB memory
```

The repository README states the intended portable execution surface as a macOS or Linux shell with Julia 1.12 or newer, Python/Pipenv for auxiliary tooling, and a TeX distribution with `latexmk` and `biber`.

**Review provenance.** The narrative literature-hardening pass was refreshed on 22 June 2026 and considered English-language records from 1973–2026. Representative targeted queries were "one-dimensional blood flow benchmark," "well-balanced blood flow finite volume," "3D 1D fractional flow reserve comparison," "cardiovascular fluid structure interaction verification," and "hemodynamics identifiability surrogate." The working evidence base consisted of 71 BibLaTeX entries, 77 logged source records, a 136-row evidence matrix, and a 169-row derived review-depth matrix generated from the repository tooling. After triage, 42 source records were treated as current cited evidence, 8 as report-adjacent scope checks, 20 as future-work context, 2 as background only, and 5 as deferred pending metadata review. Blank `bib_key` records were not used for citation. The bounded claim register for the main editorial pass is stored at:

```
public/reproducibility/manuscript-claim-register.tsv
```

**Build command.** The report build command is:

```
pipenv run ops-build-report --outdir /tmp/masters-report-build --no-sync-final-pdf
```

Julia package tests are run through the Python ops validation wrapper:

```
pipenv run ops-julia-check
```

**Case-study reproduction commands.** The Section 7 verification tables are regenerated with:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics verify mms \
  --nxs 50,100,200,400 \
  --dt-values 2e-5,1e-5,5e-6,2.5e-6 \
  --tfinal 0.002 \
  --output-dir report/assets/tables/verification \
  --overwrite

packages/stenotic-hemodynamics/bin/stenotic-hemodynamics verify rest \
  --severities 23,40 \
  --nxs 100,200,400,800 \
  --elapsed-times 0,0.001,0.01,0.1,1.0 \
  --output-dir report/assets/tables/verification \
  --overwrite
```

The rest command writes both the drift-history tables and the initial geometry-rest residual-component CSV/TeX assets consumed in Section 7.2.3. The resolved-velocity comparison uses the 23% and 40% stenosis-severity descriptions in the main text. These correspond to local source IDs 77 and 60, respectively. The raw 3D fields are optional local inputs rather than archived thesis assets; the detailed input root and upstream pointer are recorded below.

**Comparison-asset reproduction commands.** When the optional raw 3D inputs are restored locally, the tracked comparison assets are regenerated with:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics compare-3d \
  --data-root public/var/data/simulations/canic_case3 \
  --output-dir tmp/simulations/output/3d_comparison/reference_wall_evidence_nx400 \
  --nx 400 \
  --coordinate-mode reference \
  --target-time 0.9995 \
  --time-atol 1e-6 \
  --section-count 200 \
  --profile-slices 1.951,2.451,2.951 \
  --radial-bin-counts 20 \
  --radial-radius-modes current,reference \
  --overwrite \
  --publish-report-assets \
  --report-assets-dir report/assets/data/stenosis-comparison
```

The deformed-coordinate companion run changes only the coordinate mode and scratch output directory:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics compare-3d \
  --data-root public/var/data/simulations/canic_case3 \
  --output-dir tmp/simulations/output/3d_comparison/deformed_wall_evidence_nx400 \
  --nx 400 \
  --coordinate-mode deformed \
  --target-time 0.9995 \
  --time-atol 1e-6 \
  --section-count 200 \
  --profile-slices 1.951,2.451,2.951 \
  --radial-bin-counts 20 \
  --radial-radius-modes current,reference \
  --overwrite \
  --publish-report-assets \
  --report-assets-dir report/assets/data/stenosis-comparison
```

These commands reproduce the retained reference-coordinate and deformed-coordinate comparison assets. The remaining timing limitation is transient-history matching, not the final timestamp.

The package also contains the retained quasi-static membrane-FSI comparator. It solves the radial membrane update on the current lumen geometry and publishes the report-owned wall profile, fixed-point history, and summary table:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics fsi validate \
  --wall-mode quasi-static \
  --severities 23,40 \
  --meshes 8x2x8,16x4x16 \
  --parallel-workers 0 \
  --output-dir tmp/simulations/output/membrane_fsi_validation \
  --publish-report-assets \
  --report-assets-dir report/assets \
  --overwrite
```

**Optional operator and grid-sensitivity evidence.** The synthetic cross-section operator validation table is regenerated without raw 3D inputs:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics operator-validation \
  --output-dir report/assets/tables/stenosis-comparison \
  --summary-csv report/assets/data/stenosis-comparison/cross-section-operator-validation.csv
      \
  --summary-tex report/assets/tables/stenosis-comparison/cross_section_operator_validation.
     tex \
  --overwrite
```

This run uses an in-memory tetrahedron with constant and affine axial fields. It verifies the synthetic plane-cut reduction used by the cross-section quadrature workflow and is independent of the optional HDF5/XDMF velocity fields.

When the optional raw 3D inputs are available, the publication-scope grid sensitivity is:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics compare-3d \
  --data-root public/var/data/simulations/canic_case3 \
  --output-dir tmp/simulations/output/3d_comparison/grid_sensitivity_severity23_severity40 \
  --target-time 0.9995 \
  --time-atol 1e-3 \
  --nxs 200,400,800 \
  --section-count 200 \
  --radial-bins 20 \
  --no-svg \
  --overwrite \
  --grid-summary-csv report/assets/data/stenosis-comparison/grid-sensitivity-summary.csv \
  --grid-summary-tex report/assets/tables/stenosis-comparison/grid_sensitivity_summary.tex
```

The manuscript treats this as output-sensitivity evidence for the retained 23% and 40% severity section means and physical flows. It does not change the matched-data boundary: wall, boundary, material, current-geometry, and transient-history metadata for the resolved cases remain unmatched or unpersisted.

Package-benchmark assets are regenerated separately from the PDF:

```
packages/stenotic-hemodynamics/bin/stenotic-hemodynamics benchmark \
  --profile overnight \
  --output-dir tmp/simulations/output/package_benchmark/overnight-YYYYMMDD \
  --overwrite \
  --include-resolved3d \
  --publish-report-assets

pipenv run ops-render-package-benchmark-figures \
  --benchmark-dir tmp/simulations/output/package_benchmark/overnight-YYYYMMDD
```

Those rows are secondary numerical diagnostics, not the principal comparison method.

**Raw 3D inputs and manifest.** The raw upstream XDMF/HDF5 velocity inputs used for Section 7.3 are not archived in this report package. The expected local input root is:

```
public/var/data/simulations/canic_case3/
```

When available, that directory is restored from this upstream repository, commit, and subtree:

```
https://github.com/qcutexu/Extended-1D-AQ-system
056a9da2b36b480691f18025d242d2c00f6e7180
case3_all_3d_results/
```

For each local case used by the resolved-FSI loaders, the expected companion files are `velocity.xdmf`/`velocity.h5`, `pressure.xdmf`/`pressure.h5`, and `displace.xdmf`/`displace.h5`. The report archives generated CSV and PGFPlots-ready static assets, not the raw fields. The retained comparison assets include the reference-coordinate and deformed-coordinate section summaries consumed by Section 7.3.4; wall, boundary, material, and earlier transient histories remain unresolved unless a later release entry adds those data.

The tracked release manifest is stored outside the thesis PDF:

```
public/reproducibility/release-manifest.json
```

That manifest records the build command, raw-input policy, upstream 3D input pointer, comparison-asset reproduction command, and the absence of a declared public repository URL or archival DOI. No populated standalone SHA-256 file-hash manifest was present in the tracked source tree inspected for this revision; per-run workflow manifests record SHA-256 output hashes when those workflows are regenerated.

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